

School of Electrical Engineering, Computing and Mathematical Sciences

Evaluating Methods for Reliable Question Answering over Governance Documents

by

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Abstract

Retrieval-Augmented Generation (RAG) has emerged as a promising approach for improving the accuracy of large language models by grounding generated responses in external documents. However, challenges involving hallucinations, incomplete information, and incorrect interpretation remain, particularly when answering questions over complex governance documents. This research evaluates methods for improving the reliability of RAG-based question answering using governance documents from the IEEE Systems, Man, and Cybernetics Society.

Five architectures were evaluated: Baseline RAG, RISE-inspired refinement, RLVR-inspired verification and correction, Multi-Model Verification (MMV), and Explainable AI (XAI). A benchmark comprising 110 questions across 16 categories was developed to assess answer correctness, textual and semantic similarity, and computational performance.

The baseline architecture achieved an accuracy of 94.55%, while RISE-inspired refinement achieved the highest measured accuracy of 99.09%, followed by RLVR at 97.27%, MMV at 96.36%, and XAI at 95.45%. However, the results also highlighted limitations in automated evaluation, verification consistency, and explanation reliability. Furthermore, all enhanced architectures introduced additional computational overhead, demonstrating trade-offs between reliability-related capabilities and response efficiency.

Overall, the findings suggest that selective refinement and evidence-based verification can support more reliable governance-document question answering. However, the effectiveness of these approaches depends heavily on the accuracy of their evaluation and verification mechanisms, highlighting the importance of reliable assessment alongside answer generation.

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Dear Dr. Ehsan Pashajavid,

I, Lucas Zheng Yi Lim, hereby submit my thesis entitled “Evaluating Methods for Reliable Question Answering over Governance Documents.” as part of my requirements for completion of the Bachelor of Engineering (Honours) in Electrical and Electronic Engineering.

I declare that this thesis is entirely my own work with the exception of the acknowledgements and references mentioned.

Yours sincerely,



Lucas Zheng Yi Lim

20841085

Declaration of Published Work

I also acknowledge that parts of the progress report presented at the end of the first semester of this project have been used in the following chapters of this thesis, and the report has been properly referenced:

- Chapter 1: Introduction – Background information, research motivation, aims, and objectives.
- Chapter 2: Literature Review – Relevant literature, theoretical background, and previous research.
- Chapter 3: Methodology – Research approach, experimental design, and evaluation methods.
- Chapter 4: System Design and Implementation – System architecture, technical specifications, and implementation details.

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1 Introduction

1.1 Background and Context

Governance documents play an important role in defining the organisational structure, responsibilities, procedures, and decision-making processes of institutions and professional organisations. Documents such as constitutions, bylaws, operational manuals, and technical manuals establish the rules that govern organisational activities and to provide guidance for members, committees, and administrative bodies. The accurate interpretation of these documents is essential for ensuring that the decisions and actions made by the organisation remain consistent with the established policies and procedures. However, retrieving information from governance documents can be challenging in itself. Relevant provisions may be distributed across multiple documents, expressed using specialised terminology, or dependent on exceptions and conditions defined elsewhere. In addition, the documents may differ in authority, meaning that the interpretation of a particular provision requires an understanding of the relationship between different all governance sources. These characteristics can make answering even seemingly straightforward questions difficult, particularly when information must be combined or interpreted across several of the documents.

Recent developments in large language models (LLMs) have created opportunities to improve access to document-based information through natural-language question-answering (QA) systems. Rather than manually searching through lengthy documents, users can submit questions and receive responses generated in natural language. However, LLMs may produce incorrect, incomplete, or unsupported information, commonly described as hallucinations [1]. Such limitations are particularly concerning in governance-related applications, where inaccurate responses may lead to misunderstandings of organisational rules or procedures. Retrieval-

Augmented Generation (RAG) has emerged as an approach for addressing some of these limitations by incorporating information retrieved from external documents into the answer-generation process. Instead of relying solely on knowledge acquired during model training, a RAG system retrieves relevant passages from a document collection and provides them as supporting context for generating an answer. This allows responses to be grounded in the documents relevant to a particular question.

Nevertheless, access to retrieved evidence does not guarantee that a generated answer will always be correct. A system may retrieve incomplete or irrelevant information, misinterpret a provision, overlook an exception, or generate conclusions that are not adequately supported by the available evidence. Hence, additional mechanisms may be required to assess and improve the reliability of RAG-generated responses. Several approaches have been proposed to address these challenges, including iterative refinement, verification-guided correction, the use of multiple language models for independent assessment, and explainable AI techniques. Although these approaches target different aspects of reliability, their effectiveness may vary depending on the document collection, question characteristics, and computational resources available.

1.2 Problem Statement

The application of RAG to governance-document question answering presents a fundamental challenge. It is that generating a plausible response is not equivalent to generating a correct and sufficiently supported response. Governance documents frequently contain detailed provisions, conditional requirements, and relationships between different levels of organisational authority. A generated answer may accurately be able to identify a relevant provision while omitting an important qualification or incorrectly combining information from multiple sources. These errors can be difficult to detect as the resulting responses may still remain fluent and appear to be authoritative [1].

Although conventional RAG provides retrieved documentary context to support answer generation, it does not independently guarantee that the generated response correctly interprets that context. Furthermore, enhancement techniques designed to improve reliability may introduce additional language-model judgments that they themselves are susceptible to creating errors. For example, an iterative refinement mechanism may incorrectly modify an initially correct response, while a verification model may classify a supported answer as unsupported. Similarly, an explanation mechanism may provide apparently relevant evidence without being able to correctly establish the relationship between that evidence and the generated claim.

These challenges raise questions regarding the effectiveness of different RAG enhancement techniques, the types of errors they can address, and the additional computational costs that they introduce. A systematic evaluation is therefore required to determine how these techniques perform when applied to a consistent governance-document corpus and question set. Such an evaluation must consider not only the overall answer correctness but also performance across different question categories, the reliability of verification and explanation mechanisms, and the computational resources required by each approach.

1.3 Research Aim and Objectives

The primary aim of this research is to evaluate the effectiveness of different methods for improving the reliability of retrieval-augmented QA over governance documents.

To achieve this aim, the study establishes a baseline RAG architecture and compares it against four enhanced configurations incorporating different reliability mechanisms.

The specific research objectives are:

1. Develop a baseline RAG question-answering system using a selected collection of governance documents.
2. Implement and evaluate RISE-inspired refinement, RLVR-inspired verification and correction, Multi-Model Verification (MMV), and Explainable AI (XAI) as enhancements to the baseline architecture.
3. Construct a benchmark of governance-related questions covering different retrieval, reasoning, and interpretation requirements.
4. Compare the architectures using answer correctness, textual and semantic similarity, and question-category performance.
5. Examine the effectiveness and limitations of the enhancement mechanisms in identifying, correcting, or explaining potentially incorrect responses.
6. Evaluate the computational requirements of each architecture, including response latency and language-model calls.
7. Identify the practical implications, limitations, and trade-offs associated with applying these approaches to reliable governance-document question answering.

1.4 Research Questions

This study seeks to address the following research questions:

RQ1: How do different RAG enhancement techniques affect the correctness of answers generated from governance documents, both overall and across different question categories?

This question is to investigate whether the enhanced architectures achieve different levels of answer correctness compared with the baseline RAG system. It also examines how their performance varies across the different question categories requiring diverse forms of retrieval, reasoning, and interpretation.

RQ2: To what extent can refinement, evidence-based verification, multi-model verification, and explainability mechanisms identify, correct, or expose errors in governance-document QA?

This question examines the effectiveness of the enhancement mechanisms beyond overall accuracy. It considers the correction of initially incorrect responses, the reliability of verification judgments, the usefulness of explanations, and the limitations associated with automated assessment.

RQ3: What computational costs are associated with the evaluated RAG enhancement techniques, and how do these costs compare with their observed reliability benefits?

This question evaluates the additional computational requirements introduced by each of the enhancement techniques, particularly the response latency and language-model calls. It considers whether the observed benefits justify the additional processing requirements and how these trade-offs may influence the suitability of different architectures for practical applications.

1.5 Scope of the Research

This study focuses on question answering over a selected collection of governance documents associated with the IEEE Systems, Man, and Cybernetics Society (SMCS). The document corpus includes constitutional and bylaw provisions, an operational manual, and a technical manual describing organisational procedures and responsibilities.

The experimental system uses a retrieval-augmented generation pipeline incorporating document preprocessing, text chunking, embedding-based retrieval, and local language-model inference. A baseline configuration is established and extended through four enhancement approaches: RISE-inspired refinement, RLVR-inspired verification and correction, MMV, and XAI.

The architectures are evaluated using a benchmark containing 110 questions distributed across 16 categories. These categories include direct retrieval, cross-document reasoning, semantic retrieval, ambiguity, contradictions, temporal interpretation, negation, conditional reasoning, and other question types relevant to governance-document interpretation.

The evaluation considers answer correctness, textual and semantic similarity metrics, category-level performance, and computational performance. Architecture-specific results are also examined to assess the behaviour of refinement, verification, and explanation mechanisms.

The study is limited to the selected governance-document collection, implemented model configurations, and experimental environment. It does not involve fine-tuning the underlying language models or deploying the system for operational governance decisions. The findings are therefore intended to only provide comparative experimental evidence rather than establish the suitability of any architecture for unrestricted real-world use.

1.6 Significance of the Research

This research contributes to understanding how different reliability enhancement mechanisms behave when applied to retrieval-augmented QA over governance documents. First, it provides a controlled comparison of multiple enhancement approaches using a consistent document corpus, question benchmark, and evaluation framework. This allows differences in answer correctness and computational performance to be examined within a common experimental setting. Second, the study considers reliability beyond overall answer accuracy by examining refinement outcomes, verification judgments, and explanation quality. This is particularly relevant because an architecture may provide additional verification or interpretability without directly improving the correctness of its generated answers. Third, the research investigates the computational trade-offs associated with introducing additional language-model processing stages. Understanding these trade-offs is important when considering the practical feasibility of reliability-enhanced RAG systems. Finally, the study highlights the importance of evaluating the reliability of the assessment mechanisms themselves. Verification, refinement, and explanation techniques may introduce additional opportunities for incorrect judgments, meaning that increased architectural complexity does not necessarily guarantee more trustworthy responses. By examining these considerations within a governance-document setting, the research provides insights that may inform the development and evaluation of future document-grounded question-answering systems.

1.7 Thesis Organisation

The remainder of this thesis is organised as follows.

Chapter 2: Literature Review examines relevant research on large language models, retrieval-augmented generation, hallucination and reliability challenges, and techniques for refinement, verification, and explainability.

Chapter 3: Methodology presents the experimental design, governance-document corpus, benchmark construction, evaluation framework, performance metrics, and procedures used to compare the architectures.

Chapter 4: System Design and Implementation describes the baseline RAG pipeline and the implementation of the RISE-inspired, RLVR-inspired, MMV, and XAI configurations.

Chapter 5: Results present the experimental findings, including overall question-answering accuracy, category-level performance, textual and semantic similarity, computational requirements, and architecture-specific outcomes.

Chapter 6: Discussion interprets the findings, examines the effectiveness and limitations of the enhancement techniques, evaluates reliability and computational trade-offs, and considers the implications for governance-document question answering.

Chapter 7: Conclusion and Future Work summarises the main findings, addresses the research questions, and identifies opportunities for further investigation.

2 Literature Review

2.1 Introduction and Background

Large language models (LLMs) are very powerful AI tools that are capable of natural question answering. There are many of these LLMs that can be found online or as a part of various societies. Some of these may include ChatGPT (Open AI), Gemini (Google), Grok (xAI), Claude (Anthropic), etc. These LLMs are used in a variety of industries both in professional and personal spaces. However, it can be argued that their consistency and reliability are a point of concern, particularly in areas where accuracy and correctness are of upmost importance. Due to the manner in which LLMs are processed and trained, certain errors may occur. One such error are the possibilities of hallucinations occurring in the generated responses [1].

Artificial Intelligence (AI) hallucinations are when the AI model has presented false, misleading, or fabricated content in its responses during question answering (QA). The degree of these errors may range from a slight error to a completely falsified statement made by the AI model. Hallucinations may occur from a variety of reasons. Dataset inaccuracies, where large datasets contain information that are not entirely true or are biased which affects the training of the model. The generation process of the model, since LLMs work by prediction of the next word in a sequence, it can lead to divergence from accuracy of content. There is also the issue of a lack of grounding, since most models do not have any basis on any grounded documents, it leads to outputted generations that are far from the truth [1].

A wide range of techniques have been proposed in the literature to mitigate hallucinations in responses generated by large language models. These include approaches such as Reinforcement Learning with Self Verification (RISE), Reinforcement Learning with Verifiable Rewards (RLVR), extractive methods, multi-modal verification frameworks, and Explainable AI (XAI) [1]. Collectively, these methods target different stages of the model lifecycle, from training and

fine-tuning to inference and post-generation validation. Rather than relying on a single solution, existing research emphasises a combination of strategies designed to either reduce the likelihood of hallucinations occurring or to detect and limit their impact when they do arise. This highlights the complexity of the problem and the need for context dependent approaches when deploying large language models in high-stakes environments.

Hallucinations in generative AI present a significant challenge to the accuracy and reliability of question answering systems, particularly in high-stakes contexts such as organisational governance. Many institutions rely on formal documents, including constitutions, bylaws, and operational manuals, where precise interpretation and accurate citation are essential. Consequently, ensuring that QA models remain faithful to source material and do not produce unsupported or fabricated responses is critical for their safe application. Existing research has explored a range of technical approaches to mitigate hallucinations in generative AI, focusing on improving response accuracy, source attribution, and model reliability. This literature review examines and analyses these approaches, highlighting their effectiveness and limitations, and identifying the contexts in which they are most appropriately applied.

2.2 Review of Existing Work/State of the Art

2.2.1 Why Hallucinations Occur

Over the past decade, there have been many advancements in the fields of AI development and research, as well as increasing implementations of AI and AI-related models being used in a variety of industries and disciplines. As such, many improvements have been made to the smoothness of AI and its ability to generate human-like responses to questions due to the increasing use and demands of AI-related content. However, rising usage of AI in critical spaces, such as governance documents, creates concerns on the accuracy of Generative Pre-trained Transformers (GPTs) as they are at risk of hallucinating in their responses which is of upmost

importance when information must be exact. Hallucination occurs when a model generates information that may seem to be accurate and correct to the reader but in actuality, is false, usually fabricated, and unsupported in any source data. This is usually due to the way LLMs like GPTs operate, as they generate responses by predicting the most likely next word in a sentence rather than verifying its responses in any form of truth. Hence, models will hallucinate when issues like, incomplete training and overgeneralisations occur.

2.2.2 Hallucinations and Types of Hallucinations

The hallucinated errors can appear and present itself in a variety of ways. These might include overfitting, logic errors, reasoning errors, mathematical errors, unfounded fabricated, factual errors, and text output errors [2]. One of the most common and significant problems in model development is overfitting. Overfitting is when the model learns too well on the training data, which leads to it memorising the patterns of the dataset rather than learn the general rules. It also causes the model to be more sensitive to other unwanted relationships, including noise and other irrelevant patterns. Overfitting generally occurs in the training stage, where the model could be too complex for the data set or there are a lot of noise and inconsistencies in the data. Hence, whilst the model may perform very well on the training data, it will perform poorly on any new or unseen data. Other errors may be logic errors, these are when there is no causal relationship between the generated response and the inputted query. There are reasoning errors, which is when the model fails to understand the idiosyncrasies of human life. This would be humour, sarcasm, metaphors, interpersonal relationships, human behaviour, mental processes, etc [2].

There exist mathematical errors which is simply when the model fails to comprehend mathematical concepts such as fractions, decimals, imaginary numbers, geometry, positive/negative numbers, etc. It may also fail to understand the usage of the imperial and metric systems, conversion, and representation. A model may simply present entirely false information in its generated response, these are called unfounded fabricated errors. Examples of these may

be false opinions or cases in health information and fictitious papers or references. A model may also produce factual errors, which is when the model produces information that are contrary from the truth or common sense. Lastly, text output errors are when the model has made an error regarding linguistics. This would include translation errors, where it mistranslates information between languages, spelling errors, the failure to correctly spell or generate the correct text, and grammatical errors, which is the non-conformity to language rules such as lexical or syntax errors [2].

2.2.3 Long-Context and Multi-Document Reasoning

In addition to the issue of hallucination, the ability of LLMs to be able to utilise information effectively withing long contexts presents a further challenge for retrieval-augmented QA. A study investigated this limitation through multi-document QA and key-value retrieval tasks. An examination of how the position of relevant information within the input context affected model performance is then performed [3]. The results from their findings reveals that the presence of relevant information within the context does not guarantee that it will be effectively utilised by the LLM. It was found that the performance was generally highest when relevant information appeared near the beginning or end of the input context, while the performance degraded when relevant information was positioned within the middle of longer contexts.

These results have a particular implication for RAG systems, where numerous retrieved passages may be supplied to the language model simultaneously. When increasing the number of retrieved documents, additional potentially relevant information can be acquired. However, it also increases the amount of context that a model must process through. It was found that model performance could become saturated before the retrieval recall [3]. This would suggest that the retrieval of additional relevant information may not necessarily result in a corresponding improvement in the performance of the model's QA. Hence, the effectiveness of the RAG system

does not only depend on whether relevant evidence was retrieved, but also on the model's ability to identify and appropriately utilise the evidence within the supplied context.

This limitation becomes relevant to the question answering over governance documents, where the information required to answer questions can be found across multiple documents, sections, or provisions. Any relevant information may therefore appear alongside various other retrieved passages, increasing the difficulty of consistently identifying and integrating the appropriate evidence. These findings highlights the importance of retrieval quality, context construction and mechanisms for verifying that generated responses are appropriately grounded in the retrieved evidence.

2.2.4 Model Designs that Counter Hallucinations

Due to the increasing demand of AI usage in critical spaces over the past few years, many advancements have been made to tackle the emergence of hallucinations in the generated responses made by a generative AI model. One of the most notable developments is the design and implementation of the Retrieval Augmented Generation (RAG). A RAG works by applying a retrieval step in its process of generating a response in QA. In this retrieval step, the model accesses the necessary documents associated with the query and generates the response based on the retrieved information [4]. This reduces the risks of hallucinations as all responses generated by the model must be grounded in some form of document in its database [5]. Therefore, it can be said that all information produced by a RAG must contain some form of 'truth', advocating for reliability and confidence in its responses. However, as with many of the notable developments to be mentioned, RAGs have certain niche concerns attached to it due to its operation. RAGs lack a certain flexibility that general GPTs have. This is due to the nature of RAG systems, any queries from the QA that are poorly specified of require complex contextual understanding can lead to ineffective document retrieval, resulting in less accurate or incomplete responses [5]. Other issues concerning RAGs might pertain to privacy concerns of generated

information. Since RAGs must have access to documents related to associated topics to generate a pragmatic response, there lies a possibility of private information being revealed through the produced response [4]. This creates issues when RAGs are implemented/used in fields such as medicine where the RAG has accessed a patient's medical records and disclosed said information. Other fields like law, business, and even private businesses face the same concerns when trying to integrate RAGs into their scopes. Hence, while RAGs greatly reduce the chance of hallucinations in generated outputs by LLMs, it comes with their own set of risks and problems in turn as well.

There have been a considerable number of other strategies developed used to reduce hallucinations in generated responses. Some of these include Reinforced Learning with Self Verification (RISE), Reinforced Learning with Verifiable Rewards (RLVR), Extractive AI, Multi-model verification, and Explainable AI (XAI) [6]. RISE is an AI model where every response generated by the model is scrutinised and investigated on its accuracy and truthfulness to improve further response generation [7]. Hence, over time and through thorough repetition of this process, it trains the model to make more accurate responses to queries. This differs from how RAGs work as instead of influencing the generating step of the model, it works on the finished generated responses.

RISE is useful as it does not rely on external sources of information and is therefore more applicable to a more general usage across different domains. It does, however, greatly increase the complexity of the model as well as the computational resources required to operate the model. In addition to this, there is a dependency on the model's verification ability. The model must be able to note a clear 'correct' answer which means it might have trouble with more ambiguous queries that the model is unable to specify a 'correct' answer. RISE can be a powerful tool to

reduce hallucinations in generated responses with the condition that the self-verification ability of the model is precise and accurate in its ability to examine the quality of the responses.

RLVR is similar to RISE in the sense that the operate to judge the accuracy of the responses generated by the model. It differs from RISE in the sense that RISE does its examinations of the responses internally whereas RLVR does it externally. Besides this distinction, RISE and RLVR share many similarities in their operations. RLVR optimises its responses through its external verification that judges its accuracy and truthfulness to reduce the chance of hallucinations [8]. Due to using external verification, RLVR is especially useful in structured tasks such as mathematical reasoning, coding, etc. However, similarly to RISE, RLVR has a reliance on its external verification in its ability to inspect responses. Hence, RLVR is constrained by the ability of its external verifications [8]. Increased computational cost and complexity are additional downsides of the implementation of RLVR.

Other developments like extractive AI work based on the same concept as RAGs. The key difference between extractive AI and RAGs is that extractive AI do not generate responses. Instead, it will locate key pieces of information within its dataset and presents it [6]. This completely removes the risk of hallucinations as the model does cannot generate any false information. Hence, extractive AI is useful in ensuring the upmost accuracy and reliability in data extraction and information retrieval [6]. However, it is this very aspect of extractive AI that is also its greatest flaw. Extractive AI are incredibly inflexible and unable to handle convoluted queries. This is because the model can only present the source material found in its datasets. It cannot rephrase or generate new content beyond what it already has access to. It also lacks the ability to generate human-like responses which reduces the familiarity between the user and the model in QA. Therefore, extractive AI should be considered as a useful design model only in niche situations where queries are not vague, and it is being used mostly in contexts where it is mainly retrieving information.

There have also been approaches that have been developed that validates responses through cross checking, this is referred to as multimodal verification. Multimodal verification reduces the risk of hallucinations in generated responses since inconsistencies between modalities can be detected which helps to identify any fabricated or false outputs [9]. This approach greatly increases the robustness of the model as it is less reliant on any one single source of information, reducing any errors in any one modality. However, multimodal verification has an extremely high computational cost that comes from the processing, aligning and optimisation of multiple modalities, that is in itself a complex task that is technically challenging. Different modalities may not always be applicable depend on the context on which it is used, thereby restricting its usage across different fields. There is also the risk of the different modalities producing conflicting signals which would make it difficult to consolidate a final response or verify a correct answer.

Other model designs such as XAI attempt to build trust with users on its responses through other means besides improving its generative capabilities. XAI aims to explain its reasoning and thought processes to make model decisions more transparent, allowing users to understand how responses are generated and how the model got to that decision [10]. This enhances trust and accountability with the response as it allows the user to assess whether it was supported by some form of reasoning or evidence. This is critical in high-stakes situations as although it might not actually reduce the chance of a hallucination, it will help the user by revealing any inconsistencies or unsupported claims in the generated output. By showing its process, it aims to reduce the effect of the black box problem of AI, and help developers identify sources of error, including patterns that may lead to hallucinations in generated outputs [10]. Therefore, while XAI does not directly reduce the chance or effect of hallucinations in generated responses, its aims to reveal the models process to improve trust and interpretability, revealing any inconsistencies and sources of error that is usually difficult to uncover.

2.2.5 Self-Reflective Retrieval-Augmented Generation

While the conventional RAG can improve the factual accuracy of LLMs by providing access to external information, it is a fact that the generation process cannot inherently guarantee that retrieved passages are relevant or that the generated responses will remain consistent with the provided evidence. A research study proposed Self-Reflective Retrieval-Augmented Generation (Self-RAG), a framework designed to improve the quality of generations and factuality through adaptive retrieval and self-reflection [11]. Unlike the conventional RAG approaches that indiscriminately retrieve a fixed number of passages, the Self-RAG enables the model to determine when retrieval is required and to evaluate both the retrieved information and its own generated responses.

Self-RAG incorporates specialised reflection tokens representing retrieval and critique decisions throughout its generation process. The framework first determines whether additional information should be retrieved for a given input. When a retrieval is required, retrieved passages are then evaluated for relevance before being incorporated into the response. The model would the generate critique tokens to allow it to assess the output according to factors including factual support and overall response quality. This allows for the retrieval, generation and evaluation to operate as interconnected stages rather than treating the retrieved information as automatically relevant and reliable.

Experimental evaluation demonstrated that the Self-RAG model was able to outperform RAG models across several knowledge-intensive tasks, including open-domain QA, reasoning and fact verification [11]. Improvements were also observed in factuality and citation accuracy for long-form generation. These findings were able to demonstrate the potential value of incorporating verification and self-reflection mechanism into the RAG architecture, particularly in cases where the generated responses must be grounded in retrieved evidence.

For QA in governance documents, these principles become particularly relevant because the retrieval of a passage alone does not directly guarantee that the passage is relevant to the question or that the generated response was able to accurately reflect its contents. The incorporation of mechanisms that evaluate retrieved evidence and critique generated responses may therefore be able to provide additional safeguards against any unsupported or inaccurate answers. Self-RAG consequently provides an important example of how conventional RAG architectures can be extended beyond retrieval and generation processes to incorporate explicit verification and reflection processes.

2.2.6 Corrective Retrieval Augmented Generation

Although RAG can improve the factual grounding of LLM, its effectiveness still remains dependant on the relevance and quality of the retrieved information. Therefore, incorrect or irrelevant retrieval can provide misleading context to the language model and also reduce the reliability of the generated response. One such study proposed Corrective Retrieval Augmented Generation (CRAG) to address this limitation by introducing mechanisms for evaluating and correcting retrieved information before the response is generated [12].

The CRAG model incorporates a lightweight retrieval evaluator that assesses the quality of retrieved documents and assigns a confidence level that determines the subsequent retrieval actions. When information retrieved by the model is deemed to be insufficient or unreliable, the framework can obtain supplementary information through external web searches. CRAG additionally applies a decompose-then-recompose strategy that extracts any relevant information from the retrieved documents whilst filtering unnecessary or potentially misleading content [12]. This allows for the generation stage to operate ion a more focused and reliable evidence set.

The framework demonstrates that there is an important distinction between the model retrieving information and verifying whether that information is actually suitable to be used for generating

an answer to a query. The experimental evaluation across multiple QA datasets showed that there was an improvement over the conventional RAG approaches. This indicates that the corrective retrieval mechanisms can improve robustness when initial retrieval is unreliable. This becomes particularly relevant to reliable QA over governance documents, where passages of similar semantics may not necessarily contain the authoritative or applicable provision. Hence, retrieval evaluation and evidence refinement provides useful mechanisms for reducing the propagation of retrieval errors into generated responses.

2.2.7 Evaluation Metrics

An evaluation into a useful metric to judge and analyse the correctness of a generated response and predictions of a model must also be criticised. This is to assess and evaluate the utility and practicality of a model in a QA scenario. These metrics include accuracy, precision and recall, and F1-score [13]. Accuracy, is mathematical defined as:

$$\text{Accuracy} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \qquad (1)$$

Where TP is true positive, TN is true negative, FP is false positive, and FN is false negative [13].

In this way, a high accuracy value would be an indicator of the model's reliability and ability to produce correct generated responses and vice-versa. However, this metric is generally unreliable in many use cases. This is due it not being able to observe the imbalance in classes. This refers to when it cannot understand the difference in quantities of its classes in its dataset, hence being unable to properly reflect real-world properties. Another issue with using accuracy as an evaluation metric is that it fails to properly account for outliers such as noise, incomplete data, and changing environments [13]. Hence, whilst accuracy is an effective metric to provide a base estimate of a model's capabilities, it fails to provide a comprehensive check of a model's ability to manage real-world applications and will tend to overestimate its performance which might lead to overconfidence in the model [13].

There is other more detailed evaluation metrics as compared to accuracy, such as precision and recall. Precision is the measurement of how accurate the positive predictions made by the model are. Precision is calculated by the following formula:

$$\text{Precision} = \frac{\text{TP}}{\text{TP+FP}} \qquad (2)$$

Where TP is true positive, and FP is false positive [13].

A higher precision value assumes the model's accuracy in judging positive predictions correctly and produces less false positive results. This metric is useful in cases where false positives are detrimental to the operations of a system. Examples of these may be fraud detection, autonomous vehicles, facial recognition, etc.

Recall, also known as sensitivity, is the model's ability to determine the proportion of positives across all actual positive results. Recall is defined as:

$$\text{Recall} = \frac{\text{TP}}{\text{TP+FN}} \qquad (3)$$

Where TP is true positive, and FN is false negative.

Hence, recall is a measurement of the ability of a model to identify all relevant instances by quantifying the proportion of true positives detected out of all the actual positives. Recall becomes a useful metric in cases such as medical diagnosis security systems.

The difference between precision and recall is that precision measures how many of the predicted positive results are correct whereas recall measures how many positive results are successfully identified and predicted. Precision and recall will generally exhibit trade-offs, where increasing precision will lower recall and increasing recall with lower precision. Finding an appropriate balance between precision and recall would generally depend on the situation the model is being applied in. However, despite providing a more insightful evaluation on the model's capability as

compared to accuracy, similarly to accuracy, it is still unable to account for real-world and practical factors such as robustness and contextual relevance.

To find an appropriate balance between precision and recall, the F1-score is a composite evaluation metric that is a combination of the two aforementioned metrics. It is used to find a balance between the number false positives to the number of false negatives and is useful when class distributions are imbalanced [13]. The F1-score is calculated using this formula:

$$\mathrm{F1} = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \qquad (4)$$

Where Precision and Recall are calculated as shown by equations (2) and (3).

By using the F1 score formula, it will only produce a high value when both the precision and recall values are reasonably strong. This allows for a more comprehensive and equal assessment of a model's capabilities compared to using the individual metrics separately. Despite all this, the F1-score still fails to capture many real-world elements and will ultimately fail to capture the resulting factors of incorrect predictions in practical scenarios.

While accuracy provides a general measure of correctness, precision and recall capture different aspects of model performance by focusing on false positives and false negatives respectively, with the F1-score offering a balanced metric that is particularly useful in evaluating QA systems where both correctness and completeness are critical.

Evaluation metrics that are able to measure more useful quantities such as faithfulness and hallucination rate would be more effective in judging the applicability of a model in situations of reducing hallucinations. Faithfulness is a measure of how grounded an answer is in the source documents. Simply put, it is whether the generated response by the model is supported by some form of formal documentation. Some of the most commonly used faithfulness metrics include Decision Flip - Most Informative Token (DFMIT), Decision Flip - Fraction of Tokens (DFFOT),

Comprehensiveness (COMP), Sufficiency (SUFF), Correlation between Importance and Output Probability (CORR), and Monotonicity (MONO) [14].

DFMIT is a metric that only focuses on the most important token in a batch. If the prediction label is changed when the most important token is removed, it will assume that the interpretation made by the model is faithful [14].

$$\mathrm{DFMIT} = \begin{cases} 1, & \text{if } c(x) \neq c(x \backslash x_{:1}) \\ 0, & \text{if } c(x) = c(x \backslash x_{:1}) \end{cases} \qquad (5)$$

Where a score of 1 would imply that the interpretation has been faithful.

DFFOT measures faithfulness as the minimum fraction of important tokens needed to be erased in order for the model's decision to be changed [15].

$$\mathrm{DFFOT} = \begin{cases} \min \frac{k}{l_x}, \ \text{s.t. } c(x) \neq c(x \backslash x_{:k}) \\ 1, \ \text{if } c(x) = c(x \backslash x_{:k}) \text{ for any } k \end{cases} \qquad (6)$$

A low value for DFFOT would mean that the interpretation has been more faithful.

COMP assumes the faithfulness of an interpretation based on if the important tokens generally represent the entire input sequence. The faithfulness score, in this case, is measured by the original predicted classes output probability changing due to important tokens being removed [14].

$$\mathrm{COMP} = \frac{1}{|B|} \sum_{q \in B} \left(p_{c(x)}(x) - p_{c(x)}\left(x \backslash x_{:q\%}\right) \right) \qquad (7)$$

The higher the COMP score, the more faithful the interpretation made by the model would be.

SUFF is a metric that decides if the important tokens are comprised of enough information to retain the prediction. Only the important tokens are retained and the change in output probability is calculated compared to the original specific predicted class [15].

$$\mathrm{SUFF} = \frac{1}{|B|}\sum_{q\in B}\left(p_{c(x)}(x) - p_{c(x)}\left(x_{:q\%}\right)\right) \qquad (8)$$

The lower the SUFF score, the more faithful the interpretation made by the model would be.

CORR assumes that the interpretation is faithful there is a positive correlation between the importance of the token and the corresponding predicted probability when the most important token is continuously removed [14].

$$\mathrm{CORR} = -\rho(u, p) \qquad (9)$$

Where u is the importance of the token in descending order and $p = [p_{c(x)}(x\backslash x_1), p_{c(x)}(x\backslash x_2), \ldots, p_{c(x)}\left(x\backslash x_{l_x}\right)]$. p(.) denotes the Pearsons correlation [15]. The higher the CORR value, the more faithful the interpretation made by the model would be.

MONO will assume that an interpretation is faithful under the condition that when incrementally adding more important tokens, the probability of the predicted class will increase monotonically. Features are added in an ascending order of importance, starting from an empty vector, to which any corresponding classification probabilities would be recorded down. MONO calculates the relationship between the importance of the features and the probability after adding each of the features.

$$\mathrm{MONO} = \rho(u, p) \qquad (10)$$

Where u is the importance of the token in descending order and $p = p_{c(x)}(x), p_{c(x)}(x\backslash x_{:1}), p_{c(x)}(x\backslash x_{:2}), \ldots, p_{c(x)}\left(x\backslash x_{:(l_x-1)}\right)]$. p(.) denotes the Pearson correlation [15]. The higher the value of MONO, the more faithful the interpretation of the model would be.

In many faithfulness metrics, they tend to suffer from a variety of issues that tend to affect the overall functioning of the model. One of these are Out-of-distribution (OOD) inputs from perturbation methods. Many faithfulness evaluation metrics work by identifying important words

and features, remove or mask them, and then checks how the predictions of the models have changed. This works based on the assumption that if the feature has been removed and the performance of the model has decreased, it means that that feature was important. However, it is in this same way that OOD inputs are created. The issue with this is that it is testing the capability of a model based on broken inputs rather than the true reasoning competence of the model. This is detrimental because it may lead to false important signals, as unimportant signals may seem important as removing them will break the input, or unstable results which occur when the faithfulness metrics use different masking strategies which create different OOD inputs and therefore different scores.

Fine-tuned Fidelity (F-Fidelity) is a faithfulness evaluation metric that overcomes the issue of OOD inputs. F-Fidelity works in two parts, the first is "an explanation-agnostic fine-tuning strategy to prevent information leakage and second is a controlled random masking operation to overcome the OOD issues observed in the application of prior evaluation metrics" [15]. The fine-tuning process of F-Fidelity employs a variety of random masking techniques, such as randoms placing pixels in different locations of an image or time gaps in time series datasets to create improved training samples which is then used to optimise the model. F-Fidelity reduces the risk of information leakage and label bias by using explanation-agnostic stochastic masks, whilst improving the efficiency of the model by eliminating the tediousness of retraining models for each individual explainer. At the evaluation stage of the model, F-Fidelity applies a removal strategy that generates indiscriminate masking that are developed through the output of the explainer, designed to be in-distribution in regard to the masks that are being used in the fine-tuning stages of the model [15]. This effectively cancels out the OOD issues that can be observed in previously developed faithfulness evaluation metrics

2.2.8 Evaluating Correctness and Faithfulness in QA

Reliability of the evaluation of QA systems requires consideration of both the correctness of generated responses and their grounding in any provided information. A study found that properties such as correctness with respect to the information that a user needs and faithfulness in respect to the supplied knowledge [16]. Responses are able to produce the correct answer in addition to supplementary information that is unsupported by the retrieved evidence. This demonstrates that using correctness as an evaluation criterion by itself is insufficient for reliable QA.

The study further identifies that there are many limitations in traditional QA metrics such as exact match and token-level F1 when applied to instruction-following language models [16]. Responses generated by the model end up becoming more verbose in comparison to the reference answers and may end up expressing equivalent information in a different wording. This causes metrics that use lexical comparison to be unable to accurate evaluate the quality of the model's response. The study found that, through human judgement and evaluation, the recall correlated strongly with the correctness metric, while K-Precision, which measured the proportion of response tokens supported by the provided knowledge, was effective for assessing faithfulness [16]. It was also found that LLM-based evaluation using GPT-4 was strongly correlated with human judgements, with the limitations of potential systematic biases and computational costs.

These findings support a multidimensional approach to evaluating retrieval-augmented QA rather than relying on a single automated metric. For governance-document specific QA, distinguishing between correctness and faithfulness becomes particularly important, as a response generated by a model should not only provide the required information but also remain grounded in the authoritative documents that are supplied to the system.

2.2.9 Automated Evaluation of RAG

The evaluation of RAG systems introduces new challenges beyond evaluating the generated response alone, as the performance of the model is dependent on both the retrieval and generation stages. A group from Cardiff University proposed Retrieval Augmented Generation Assessment (RAGAS), a framework designed for the evaluation of different component of the RAG systems without having to rely on human annotated reference answers [17]. Rather than representing performance through a single measure, RAGAS evaluates the distinct characteristics of the retrieval and generation processes.

The framework evaluates the RAG system through a variety of metrics, such as faithfulness, answer relevance, and the quality of the retrieved context. Faithfulness assesses whether the generated response of the model can be supported by the retrieved context, while answer relevance considers whether the original query was actually answered by the generated response. Evaluation focused on the context examines whether the information retrieved is actually relevant to the question and whether sufficient information has been retrieved information to support a response appropriate to the query. By separating the evaluation into these sections, the framework helps to distinguish failures originating from retrieval from those introduced during response generation.

RAGAS use of its LLMs to perform reference-free evaluation, reducing dependence on manually annotated ground-truth answers and enabling more scalable evaluation of RAG architectures. However, this automated evaluation of the model introduces its own set of limitations, as the results may depend on the capabilities of the evaluator model and the prompts used to perform the assessment. Hence, RAGAS demonstrates the value of multidimensional evaluation while also highlighting the importance of considering the reliability of evaluation metrics when assessing RAG systems.

2.2.10 LLM-based Evaluation and G-Eval

As LLM outputs become increasingly complex, there has been an increasing usage of automated evaluators for generated responses. The traditional reference-based metrics such as BLEU and ROUGE primarily measure lexical similarity and may exhibit limited alignment with human judgement when the semantically correct responses differ from reference wording. Due to this, a study proposed G-Eval, an LLM-based evaluation framework designed to provide automated assessments that more closely correspond with human evaluation.

G-Eval uses LLMs alongside explicitly defined evaluation criteria and a structured evaluation procedure. The framework employs chain-of-thought reasoning to generate evaluation steps from the specified criteria before assessing generated responses using a form-filling approach [18]. Experimentation on text summarisation and dialogue generation demonstrates a stronger correlation with human judgement than several existing automated evaluation approaches, with GPT-4-based G-Eval achieving a Spearman correlation of 0.514 on the evaluated summarisation task [18].

Despite these results, LLM-based evaluation introduces many limitations concerning evaluator reliability. The study identifies potential bias towards LLM-generated text, demonstrating that an LLM evaluator should not necessarily be treated as an objective ground truth [18]. Nevertheless, G-Eval demonstrates the potential of a criterion-based LLM evaluation for assessing characteristics that are difficult to capture through lexical matching metrics alone. This is applicable to governance QA where dimensions such as correctness, completeness, and faithfulness require assessment of the meaning and evidential support of generated responses rather than measuring solely the similarity of the generated answer and the ground truth.

2.3 Sustainability Considerations

The development and deployment of AI-based QA systems introduce a range of sustainability considerations that spans across the environmental, economic, and social regions. These factors are particularly important in the context of governance documentation applications, as it is expected and required for the systems to be operating in a reliable, safe and at scale manner.

2.3.1 Environmental Concerns

From an environmental perspective on the matters on introducing AI in governance documents, the exists the significance of the cost of production, development, and deployment of the AI model in such contexts. For most of the lifecycle of an AI model, it requires copious amounts of computational resources, which leads to increased energy consumption and carbon emissions. An example of the impact is GPT-3 by OpenAI, which used 700,000 litres of water during just its training phase [19]. This is a substantial amount of water, especially when it is considered that this value only considers the training phase of the model and not the amount of water necessary to keep the AI operational. GPT-3 is only one of many AI models being trained and developed in the world, signifying that the total amount of water being used in AI expansion and applications is vastly greater. Whilst the water consumption due to AI fields is minor compared to fields such as agriculture and energy, it is still an important consideration in the development of AI because even small changes to the world's consumption of water may lead to significant environmental and social consequences.

Water is just one of the many environmental factors that the development and expansion of AI is affecting. Energy consumption and carbon footprint is one of them. From the training stage to the commissioning stages of the AI model, the energy consumption and carbon footprint must be studied and understood. Approximately 50% of the operational carbon footprint of an AI model (location specific), is due to the manufacturing costs [20]. The operational carbon footprint of

the model can be offset through renewable energy sources, in which case the carbon footprint from the operation of the AI model can be reduced significantly. However, this leaves the majority of the carbon footprint to be from the manufacturing stage of the AI [20]. Understanding the implications of societal and environmental impacts is an important consideration in AI technology development.

It is necessary to address these concerns by prioritising efficient model architectures and optimised inference processes. There are many approaches that can be applied in this instance such as reducing any unnecessary model complexity, limiting the existence of redundant computations, and leveraging retrieval-based methods instead of purely generative response model architectures. These approaches can significantly reduce the computational overhead. Additionally, reusing pre-existing models that have been already trained on a dataset and applying fine-tuning to said model can further reduce any resource consumption rather than starting from the beginning and having to perform full-scale development of a model

2.3.2 Economic Concerns

Governance documents are a huge expenditure that many companies allocate resources to. Organisations that tend to rely on assets such as constitutions, bylaws, operational policies, and documentation tend to require significant human resources and effort to interpret and verify the information. The implementation of a grounded QA AI model would be of great benefit to such an organisation as it can substantially reduce the time and costs associated with having to manually investigate documents for review.

Although it would be highly dependent on the requirements of the organisation, where the costs saved by implementing such a system would outweigh the cost of development and deployment of the system, improving the reliability and accuracy of QA systems contributes to increasing the operational efficiency of the organisation and reduce their expenditure on labour costs.

Furthermore, having a more efficient model design also lowers any infrastructure and deployment costs, particularly when it is used in large-scale scenarios or when it is being continuously used.

Errors and hallucinations in high-stakes contexts such as governance documents can be detrimental to the organisation. Incorrectly generated QA responses can lead to a variety of issues that may end up being extremely costly to the organisation. Such issues include the misinterpretation of governance rules, issues with legal and compliance rules, or financial penalties. Misinterpretation of governance rules may lead to inadvertently violating policies or regulations that are in place and breaching contractual or organisational requirements. Infringing on these can lead to economic damages such as fines, penalties, and legal disputes. If the system is used to support or inform decisions on governance documents and the QA system produces a hallucinated response, decisions may be made on false premises which may lead to the organisation taking inappropriate actions against their best interests. These may result in financial losses, inefficient allocation of resources, and disruptions of the organisation's operations.

Hallucinations may also produce false positives and false negatives. In the context of governance documentation, false positives may introduce rules that are entirely fabricated or false whilst false negatives may lead to missing rules. False positives may result in unnecessary restrictions to actions made or limitations to any processes. False negatives may unintentionally force non-compliance risks being taken. Both false positives and false negatives carry economic consequences, however in the context of governance, false positives are particularly more damaging to the organisation.

Reliable QA systems play a critical role in reducing organisational risk and supports long term financial stability. By ensuring that the responses generated by the model are accurate and grounded in source documents, the system will minimise the risk of disputes arising from the

misinterpretation of rules or inconsistent application of policies. Miscommunication is another major risk that results in substantial economic consequences. Inaccurate interpretations of governance documents may lead to different departments in an organisation making conflicting actions and delays in decision making. These inefficiencies not only increase the overall operational costs but also leads to disruptions in the organisational workflow.

By improving the reliability and faithfulness of the generated responses made by QA systems, it contributes to mitigating these risks. This has long-term economic implications, as the reduction in uncertainty and error in decision-making processes would enhance the organisation's stability and resilience. Rather than providing short-term economic savings, reliable and faithful QA systems help to establish consistent and trustworthy processes that aims to support sustainable financial performance.

2.3.3 Social Concerns

The social implications of AI based QA systems are particularly significant in the context of governance documentation, where the access to accurate and understandable information directly affects the organisation and the people part of it. Reliable and faithful QA systems can enhance the transparency and accessibility by enabling the users of the system to efficiently be able to interpret complex governance documents such as constitutions, bylaws, and operational policies. This reduces the reliance and need for specialised knowledge and allows for a broader range of stakeholders that may be able to engage with and understand the organisational rules.

One of the key aspects of social sustainability is trust. If QA systems produce hallucinated, false, or unsupported responses, users may lose confidence in both the system and the underlying governance processes. In high-stakes applications, this could lead to the undermining of the perceived legitimacy of decisions and may lead to the creation of uncertainty among the

stakeholders [21]. By improving the faithfulness and reliability of generated responses, it contributes to building trust in the decisions being made by the AI QA system and supports more consistent and transparent governance practices.

In addition, reliable QA systems promote fairness and consistency in the interpretation of the governance documentation. In many manual processes, the interpretation of governance documents may slightly vary depending on the individual or the context. AI systems that are properly designed and evaluated allows for the standardisation of responses generated, which reduces the ambiguity and the risk of biased or uneven application of different policies.

However, social sustainability also calls for the careful consideration on the over-reliance on AI systems. Users may accept responses generated by the AI without any sufficient verification, particularly in the situation where the system appears authoritative. This highlights the necessity to develop and designing a system that supports explainability, source attribution, and human oversight to ensure that the user is able to critically evaluate the outputs made by the model rather than blindly trusting its responses. Overall, the social sustainability of AI QA systems depends not only on improving its technical performance but also focusing on fostering trust, accessibility, and responsible usage. By reducing hallucinations and ensuring that the responses made by the system aligns with the source material, it supports a more transparent, equitable, and accountable access to governance documentation.

2.4 Discussion

2.4.1 Key Findings

The literature reviewed in this study demonstrates that hallucinations remain to be a persistent and complex limitation that LLMs face, particular when in high-stakes QA applications and contexts such as governance documentation. These hallucinations generally arise due to the probabilistic nature of LLMs, which generate their outputs based on likelihood rather than actual

factual verification, as well as limitations in the data quality in the training stages, model generalisation, and lack of grounding in authoritative sources. As a result, highly advanced models are susceptible to producing outputs that may appear to be coherent and convincing to the reader but are factually incorrect or unsupported by any evidence.

A key finding from this review is that the mitigation of hallucinations in generated responses is not a problem that can be resolved through the use of a single technique. Instead, the current approaches address the different stages of the model's lifecycle and must be understood as complementary to rather than competing solutions. Retrieval-based methods, particularly RAG, significantly improves the factual accuracy by grounding documents in external documents. However, their effectiveness is highly dependent on the quality of the document retrieval and query formations. Reinforcement learning-based approaches such as RISE and RLVR introduce verification mechanisms that evaluate and refine model outputs, but also comes at the cost of increased computational complexity and reliance on the quality of internal and external evaluators. Extractive methods offer the highest level of factual accuracy and reliability by eliminating the need for a generative process altogether, yet they lack the flexibility to deal with and are often limited in situations involving complex or ambiguous queries.

Another important observation is that improving reliability is not solely regarding reducing hallucinations but also about increasing the transparency and interpretability of the model's responses and thought processes. Approaches such as XAI do not directly prevent hallucinations but instead provides insights into the reasoning processes of the model, which allows users to be able to identify potential inconsistencies or unsupported claims. This is particularly valuable in the context of governance documentation, where the understanding of how a decision or answer was derived is as important as the answer itself.

The evaluation of LLM performance also emerges as a critical issue. Traditional metrics such as accuracy, precision, recall, and F1-score are able to provide a general indication of performance but fail to capture whether the generated responses are actually grounded in the source material. Faithfulness-based evaluation metrics offer a more targeted evaluation by measuring the alignment between the outputs generated by the model and the source documents. However, these metrics introduce their own limitations, particularly the generation of out-of-distribution (OOD) inputs during perturbation-based evaluation. This may lead to misleading assessments of the performance of the model, as the models are tested on inputs that do not reflect realistic usage scenarios. Emerging approaches such as F-Fidelity attempts to address these limitations as it ensures that the evaluation inputs remain within the distribution of the training data, representing an important progress being made in the development of evaluation methodologies.

Lastly, the integration of sustainability considerations highlights that the challenges associated with LLMs extend beyond that of the technical performance of the model. The computational demands of training, fine-tuning, and deploying these models contribute to environmental impacts such as energy consumption, carbon emissions, and water usage. Concurrently, unreliable model outputs can lead to economic inefficiencies and social consequences, particularly in high-stakes applications such as governance documents. Therefore, improving the reliability and faithfulness of QA systems should not only consider technical improvements but also a broader sustainability consideration that affect environmental, economic, and social outcomes.

2.4.2 Knowledge Gaps

Despite the significant advancements that have been made in mitigating hallucinations and improving the reliability of LLMs, several critical gaps remain in the current body of research. One of the most prominent gaps is the lack of an integrated framework that combines multiple mitigation strategies into a single unified system. Most of the current approaches have been

studied and evaluated independently of one another. This limits the understanding of how these techniques interact in practical deployments, where multiple strategies may be required to be simultaneously deployed to achieve acceptable levels of reliability and faithfulness.

Another key limitation is the lack of domain-specific research, particularly involving contexts such as governance documentation. Most studies evaluate the performance of the LLMs on general-purpose datasets, which do not reflect the structured, formal, and legally sensitive nature of governance related content. Governance documents often require precise interpretation, with strict adherence to source material, and accurate citations. However, the current models and evaluation method fail to meet most of these standards. This creates a gap between the experimental performance vs real-world applicability in high-stakes organisational settings.

The evaluation methodologies themselves present another significant gap. While faithfulness evaluation metrics represent an improvement over the traditional evaluation methods, they are still limited by issues such as OOD inputs, instability across different perturbation techniques, and overall sensitivity to masking strategies. Although there are newer strategies such as F-Fidelity that aim to address these challenges, they are still in the early stages of adoption and lack comprehensive validation across various diverse datasets and applications to be considered established. Furthermore, there is no widely accepted standard for evaluation hallucinations, making it difficult to compare results across multiple models and studies.

In addition to the technical limitations of hallucination mitigation, there is a lack of research involving the trade-offs between reliability and computational efficiency. Many of the hallucination mitigation strategies, particularly the ones involving multi-modal verification or reinforcement learning, will lead to the significant increase in computational requirements. This has direct implications on for environmental sustainability, as the higher computational demands will inevitably lead to increased energy consumption and resource usage.

2.5 Conclusion

LLMs are powerful tools for QA, but hallucinations remain to be a significant limitation, particularly in high-stakes contexts such as governance documents. This review shows that various mitigation strategies can reduce the risk of hallucinations in generated responses, however, no single approach fully resolved this problem. It also highlights that traditional evaluation metrics are insufficient, with many faithfulness-based evaluation metrics providing more relevant insights despite existing limitations. Additionally, the deployment of this system introduces environmental, economic, and social considerations that must be addressed. Overall, improving the reliability of QA systems requires a balanced approach that combines effective model design, robust evaluation, and responsible implementation to ensure accurate and trustworthy outcomes.

3 Research Methodology

After conducting research into the existing literature related to the thesis topic, changes were made to the methodology and the approach used to test hallucination reduction. This research adopts an experimental comparative methodology to investigate techniques for reducing hallucinations in Large Language Model (LLM)-based question-answering systems operating on governance documents. The study focuses on improving the factual reliability and truthfulness of responses generated from existing organisational documents by integrating hallucination mitigation strategies into a Retrieval-Augmented Generation (RAG) framework.

A standard RAG architecture is selected as the baseline system due to its widespread adoption and demonstrated effectiveness in grounding generated responses using externally retrieved contextual information. Since the effectiveness of RAG systems has already been extensively investigated within existing literature, this study does not seek to validate the RAG architecture itself. Instead, the baseline RAG model is used as a controlled configuration against which additional hallucination reduction techniques are applied and evaluated. This allows the research to focus specifically on whether supplementary mitigation strategies can improve the reliability of the final generated responses beyond that achieved through conventional RAG alone.

3.1 Governance Document Corpus

The document corpus used throughout the experimentation consists of governance and operational documents sourced from IEEE and associated organisations. These include the IEEE Constitution and Bylaws, Draft Operational Manual V5, SMC Constitution and Bylaws (2023), and the SMCS Technical Manual (2023). These documents were selected due to their highly structured and policy-oriented nature, where factual accuracy and correct interpretation of organisational information are particularly important.

Governance documentation presents an appropriate environment for hallucination analysis because incorrect responses may result in procedural misinterpretations, policy misunderstandings, incorrect attribution of responsibilities, or misinformation regarding organisational rules and regulations. The presence of related but distinct governance entities, procedures, responsibilities, temporal requirements, and policies additionally create scenarios in which an LLM may generate plausible but unsupported responses.

3.2 Document Pre-processing and Baseline RAG Architecture

Prior to experimentation, all documents undergo several pre-processing steps to prepare them for retrieval and generation tasks. This process includes text extraction, segmentation into smaller contextual chunks, and vector embedding generation. The processed embeddings are stored within a ChromaDB vector database to enable semantic similarity retrieval during query execution.

The baseline RAG architecture follows a conventional retrieval-generation pipeline. User queries are converted into vector embeddings and compared against the stored document embeddings to identify relevant contextual information. The retrieved document chunks are subsequently appended to the prompt supplied to the LLM, enabling a response to be generated using information retrieved from the governance corpus.

The baseline implementation uses nomic-embed-text through Ollama for embedding generation and Llama 3 as the primary generation model. Documents are divided using a chunk size of approximately 800 characters with an overlap of 200 characters between consecutive chunks. During retrieval, the twelve highest-ranked document chunks are supplied as contextual information for response generation.

The retrieval configuration is maintained as a controlled component throughout the experiments rather than being independently evaluated. The same governance corpus, embedding model,

vector database, chunking configuration, and retrieval parameters are maintained across experimental architectures wherever applicable. Consequently, the primary focus of the comparative evaluation is whether each hallucination mitigation strategy improves the correctness and reliability of the final generated response when operating from a consistent retrieval environment.

3.3 Experimental Model Configurations

Several enhanced model configurations are developed by integrating additional hallucination mitigation strategies into the baseline architecture. Each experimental model builds upon the same underlying RAG pipeline while maintaining the same document corpus, benchmark questions, retrieval environment, and core generation configuration. This enables performance differences to be associated more directly with the applied mitigation strategy rather than changes to the underlying dataset or retrieval system.

The experimental model configurations investigated within this research are:

- Baseline RAG: A standard retrieval-augmented generation pipeline used as the experimental control.
- RAG + RISE-inspired Refinement: Introduces inference-time self-verification and iterative refinement of responses identified as requiring improvement.
- RAG + RLVR-inspired Verification: Introduces inference-time document-grounded verification using explicit verification criteria and targeted correction.
- RAG + XAI: Integrates explainability mechanisms to provide greater transparency regarding generated responses and their evaluation.
- RAG + Multi-model Verification: Uses independently developed LLM families to verify the same generated response against the same retrieved evidence.

- RAG + Combined Techniques: A hybrid implementation combining complementary hallucination mitigation strategies.

The RISE- and RLVR-based configurations used within this research are described as RISE-inspired and RLVR-inspired rather than complete reinforcement-learning implementations. The locally deployed models are accessed through Ollama without fine-tuning or modification of model parameters. Hence, the principles underlying these techniques are adapted into inference-time verification and refinement architectures rather than being implemented through reinforcement-learning-based model training.

3.3.1 RISE-Inspired Refinement

The RISE-inspired configuration introduces iterative refinement following the initial RAG generation and evaluation stages. Rather than applying reiteration indiscriminately to every generated response, responses identified as problematic are targeted for additional refinement. This approach was selected to reduce the possibility of modifying already-correct responses and introducing degradation.

The refinement process operates using the original question, initial generated response, and retrieved governance evidence. The model is instructed to reconsider the initial response using the available documentary evidence and generate an improved response where appropriate. The revised response can subsequently be evaluated to determine whether the refinement process successfully improves factual correctness and reliability.

This configuration investigates whether self-verification and iterative reconsideration of an initial response can reduce hallucinations without requiring modification of the underlying language model.

3.3.2 RLVR-Inspired Verification

The RLVR-inspired configuration investigates whether externally defined and document-grounded verification criteria can provide a more reliable correction signal than self-reflection alone. As no model-weight training or reinforcement learning is performed, the principles of verifiable rewards are adapted into an inference-time verification architecture.

Following generation of the initial RAG response, material claims within the response are evaluated against the retrieved governance evidence. Verification follows a strict entailment principle in which evidence must directly establish the same material proposition as the generated claim. Evidence that is merely related, analogous, or dependent upon additional unsupported assumptions is not considered sufficient verification.

Responses containing claims identified as unsupported, contradictory, or unverifiable can trigger a targeted correction attempt. The correction process uses the question, initial response, retrieved evidence, and identified verification failure to generate a revised response. The revised response is subsequently subjected to the same verification process before being considered as a replacement for the original response.

The RLVR-inspired configuration therefore investigates whether an external evidence-grounded verification signal can improve the reliability of RAG-generated responses without modifying the underlying LLM parameters.

3.3.3 Explainable Artificial Intelligence

The XAI configuration investigates the use of explainability mechanisms alongside the RAG architecture. Rather than focusing exclusively on whether a generated response is classified as correct or incorrect, the XAI implementation provides additional information explaining the basis of the evaluation.

The approach is intended to provide greater transparency regarding the relationship between the generated response and the retrieved governance evidence. Explanations can identify why a response is considered correct or incorrect, which evidence contributes to that determination, and where unsupported or contradictory information may occur.

The XAI configuration is therefore evaluated not only in relation to final response correctness but also in terms of its ability to provide interpretable information regarding the behaviour and reliability of the QA system. This is particularly relevant to governance applications, where users may require justification for responses concerning organisational policies, procedures, and responsibilities.

3.3.4 Multi-Model Verification

The multi-model verification configuration investigates whether verification diversity across independently developed language-model families can reduce errors associated with a single model evaluating its own generated responses.

The initial RAG response is generated using Llama 3. Following generation, three independently developed LLMs—Llama 3, Qwen 3 8B, and Mistral 7B—independently evaluate the generated response using the same original question and the same twelve retrieved contextual chunks available during generation. All verifier models use the same generic verification instructions and deterministic generation settings to maintain consistency.

The individual verifier decisions are combined using an ensemble majority-voting mechanism. This enables comparison between self-verification performed by the original Llama 3 model and verification performed by independently developed model families. Individual verifier decisions are retained to allow analysis of agreement and disagreement between models.

In its initial experimental configuration, multi-model verification operates exclusively as a verification mechanism and does not modify the generated response. This separates verification

performance from correction performance and allows the experiment to determine whether model diversity itself improves the ability to distinguish reliable responses from incorrect or unsupported responses.

3.4 Benchmark Question Set

To evaluate performance, a fixed benchmark consisting of 110 governance-related questions is used across all model architectures. The questions were manually developed from the governance corpus and are designed to represent different forms of information retrieval, interpretation, and reasoning that may occur within governance-oriented QA applications.

The benchmark is divided into the following categories:

- Direct Retrieval
- Cross-Document
- Semantic Retrieval
- Hallucination Trap
- Ambiguous
- Contradicting
- Tricky
- Multi-Hop Reasoning
- Temporal
- Negation
- Confirmatory
- Clarification Required
- Conditional
- Procedural
- Summary
- Differentiation

These categories provide coverage of both straightforward factual retrieval and more challenging situations requiring reasoning across multiple sources, interpretation of temporal or conditional information, recognition of ambiguity or insufficient information, differentiation between similar governance concepts, and resistance to misleading assumptions.

Each model configuration is evaluated using the identical benchmark question set under controlled experimental conditions. Reference answers are manually established using information derived directly from the governance documents.

Importantly, these reference answers are isolated from the response generation and mitigation processes. They are used only during benchmark evaluation and are not supplied to the generation, verification, refinement, or correction stages of the experimental architectures. This separation prevents the mitigation techniques from using benchmark ground truth when determining whether a generated response is reliable.

3.5 Definition of Hallucination

A formal definition of hallucination is established to maintain consistency throughout the evaluation process. Within the context of this study, a hallucination is defined as a generated factual statement that is unsupported, contradictory, or otherwise unverifiable within the provided governance document corpus.

This includes fabricated policies or procedures, incorrect procedural details, unsupported interpretations, incorrect attribution of responsibilities, incorrect relationships between governance entities, inaccurate temporal information, and factual claims that cannot be established from the supplied governance documentation.

A response does not necessarily need to contain completely fabricated information to be considered hallucinated. A response may contain information that is generally related to the question but incorrectly attributes a responsibility, policy, procedure, condition, timeframe, or governance requirement. Such responses are also considered unreliable within the context of governance QA.

3.6 Response Evaluation

Evaluation focuses primarily on the quality and factual reliability of the final generated responses. Generated responses are compared against the governance-derived reference answers using several complementary performance measures.

Each response is evaluated for correctness, completeness, and faithfulness. Correctness measures whether the information provided is factually accurate relative to the reference answer and governance documentation. Completeness considers whether the response contains the information required to sufficiently answer the question. Faithfulness evaluates whether the response remains grounded in the supplied governance documentation without introducing unsupported information.

These measurements contribute to an overall evaluation score and allow benchmark accuracy to be calculated for each model configuration.

Standard natural-language generation metrics are additionally calculated to provide complementary measurements of similarity between generated and reference responses. These include:

- ROUGE-1 F1;
- ROUGE-2 F1;
- ROUGE-L F1;
- BLEU;
- BERTScore Precision;
- BERTScore Recall; and
- BERTScore F1.

ROUGE and BLEU provide measurements of lexical overlap between generated and reference responses, while BERTScore provides a semantic similarity measurement that is less dependent on exact wording. These metrics are treated as supplementary measurements rather than direct indicators of hallucination, as a factually correct governance response may express the same information using different wording from the reference answer.

Overall benchmark accuracy and document-grounded correctness, completeness, and faithfulness therefore remain the primary indicators of response reliability.

3.7 Architecture-Specific Evaluation

Additional measurements are collected for architectures containing verification, refinement, or correction mechanisms.

For the RISE-inspired and RLVR-inspired configurations, the evaluation considers the number of responses identified for additional processing, the number of responses modified, the number of initially incorrect responses successfully corrected, and the number of initially correct responses degraded following intervention.

This enables the mitigation strategy to be evaluated not only according to final accuracy but also according to the nature of the changes it introduces. An architecture that corrects erroneous responses but frequently degrades previously correct responses may therefore be distinguished from an architecture that produces similar final accuracy while maintaining greater stability.

For multi-model verification, each verifier is evaluated individually in addition to the combined ensemble decision. Measurements include verification classification accuracy, true positives, true negatives, false positives, false negatives, unanimous agreement, disagreement between models, and category-level verification performance. This enables analysis of whether independently developed model families exhibit different verification behaviours and whether ensemble verification provides an advantage over single-model self-verification.

3.8 Evaluation Reliability and Consistency

As part of the evaluation process, an LLM-based evaluator is used to assess generated responses against the manually developed reference answers. The evaluator assesses each response according to the predefined criteria of correctness, completeness, and faithfulness, in addition to assigning an overall score from 0 to 3. Although this approach enables the large number of generated responses to be evaluated consistently, the use of an LLM as an evaluator introduces the possibility of judgement errors, inconsistencies, and evaluator bias. Consequently, the evaluator is treated as an automated mechanism for applying the defined evaluation rubric rather than as an infallible source of ground truth.

Several measures are used to improve evaluation consistency across the experiments. The same evaluation model, evaluation prompt, scoring criteria, and reference answers are used across each model configuration. Generated responses are evaluated according to their semantic meaning rather than requiring exact textual correspondence with the reference answer. The evaluation criteria are also explicitly defined within the evaluator prompt to reduce ambiguity in the scoring process. Special evaluation rules are applied to question types where conventional factual comparison may be inappropriate, including questions requiring the model to recognise insufficient information or ambiguity rather than generate an unsupported answer.

The reference answers are derived from the governance document corpus and serve as the primary basis against which generated responses are assessed. Where feasible, evaluation results that appear anomalous or inconsistent with the reference answer are manually reviewed to identify potential evaluator errors. This is particularly important for borderline responses where minor differences in interpretation may affect the assigned score.

Automated evaluation is supplemented by reference-based quantitative metrics, including ROUGE-1, ROUGE-2, ROUGE-L, BLEU, and BERTScore. These metrics provide additional

measurements of similarity between generated and reference answers; however, they are not treated independently as definitive measures of factual correctness. In particular, lexical similarity metrics may penalise responses that express the correct information using different wording or may produce favourable scores for responses that share terminology with the reference answer while containing factual errors. For this reason, the evaluation framework considers the LLM-based rubric assessment and reference-based metrics together when analysing model performance.

Despite these measures, some degree of evaluator uncertainty remains and is recognised as a limitation of the experimental methodology. The potential for evaluator judgement errors and bias is therefore considered when interpreting small performance differences between model configurations and is further addressed within the threats to validity of the study.

3.9 Computational Efficiency

In addition to factual reliability, the computational efficiency of each model configuration is evaluated to determine the additional processing cost introduced by the hallucination mitigation techniques. Computational measurements are collected consistently across the baseline and enhanced RAG architectures, allowing the computational overhead of each technique to be compared against the baseline system.

The evaluation records response latency and the duration of major processing stages where applicable. The number of LLM inference calls required to process each benchmark question is also recorded, providing an additional architecture-level measurement of computational overhead. This is particularly relevant for techniques involving iterative refinement, repeated verification, or multiple verifier models, as these approaches may require substantially more inference operations than the baseline RAG architecture.

Computational results are analysed alongside factual reliability measurements rather than independently. This allows each mitigation strategy to be evaluated according to the trade-off between improvements in response reliability and the additional computational resources required to achieve those improvements. Therefore, an architecture producing higher factual accuracy at the cost of substantially increased latency or inference operations can be distinguished from an architecture providing similar reliability improvements with lower computational overhead.

All computational measurements are collected under the same experimental hardware and software environment wherever possible to maintain comparability between model configurations. The resulting measurements are therefore intended primarily for relative comparison between the experimental architectures rather than as hardware-independent performance benchmarks.

3.10 Comparative Analysis

Following experimentation, comparative analysis is conducted across all model configurations. The analysis focuses primarily on reductions in hallucination frequency and improvements in factual correctness, completeness, faithfulness, and overall benchmark accuracy.

Category-level analysis is conducted to determine whether particular mitigation strategies are more effective for specific forms of governance inquiry. For example, a technique may perform strongly on direct retrieval questions while providing limited improvement on temporal, cross-document, or multi-hop reasoning questions.

The behaviour of verification mechanisms is also examined independently from final answer accuracy where appropriate. This allows the research to distinguish between an architecture failing to identify an erroneous response and an architecture correctly identifying an error but subsequently failing to correct it.

Computational performance is considered alongside factual reliability to determine the practical trade-offs introduced by each mitigation strategy. A technique that achieves a small improvement in factual accuracy while substantially increasing latency or memory consumption can be distinguished from a technique that provides similar improvements at a lower computational cost.

The combined analysis therefore considers both the reliability improvements achieved by each mitigation strategy and the computational resources required to achieve those improvements.

3.11 Methodological Limitations

Several limitations to the methodology should be acknowledged. Firstly, the document corpus is limited to a relatively small collection of IEEE and SMCS governance documentation, which may reduce the generalisability of the findings to broader organisational or knowledge domains.

Secondly, the benchmark questions and reference answers are manually developed. Although the questions are distributed across multiple categories to provide a range of difficulty and reasoning requirements, manual construction may introduce selection bias depending on how representative the benchmark is of real-world governance inquiries.

Thirdly, experimental outcomes may be influenced by model-specific characteristics. Different language-model families may exhibit different hallucination, reasoning, instruction-following, and verification behaviours. Results obtained using Llama 3 and the additional verifier models therefore cannot necessarily be assumed to apply to all available LLMs.

The RISE- and RLVR-inspired architectures also operate entirely at inference time and do not represent complete reinforcement-learning implementations involving model-weight updates. The findings relating to these architectures should be interpreted as evaluations of inference-time adaptations inspired by these techniques rather than evaluations of the complete reinforcement-learning methodologies.

Engineering decisions including embedding model selection, chunk size, chunk overlap, retrieval count, prompting structure, and model configuration may additionally influence system behaviour. Although these parameters are maintained consistently wherever possible, different retrieval or generation configurations may produce different absolute performance results.

Finally, computational performance measurements are influenced by the hardware and local inference environment used during experimentation. Latency and memory measurements therefore provide useful comparative information between architectures under the same experimental environment but may not directly represent performance on alternative hardware.

3.12 Measures to Maintain Experimental Consistency

To minimise the effect of these limitations, standardised experimental conditions are maintained across model configurations wherever possible. The same governance corpus, fixed benchmark question set, embedding model, vector database, retrieval configuration, and core generation environment are used throughout the experiments.

Generation and verification settings are standardised where appropriate, including the use of deterministic generation settings during final controlled experiments to reduce variation between repeated executions.

Reference answers remain isolated from generation, verification, refinement, and correction processes and are used only during benchmark evaluation. This prevents experimental architectures from directly accessing the expected answer when deciding whether a response should be retained or modified.

Multiple evaluation measures are used to reduce reliance on any single performance indicator. Document-grounded correctness, completeness, faithfulness, and benchmark accuracy provide the primary assessment of factual reliability, while ROUGE, BLEU, and BERTScore provide supplementary measurements of response similarity. Architecture-specific measurements and

computational performance indicators provide additional information regarding how reliability improvements are achieved and the operational cost associated with each mitigation strategy.

Through these controls, the experimental methodology aims to provide a consistent comparison of hallucination mitigation strategies and determine their effectiveness in improving the factual reliability of RAG-based question-answering systems operating on governance documentation.

4 System Design and Implementation

4.1 Overall System Architecture

The developed system was designed to evaluate and compare methods for improving reliability for question answering over governance documents. A common Retrieval-Augmented Generation (RAG) architecture was established as an experimental baseline, with each of the enhancement techniques implemented as a modification or extension of this baseline. Maintaining a shared underlying architecture allowed differences in performance to be more directly attributed to the introduced enhancement rather than changes to the document corpus, retrieval configuration, question set or evaluation procedure.

The overall system consists of four primary stages. These are document processing and storage, information retrieval, answer generation, and evaluation. During document processing, the governance documents are extracted and divided into smaller overlapping text chunks suitable for retrieval. Each of the chunks are then converted into a vector representation using the nomic-embed-text embedding model and stored within a ChromaDB vector database. This processing stage is performed prior to QA allowing the stored document representations to be reused across the experimental question set. During QA, the user's question is converted into an embedding and compared against the document representations stored within the vector database. The most relevant document chunks are retrieved and incorporated into a structured prompt alongside the original question. The resulting prompt is provided to the locally hosted LLM through Ollama, which generates an answer using the retrieved governance documents as supporting context.

The baseline architecture follows this retrieval and generation process without additional post-generation reliability mechanisms. The enhancement architectures extend different stages of the baseline pipeline depending on their intended purpose. Retrieval-based enhancements modify the selection of contextual information supplied to the language model, while verification and

refinement approaches introduce additional processing after the initial response has been generated. These mechanisms include retrieval diversification, iterative refinement, verification-based candidate selection, multi-model verification and explainability mechanisms.

Following the generation of the response, each of the responses is passed through a common evaluation pipeline. Generated responses are then compared against predefined reference answers using a consistent evaluation framework. The system records the generated responses, evaluation outcomes, automated text-similarity metrics and computational performance measurements. For architectures capable of revising an initial response, both the initial and final responses are retained to enable analysis of whether the enhancement improved, preserved or degraded the original answer.

All of the architectures operated on the same governance document corpus and question set, hence retaining the same core baseline configuration wherever possible. This common architecture provides the foundation for controlled comparison between the baseline RAG system and the implemented enhancement techniques.

4.2 Baseline RAG Implementation

A baseline Retrieval-Augmented Generation (RAG) system was implemented to provide a consisted experimental control against which each enhancement architecture could be evaluated. The baseline combines document preprocessing, vector-based retrieval and local LLM generation to produce responses grounded in the selected governance documents. The same underlying baseline configuration was kept the same across the enhanced architectures wherever possible to ensure a direct comparison of their effects on QA performance.

4.2.1 Document Processing and Chunking

The knowledge base consisted of a collection IEEE and IEEE Systems, Man, and Cybernetic Society (SMCS) governance documents selected for their relevance to organisational rules,

responsibilities, procedures and operational requirement. Before retrieval could be performed, the documents were processed into smaller textual segments suitable for embedding and storage within the vector database.

Extracted document text was divided using chunk size of 800 characters with an overlap of 200 characters between consecutive chunks. The overlap was included to reduce the likelihood that information spanning a chunk boundary would become separated during preprocessing. This was particularly relevant for governance documents, where the interpretation of an individual provision may depend on preceding or subsequent text.

Each resulting chunk was retained as an independent retrievable unit while maintaining its association with the source material. Once the preprocessing was completed, the same processed corpus used throughout the experiments to prevent any differences in document segmentation from influencing comparisons between architectures.

4.2.2 Embedding and Vector Storage

Following preprocessing, each of the document chunk was transformed into a numerical vector representation using nomic-embed-text embedding model. Embeddings represent the semantic characteristics of textual passages within a high-dimensional vector space, allowing for passages with similar semantic content to be identifies even where the wording of the query does not exactly match the wording used within the governance documents.

The generated embeddings and their corresponding document chunks were stored using ChromaDB. This provided a persistent vector database through which relevant governance information could be retrieved during QA. The same embedding model and processed document collection were retained across the baseline and enhancement experiments. This ensured that the differences observed in the generated answers were not caused by changes to the underlying embedding representation or knowledge base.

4.2.3 Information Retrieval

For each of the question, the query was embedded using the same nomic-embed-text model used for the governance document chunks. The query representation was then compared against the vectors stored within ChromaDB to identify semantically relevant passages. The baseline retrieval configuration returned the highest-ranked document chunks according to vector similarity. These retrieved passages formed the evidence supplied to the generation model. The retrieval stage therefore acted as intermediary between the complete governance corpus and the large language model, limiting the generation context to information considered most relevant to the question.

This retrieval process is important because the quality of the final response is dependant not only on the capabilities of the generation model, but also on whether sufficient and relevant evidence is made available within its context. If an applicable provision is not retrieved is not retrieved, the generation model may lack the information required to construct a correct answer. Conversely, the retrieval of irrelevant or weakly related passages can introduce additional context that may distract the model from the most applicable evidence. The baseline used a fixed retrieval configuration for all questions to provide a consistent reference point for subsequent enhancement architectures.

4.2.4 Prompt Construction and Answer Generation

After retrieval, the selected document chunks were then combined with the original question to construct the input prompt supplied to the generation model. The prompt instructed the model to answer the question using the provided governance-document context, therefore encouraging the response being generated to remain grounded in the retrieved evidence rather than relying solely on the model's internal knowledge.

Answer generation was performed locally through Ollama. Running all the models locally provided greater control over the experimental environment and model configuration while avoiding changes that may occur when relying on externally hosted model services. The baseline generation procedure was maintained consistently across the question set and provided the initial answer-generation process used by the enhanced architectures.

The generated answer was subsequent passed to the common evaluation pipeline, where it was assessed against the corresponding reference answer. The baseline architecture did not introduce additional iterative refinement, multi-model verification, reward-based candidate selection or post-generation correction mechanisms. Its output therefore represents the performance of the standard RAG procedure before the application of the reliability enhancements investigated in this study. This baseline establishes the reference point for determining whether additional retrieval, verification, refinement and explainability mechanisms provide measurable improvements in reliable QA over governance document corpus.

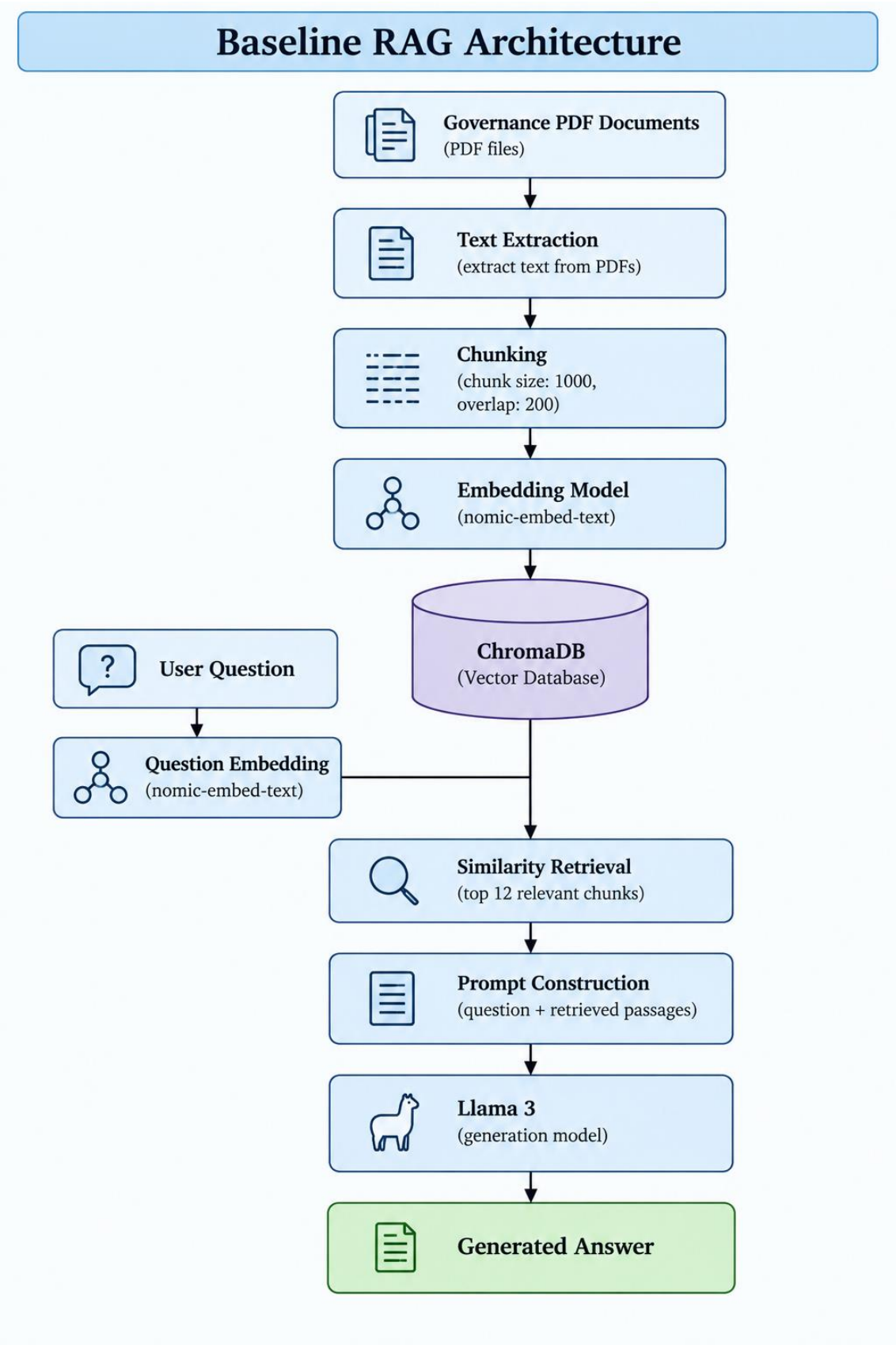


Figure 4.1: Flowchart of Baseline RAG Architecture

4.3 Enhancement Architectures

The baseline RAG architecture was extended using several enhancement techniques that targeted different potential sources of unreliability within the QA pipeline. Rather than altering the complete system for each experiment, each of the enhancements was integrated into the shared baseline architecture while retaining the same document corpus, question set and core configuration wherever possible. This allowed the effect of each of the techniques to be evaluated relative to a common experimental baseline.

4.3.1 RISE-Inspired Iterative Refinement

The first post-generation enhancement investigated in this study was an iterative refinement architecture inspired by RISE. The purpose of this enhancement was to improve the responses generated by the model that were identified as inaccurate or incomplete after the initial generation, while minimising any unnecessary modifications to the responses that were already considered correct. Rather than replacing the baseline RAG process, the refinement mechanism was applied after the initial retrieval, generation and evaluation stages.

The architecture initially follows the same procedure of the baseline RAG system. For each of the questions, relevant governance document passages are retrieved and supplied to the generation model, which produces the initial response. This response would then be evaluated using the common evaluation framework. Responses receiving an evaluation score of 0, 1 or 2 are identified as requiring improvements and are passed to the refinement stage, while responses above this threshold retain their initial generated answer without further modification.

During the refinement stage, the model is provided with the original question, the initial answer generated by the model and the retrieved governance document context. It is then instructed to reconsider the response using the supplied documentary evidence and produce a revised answer. This is to allow the model to identify any potential errors, unsupported claims or missing

information made in the initial response while maintaining grounding in the same governance corpus used during initial generation. The refinement stage does not receive the predefined reference answer. This is to prevent any direct exposure of the expected benchmark response during revision.

Each revised response is then re-evaluated using the same evaluation procedure that the initial response underwent. A revision is accepted only when its evaluation score is an improvement relative to the previous answer. If the revised response fails to achieve a higher score, the previous response is retained. A maximum of two refinement attempts is permitted, with the process terminating earlier when an acceptable improvement is obtained. This acceptance rule was introduced to reduce the possibility of iterative refinement degrading the current existing response.

The final architecture was developed following observations from earlier implementations in which refinement of generated answers occurred without first restricting the process to responses identified as requiring improvement. Under this design, the responses that were initially correct could be unnecessarily reconsidered and subsequently changed into less accurate responses. This demonstrated that additional inference-time reasoning does not inherently result in improved reliability and that unrestricted self-refinement can introduce new errors into otherwise acceptable answers.

To address this behaviour, the final implementation uses evaluation as a gating mechanism. Only the responses that were identified as incorrect or insufficient according to the defined evaluation threshold would enter the refinement process, and revised responses will replace the previous generations only when a measurable improvement had been observed. Therefore, this design attempts to preserve the benefits of iterative self-correction while reducing any risk of degradation introduced by unnecessary revision.

For every question, the system records the initial response, initial evaluation, refinement attempts, subsequent evaluations and final selected response. Retaining this information enables analysis beyond final accuracy alone, including whether refinement successfully corrected an initially incorrect response, failed to produce an improvement, or altered the evaluation score across successive attempts. These records also provide greater transparency regarding how the enhancement influences individual responses.

The RISE-inspired architecture therefore extends the baseline RAG system through selective post-generation refinement rather than modifying any of the underlying retrieval or initial generation process. Its evaluation will provide an indication of whether targeted iterative refinement is able to improve the reliability of QA over governance documents while controlling the risk of degrading initially satisfactory responses.

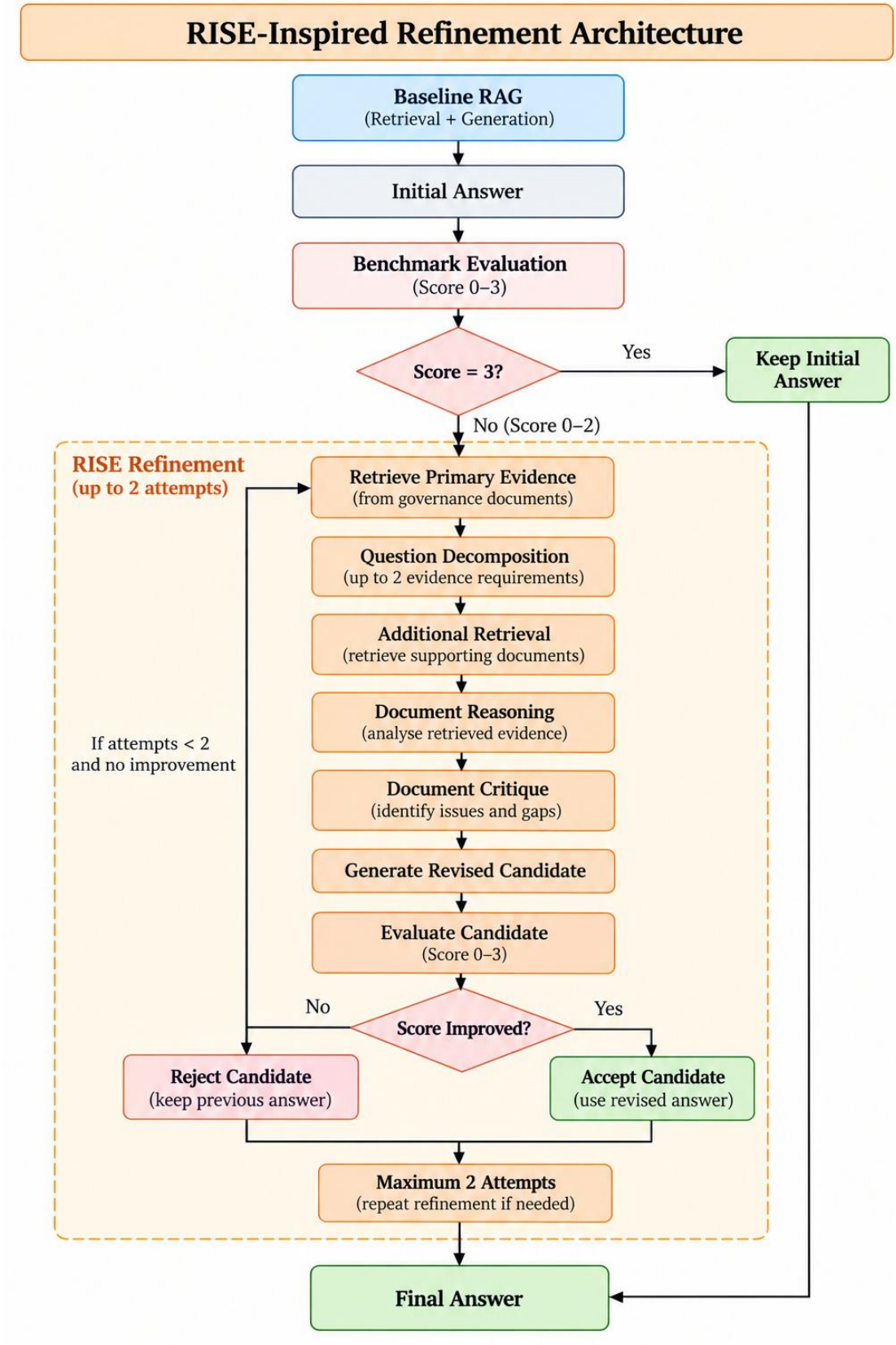


Figure 4.2: Flowchart of Rise-Inspired Architecture

4.3.2 RLVR-Inspired Verification

The second reliability enhancement investigated in this study was a verification architecture inspired by Reinforcement Learning with Verifiable Rewards (RLVR). While conventional RLVR uses verifiable reward signals during model training, the implementation developed in this study applies the broader principle of verification-guided improvement during inference. Rather than updating the parameters of the language model, generated responses are assessed against the retrieved documentary evidence to determine whether corrective generation is required. The approach is therefore referred to as RLVR-inspired rather than a direct implementation of RLVR, as it retains the general concept of using verifiable feedback to guide response improvement without performing reinforcement learning or model training.

This architecture initially follows the same retrieval and generation procedure as the baseline RAG system. Relevant passages are retrieved from the governance document corpus and supplied to the generation model alongside the question. The model then produces an initial response, which is passed to a claim-level verification stage. The verifier evaluates the material claims contained within the response against the retrieved documentary evidence and determines whether the answer contains a verification failure.

Rather than relying on a continuous scalar reward, the final implementation uses the outcome of claim-level verification as the decision signal. Each material claim is assessed against the retrieved evidence to determine whether it is supported, contradicted, or cannot be established from the available evidence. The verification stage also considers whether the generated response addresses the target of the question. The response is considered to have passed verification only when no material verification failure is identified.

When the initial response passes claim-level verification, it is retained without any additional generation. This prevents unnecessary computational processing when the original answer is

already sufficiently supported by the retrieved evidence. When a material verification failure is detected, however, the system initiates a targeted corrective-generation stage.

During corrective generation, the original question, initial response, retrieved documentary evidence, and identified verification issues are used to produce a single targeted correction. Unlike an approach that generates several independent candidate responses, the corrective response is specifically intended to address the claim-level problems identified by the verifier. The corrected response is subsequently passed through the same claim-level verification procedure used to assess the initial response.

Candidate selection is deliberately conservative. The corrected response can replace the initial answer only when it fully passes the same claim-level verifier that identified the failure in the original response. If the corrected response continues to contain a contradicted, unsupported, or otherwise failed material claim, the correction is rejected and the initial response is retained. This decision rule was included to reduce the risk that corrective generation introduces a different unsupported response simply because an additional answer has been produced.

This architecture differs from the RISE-inspired approach primarily in how response improvement is attempted. RISE performs a multi-stage iterative refinement process in which the existing response is reconsidered through additional retrieval, document reasoning, critique, and revision. The RLVR-inspired architecture instead uses claim-level verification as a gate for a single targeted correction. RISE can therefore be considered a refinement-based approach, whereas the RLVR-inspired architecture represents a verification-guided corrective approach.

The implementation also separates the verification mechanism used to guide corrective generation from the predefined reference answer used for final experimental evaluation. The benchmark reference answer is not supplied during claim verification or corrective generation. Instead, these stages operate using the question, generated response, and retrieved governance

evidence. The reference answer is introduced only during the subsequent benchmark evaluation stage. This separation prevents the enhancement architecture from obtaining direct access to the expected benchmark response and ensures that any improvement must arise from verification against the retrieved governance documents.

For each question, the system records the initial response, the results of its claim-level verification, whether corrective generation was triggered, the generated correction where applicable, the verification result of that correction, and the response ultimately selected as the final answer. This information allows the behaviour of the architecture to be analysed in addition to its final QA accuracy. In particular, the stored results can be used to determine how frequently verification failures triggered corrective generation, how frequently corrections successfully passed verification, and whether verified corrections resulted in improvements to final answer quality.

The RLVR-inspired architecture therefore investigates whether the broader principle of verifiable feedback can be adapted from a training-oriented concept into an inference-time reliability enhancement for RAG-based governance QA. By generating a targeted correction only when claim-level verification identifies a material problem, and accepting that correction only when it subsequently passes the same verification process, the architecture aims to improve answer reliability while limiting unnecessary additional model calls.

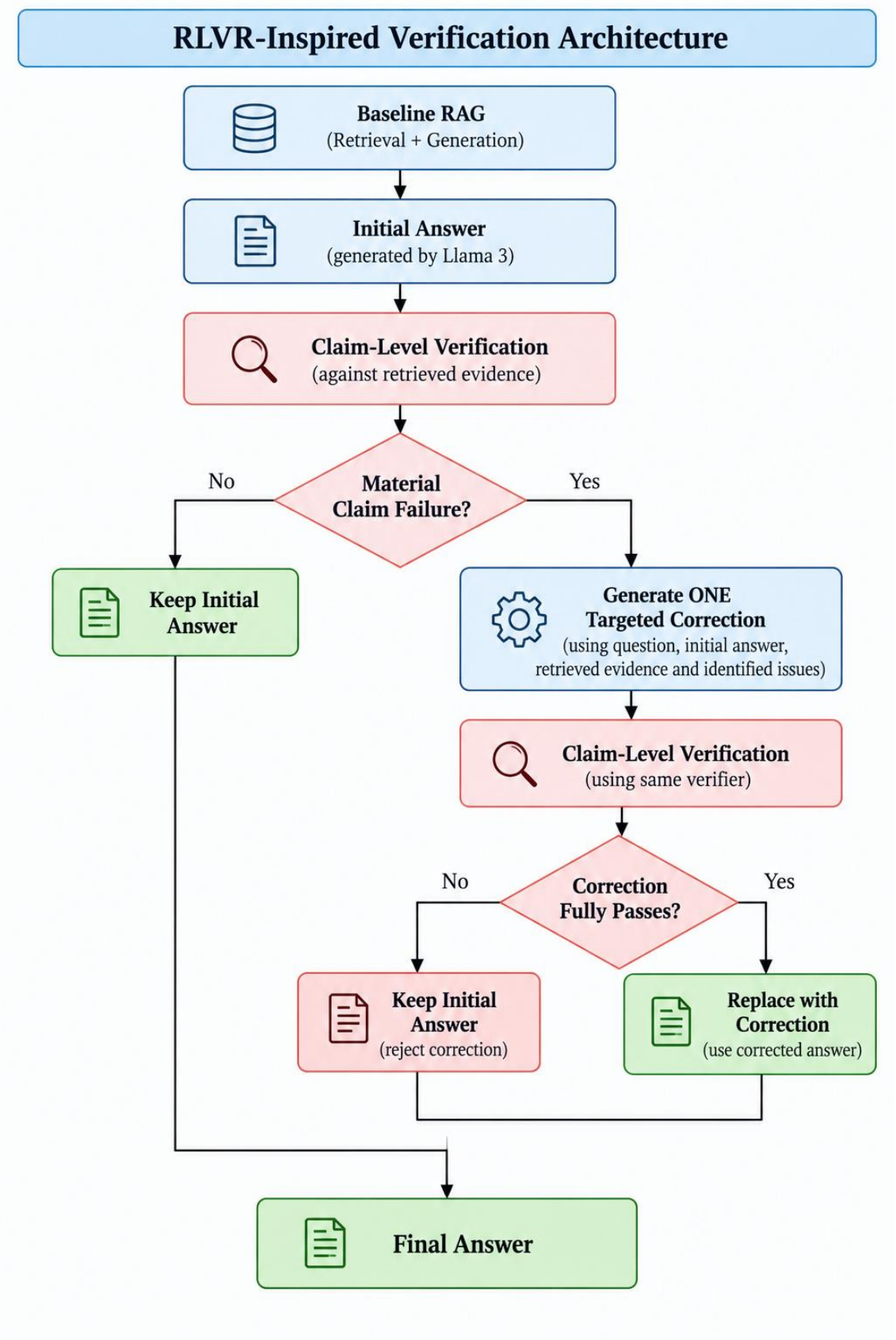


Figure 4.3: Flowchart of RLVR-Inspired Architecture

4.3.3 Multi-Model Verification

The third reliability enhancement investigated in this study was a multi-model verification architecture. Unlike the previous approaches, which have a reliance primarily on performing additional generations or refinements, this architecture instead introduces multiple independent LLMs to assess the validity of a generated response. The purpose of this approach was to investigate whether agreement between different model families could provide a more robust verification signal than relying on the judgement of a single model.

The architecture initially follows the same retrieval and answer-generation procedure as the baseline RAG system. Following generation of the initial response, the question, generated answer and retrieved documentary evidence are provided to three independent verification models. The models used for verification were Llama 3, Qwen3:8B and Mistral:7B, all executed locally through Ollama. Using different model families was intended to introduce variation in the verification process and reduce dependence on any one behaviour or judgement belonging to a single model.

Each verifier received the same question, generated response and retrieved governance document context. The verification prompt instructed each of the models to assess the generated response against the supplied evidence and produce a verdict indicating whether the answer is supported by the retrieved documents. The verifier is therefore expected to base its judgement on the provided governance information rather than external knowledge or the predefined reference answer.

To ensure consistency between all verification runs, the verifier models are configured using deterministic generation settings, including a temperature of zero. The same verification prompt and retrieved context are also supplied to each model. The verification prompt is kept independent of the benchmark question set and does not contain any examples that could have

been derived from the reference answers. This is to reduce the possibility of the verifier being directly influenced by knowledge of the expected benchmark outputs.

The individual verifier decisions are combined using majority voting. With three independent models, a final verification decision can be reached when at least two models produce the same verdict. This approach reduces the influence of an anomalous judgement from an individual verifier and provides a simple mechanism for aggregating multiple model assessments. Rather than assuming that any individual LLM can consistently evaluate generated answers correctly, the architecture therefore uses agreement between different models as the basis for the final verification outcome.

A consistent set of retrieved passages is supplied to each verifier to ensure that differences between their decisions arise from the models themselves rather than differences in available evidence. In the implemented architecture, 12 retrieved document chunks are provided during verification. This provides each verifier with access to the same evidence base when assessing whether the generated answer is supported by the governance documents.

The use of multiple verification models introduces additional computational cost compared with the baseline architecture, as a single generated response requires several additional model calls. This trade-off is recorded through the computational metrics collected by the evaluation pipeline, allowing for any improvement in reliability to be considered alongside the additional inference requirements introduced by multi-model verification architecture.

The system records the verdict produced by each verifier, the aggregated majority decision and associated verification statistics for each question. Retaining the individual model decisions allows disagreement between verifiers to be examined rather than recording only the final majority result. This provides additional insight into cases where the correctness or evidential support of a response may be difficult for automated evaluators to determine consistently.

Overall, the multi-model verification architecture investigates whether independent verification and majority voting can improve the robustness of answer assessment within governance-document question answering. By distributing the verification task across several model families, the approach aims to reduce dependence on the potential biases, reasoning errors or inconsistencies of any single verifier.

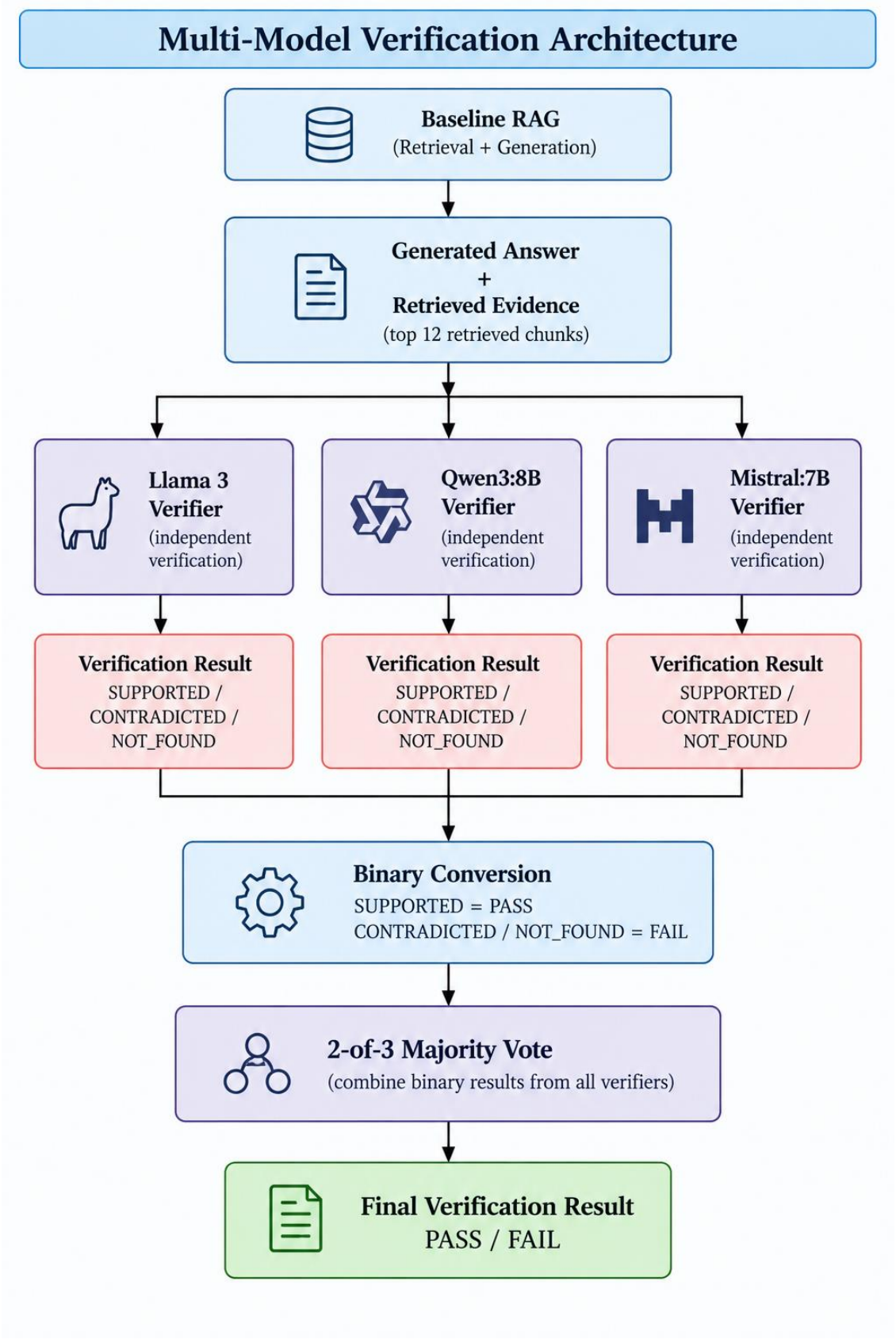


Figure 4.4: Flowchart of MMV Architecture

4.3.4 XAI Enhancement

The final enhancement investigated in this study incorporated Explainable Artificial Intelligence (XAI) principles into the baseline RAG architecture. Unlike the RISE-inspired, RLVR-inspired and multi-model verification approaches, the XAI implementation was not designed to modify generated responses or improve question-answering accuracy. Instead, its purpose was only to improve the transparency and interpretability of the system by providing additional information explaining why the generated response was evaluated to be either correct or incorrect.

The architecture follows the same retrieval and answer-generation procedure as the baseline RAG system. Relevant passages are retrieved from the governance document corpus and provided to the LLM as contextual evidence for generating a response. The generated response is then evaluated using the established evaluation framework. Following this evaluation, an explanation is produced describing the reasoning behind the evaluation outcome.

For any responses evaluated as correct, the explanation identifies how the generated answer aligns with the expected information and relevant evidence. For responses evaluated as incorrect or incomplete, the explanation identifies the aspects of the response that contributed to the lower evaluation result, such as missing information, inaccurate statements or inconsistencies with the expected answer. This is to provide additional context beyond a numerical evaluation score and allows individual model responses to be inspected in greater detail.

The explanation stage does not feed information back into the generation process. The original generated answer is preserved regardless of what the explanation produced. No additional refinement, candidate generation or answer replacement is performed. As such, the XAI implementation does not alter the accuracy of the underlying RAG system. Any differences between the baseline and XAI configurations relate to the additional explanatory information produced rather than changes to the generated answers themselves.

This distinction is important when comparing XAI with the other enhancement architectures that are investigated in this study. RISE-inspired refinement and RLVR-inspired verification attempt to improve the final response through additional inference-time processing, while multi-model verification introduces additional mechanisms for assessing generated responses. In contrast, XAI addresses a different aspect of system reliability by making evaluation outcomes more interpretable to the user or researcher.

This functionality is particularly relevant to QA over governance documents, where the understanding of why a response is considered correct or incorrect may be important when examining organisational rules, responsibilities and procedures. A numerical score alone provides very limited information regarding the specific strengths or deficiencies in a generated response. The additional explanation can therefore assist with being able to identify recurring failure patterns and performing manual analysis of individual responses.

For each of the questions, the system records the generated response, its evaluation result and the corresponding explanation. This information enables for future qualitative analysis of model behaviour without altering the quantitative performance of the underlying baseline architecture. Overall, the XAI implementation should be considered an interpretability enhancement rather than an answer correcting mechanism. Its contribution is therefore evaluated in terms of the additional transparency it provides to the QA and evaluation process, rather than improvements in final answer accuracy.

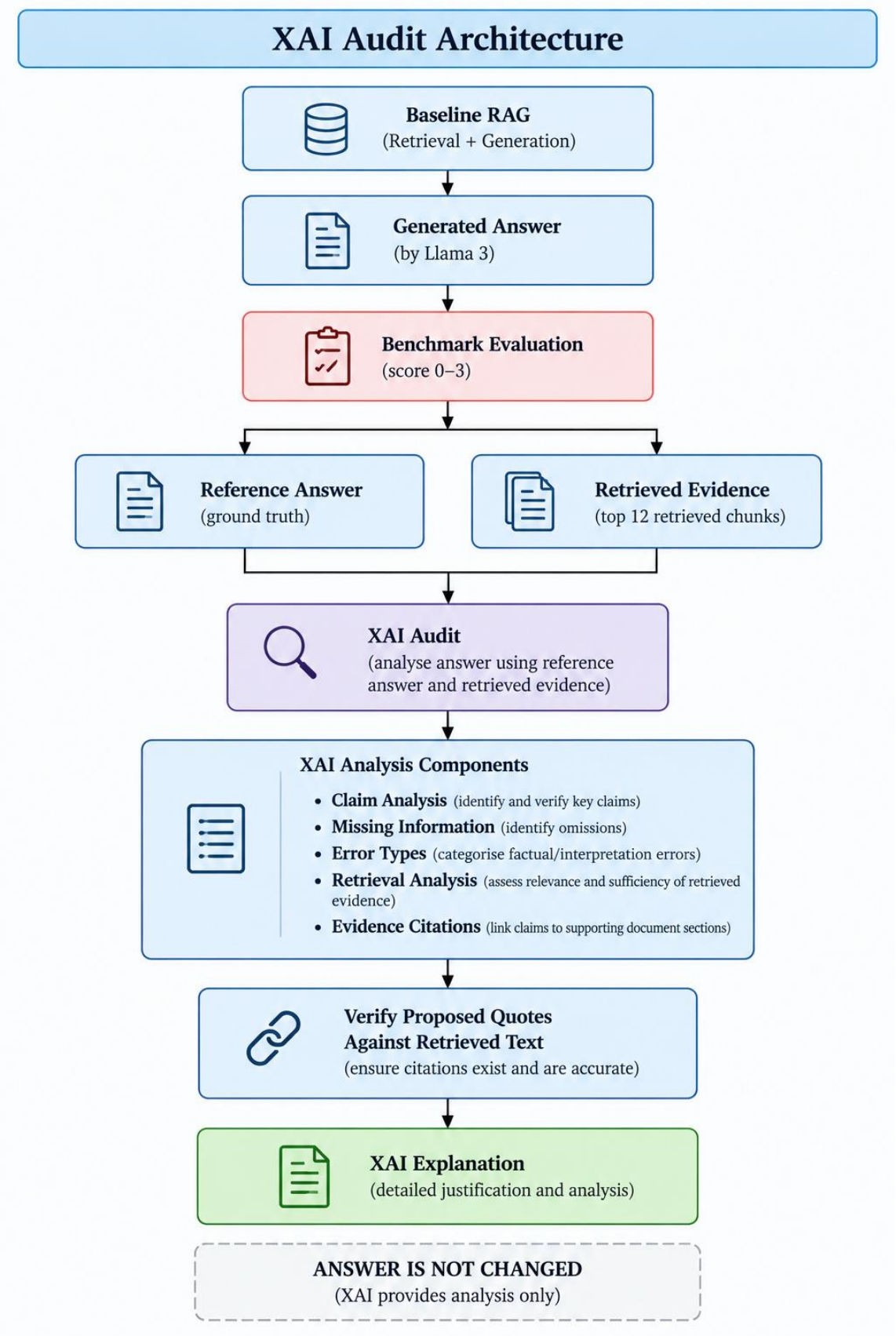


Figure 4.5: Flowchart of XAI Architecture

5 Experiment Results

This section presents the experimental results obtained from the Baseline Retrieval-Augmented Generation (RAG) system and the four reliability-enhancement architectures investigated in this study, namely RISE-inspired refinement, RLVR-inspired verification, Multi-Model Verification (MMV), and Explainable Artificial Intelligence (XAI). The results are presented comparatively to evaluate how each of the approaches affects the reliability and performance of QA over the governance document corpus.

The evaluation considers several dimensions of the systems performances. Overall QA performance is first assessed using the benchmark evaluation framework, followed by a category-level analysis so to identify any differences in performance across the various question types contained within the benchmark. Textual and semantic similarity are further examined using ROUGE, BLEU, and BERTScore metrics. Computational performance is also evaluated through measurements including response latency, processing duration, and the number of model calls required by each architecture.

In addition to these common evaluation measures, architecture-specific results are also examined to provide further insight into the behaviour of each of the different reliability enhancement. This includes the refinement behaviour of the RISE-inspired architecture, the verification and corrective-generation behaviour of the RLVR-inspired architecture, the individual and collective verifier performance of the Multi-Model Verification architecture, and the evidence-grounded explanations produced by the XAI architecture. Together, these results provide the basis for the evaluation of both the effectiveness and computational trade-offs of the investigated approaches, which are later examined in the Discussion chapter.

Table 5-1: Accuracy results across all architectures

Architecture	Correct	Incorrect	Accuracy	Δ Accuracy vs Baseline (pp)
Baseline RAG	104	6	94.55	-
RISE-Inspired	109	1	99.09	4.55
RLVR-Inspired	107	3	97.27	2.77
MMV	106	4	96.36	1.82
XAI	105	5	95.45	0.91

5.1 Overall QA Performance

The overall QA performance of each of the architectures was evaluated across the complete benchmark of 110 questions. Table 5.1 presents the number of correct and incorrect responses produced by each architecture, together with the resulting accuracy and the change in accuracy relative to the Baseline RAG system. Figure 5.1 provides a visual comparison of the overall accuracy achieved by each architecture.

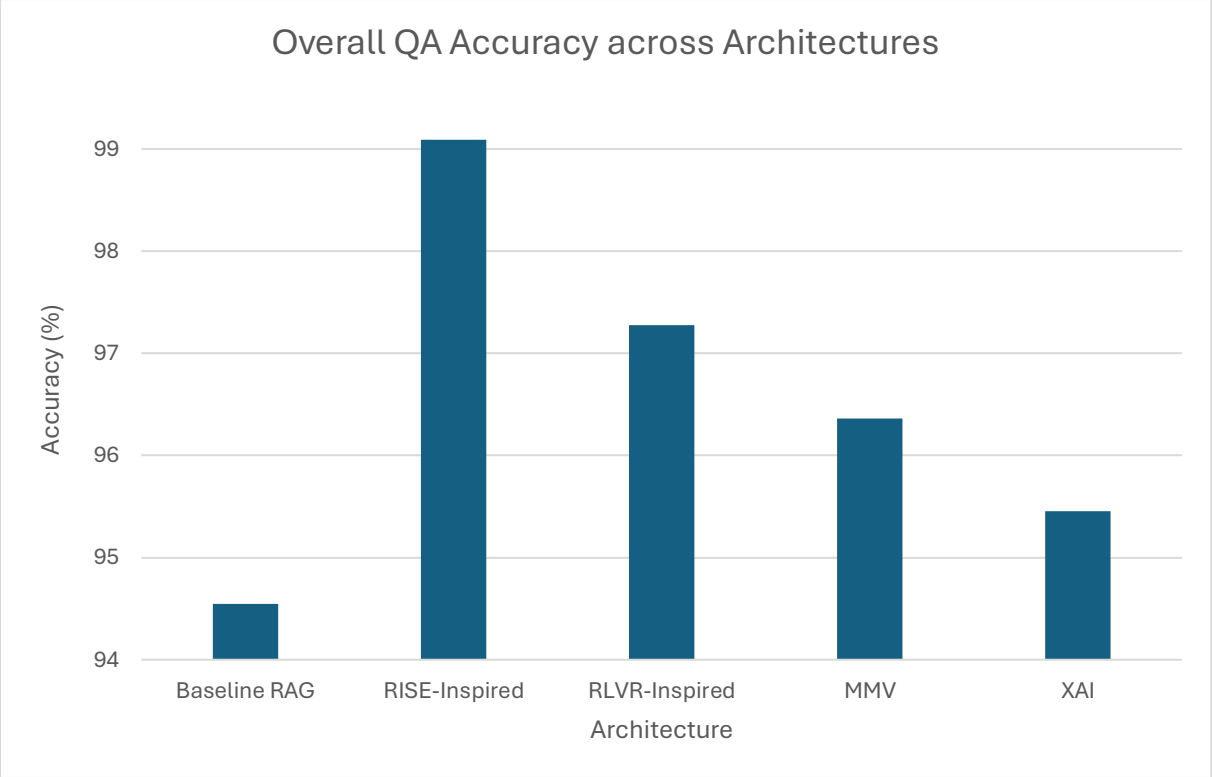


Figure 5.1: Graph of Overall QA Accuracy across all studied architectures

The Baseline RAG system correctly answered 104 of the 110 benchmark questions, corresponding to an overall accuracy of 94.55%. This result establishes the reference performance against which the reliability-enhancement architectures are compared.

The RISE-inspired architecture achieved the highest overall QA performance, correctly answering 109 of the 110 questions and producing only one incorrect response. This resulted in an accuracy of 99.09%, representing an increase of 4.55 percentage points compared with the Baseline RAG system. This was the largest positive difference in overall accuracy observed among the evaluated architectures.

The RLVR-inspired architecture achieved the second-highest overall accuracy, correctly answering 107 questions and producing three incorrect responses. This resulted in an accuracy of 97.27%, representing an increase of 2.73 percentage points relative to the baseline. The Multi-Model Verification (MMV) architecture correctly answered 106 questions, with four incorrect

responses, resulting in an overall accuracy of 96.36%. This represents an increase of 1.82 percentage points compared with the Baseline RAG system.

The XAI architecture achieved an overall accuracy of 95.45%, correctly answering 105 questions and producing five incorrect responses. This represents an increase of 0.91 percentage points compared with the Baseline RAG run. Unlike the RISE-inspired and RLVR-inspired architectures, the XAI implementation does not modify the generated response and instead operates as a post-hoc explanation and analysis mechanism. Similarly, the MMV architecture performs verification of generated responses rather than modifying them. The observed differences in final QA accuracy for these architectures should not be interpreted as improvements directly caused by their respective verification or explanation mechanisms. Their architecture-specific performance is examined separately in later sections.

Overall, the results demonstrate relatively high question-answering performance across all experimental runs, with accuracy ranging from 94.55% to 99.09%. Among the architectures capable of modifying generated responses, the RISE-inspired approach achieved the highest overall accuracy, followed by the RLVR-inspired approach. Both architectures demonstrated positive differences in accuracy relative to the Baseline RAG run. Further analysis by question category is presented in the following section to determine whether these overall performance differences are consistent across different types of governance questions.

Table 5-2: Accuracy results across individual question categories

Question Category	# of Question	Baseline RAG	RISE-Inspired	RLVR-Inspired	MMV	XAI
Direct Retrieval	12	92%	100%	100%	100%	100%
Cross-Document	10	90%	100%	90%	90%	90%
Semantic Retrieval	10	100%	100%	100%	100%	100%
Hallucination Trap	8	100%	100%	100%	100%	100%
Ambiguous	5	100%	100%	100%	100%	80%
Contradicting	5	100%	100%	100%	100%	100%
Tricky	10	80%	100%	90%	80%	80%
Multi-Hop Reasoning	10	100%	100%	90%	100%	100%
Temporal	10	100%	100%	100%	100%	100%
Negation	5	80%	100%	100%	100%	80%
Confirmatory	5	100%	80%	100%	100%	100%
Clarification Required	5	80%	100%	100%	100%	100%
Conditional	5	100%	100%	100%	80%	100%
Procedural	5	100%	100%	100%	100%	100%
Summary	3	100%	100%	100%	100%	100%
Differentiation	2	100%	100%	100%	100%	100%

5.2 Performance by Question Category

To further examine the differences in QA performance between the evaluated architectures, the benchmark results were analysed according to question category. Table 5.2 presents the accuracy achieved by each architecture across the 15 question categories contained within the benchmark. The number of questions within each of the categories is also provided, as the category sizes varied between two and twelve questions. Cell shading is used to visually distinguish the different accuracy levels and highlight categories where performance varied between architectures.

Overall, the results demonstrate consistently high performance across the majority of question categories. All five architectures achieved 100% accuracy for Semantic Retrieval, Hallucination Trap, Contradicting, Temporal, Procedural, Summary, and Differentiation questions. These categories therefore showed no observable differences in performance between the evaluated architectures. However, greater variation was observed in Direct Retrieval, Cross-Document, Ambiguous, Tricky, Multi-Hop Reasoning, Negation, Confirmatory, Clarification Required, and Conditional questions.

The Baseline RAG system achieved 100% accuracy across 10 of the 15 question categories. Its lowest performance was observed in Tricky, Negation, and Clarification Required questions, each achieving 80% accuracy. Cross-Document questions achieved 90%, while Direct Retrieval achieved 92%. These results establish the category-level reference performance against which the remaining architectures are compared.

The RISE-inspired architecture demonstrated the strongest overall category-level performance, achieving 100% accuracy in 14 of the 15 categories. Compared with the Baseline RAG system, improvements were observed in Direct Retrieval, Cross-Document, Tricky, Negation, and Clarification Required questions. In particular, Tricky questions increased from 80% to 100%, while Negation and Clarification Required questions also increased from 80% to 100%. Direct Retrieval improved from 92% to 100%, and Cross-Document improved from 90% to 100%. However, Confirmatory questions achieved 80% accuracy compared with 100% for the baseline, representing the only category in which the RISE-inspired architecture produced a lower accuracy.

The RLVR-inspired architecture achieved 100% accuracy across 12 of the 15 categories. Relative to the Baseline RAG system, improvements were observed in Direct Retrieval, Negation, and Clarification Required questions, all of which increased to 100%. However, Cross-Document and Tricky questions each achieved 90% accuracy, while Multi-Hop Reasoning questions decreased from 100% to 90%. These results indicate that the RLVR-inspired architecture achieved improvements in several categories while retaining some variation in performance across more complex question types.

The Multi-Model Verification (MMV) architecture achieved 100% accuracy in 11 of the 15 categories. Compared with the baseline, higher accuracy was observed for Direct Retrieval, Negation, and Clarification Required questions. Conversely, Tricky questions remained at 80%,

Cross-Document questions achieved 90%, and Conditional questions decreased from 100% to 80%. As the MMV architecture performs verification without modifying the generated responses, these differences reflect variation between experimental runs rather than improvements or degradations directly caused by the verification mechanism.

The XAI architecture achieved 100% accuracy in 11 of the 15 categories. Improvements relative to the baseline were observed for Direct Retrieval and Clarification Required questions, both of which achieved 100% accuracy. However, Ambiguous questions decreased from 100% to 80%, while Tricky and Negation questions remained at 80% and Cross-Document questions remained at 90%. As the XAI implementation operates as a post-hoc explanation mechanism without modifying generated responses, these differences similarly reflect variation in the underlying answer generation rather than changes introduced by the explanation process.

The category-level results demonstrate that the differences in overall accuracy reported in Section 5.1 were concentrated within a limited number of question categories rather than being uniformly distributed across the benchmark. In particular, Tricky, Negation, Clarification Required, and Cross-Document questions exhibited notable variation between architectures, while several other categories maintained perfect accuracy across all experimental runs. The RISE-inspired architecture achieved the most consistent category-level performance, although its lower accuracy for Confirmatory questions indicates that improvements were not universal across all question types. These findings provide a basis for further examining the effectiveness and limitations of the investigated reliability enhancements in Chapter 6.

5.3 Answer Similarity and Language-based Metrics

In addition to the correctness-based evaluation presented in Sections 5.1 and 5.2, textual and semantic similarity metrics were used to assess the similarity between generated responses and

their corresponding reference answers. The evaluation incorporated ROUGE-1, ROUGE-2, ROUGE-L, BLEU, and BERTScore F1 to provide complementary measurements of lexical overlap and semantic similarity. Table X presents the average scores obtained by each architecture across the 110 benchmark questions.

Table 5-3: Similarity and Language Based metrics across architectures

Architecture	ROUGE-1 F1	ROUGE-2 F1	ROUGE-L F1	BLEU	BERTScore F1
Baseline RAG	0.3127	0.1238	0.2322	0.0594	0.8640
RISE-Inspired	0.3158	0.1189	0.2269	0.0570	0.8640
RLVR-Inspired	**0.3324**	**0.1300**	**0.2421**	**0.0608**	**0.8687**
MMV	0.3098	0.1224	0.2296	0.0594	0.8646
XAI	0.3128	0.1238	0.2303	0.0585	0.8655

The RLVR-inspired architecture achieved the highest scores across all the similarity metrics, recording ROUGE-1, ROUGE-2, and ROUGE-L F1 scores of 0.3324, 0.1300, and 0.2421, respectively. It also achieved the highest BLEU score of 0.0608 and BERTScore F1 of 0.8687. These results indicate that the responses generated by the RLVR-inspired architecture exhibited the greatest overall similarity to the reference answers among the evaluated architectures.

The Baseline RAG architecture achieved ROUGE-1, ROUGE-2, and ROUGE-L F1 scores of 0.3127, 0.1238, and 0.2322, respectively, alongside a BLEU score of 0.0594 and BERTScore F1 of 0.8640. The RISE-inspired architecture produced a marginally higher ROUGE-1 F1 score of

0.3158 but slightly lower results across the remaining metrics, including a BLEU score of 0.0570 and BERTScore F1 of 0.8640. Similarly, the MMV and XAI architectures produced scores broadly comparable to the baseline, with BERTScore F1 values of 0.8646 and 0.8655, respectively.

Across all five architectures, BLEU scores remained relatively low, ranging from 0.0570 to 0.0608, whereas BERTScore F1 values were consistently higher, ranging from 0.8640 to 0.8687. This difference suggests that the generated responses exhibited a relatively limited exact n-gram overlap with the reference answers while maintaining substantial similarity under the contextual embedding-based measure. These differences are extremely relevant to governance QA, where factually equivalent responses may vary in their wording, terminology, and level of detail.

A notable observation is that the architecture achieving the highest correctness-based accuracy did not achieve the highest textual or semantic similarity scores. The RISE-inspired architecture recorded the highest QA accuracy of 99.09%, while the RLVR-inspired architecture achieved the highest similarity scores despite its lower QA accuracy of 97.27%. This demonstrates that similarity-based metrics and correctness-based evaluation do not necessarily produce identical performance rankings.

Overall, the findings indicate that textual and semantic similarity metrics provide useful supplementary information regarding response quality but should not be interpreted as direct measurements of factual correctness or faithfulness to the retrieved governance documents. The relationship between these metrics and the correctness-based results is examined further in Chapter 6.

Table 5-4: Computational efficiency metrics across architectures

Architecture	Mean Latency (s)	Median Latency (s)	Latency Standard Deviation (s)
Baseline RAG	3.3610	2.7398	2.0824
RISE-Inspired	8.1547	7.1005	4.4123
RLVR-Inspired	11.1351	9.0870	6.7935
MMV	31.6844	25.7706	27.9385
XAI	15.0662	14.2678	5.0846

Retrieval Latency (s)	Generation Latency (s)	LLM Calls	Embedding Calls
0.0176	3.3434	110	110
0.0195	3.3649	240	115
0.0230	3.1290	280	110
0.5723	2.9022	440	110
0.0199	3.2240	220	110

5.4 Computational Performance

The computational performance of the five architectures was evaluated to examine the processing overhead introduced by the implemented reliability-enhancement techniques. Performance was assessed using mean response latency, median latency, latency standard deviation, retrieval latency, generation latency, and the total number of LLM and embedding calls. Table 5.4 presents the computational measurements recorded across the 110 benchmark questions, while Figure 5.2 compares the average total latency of each architecture.

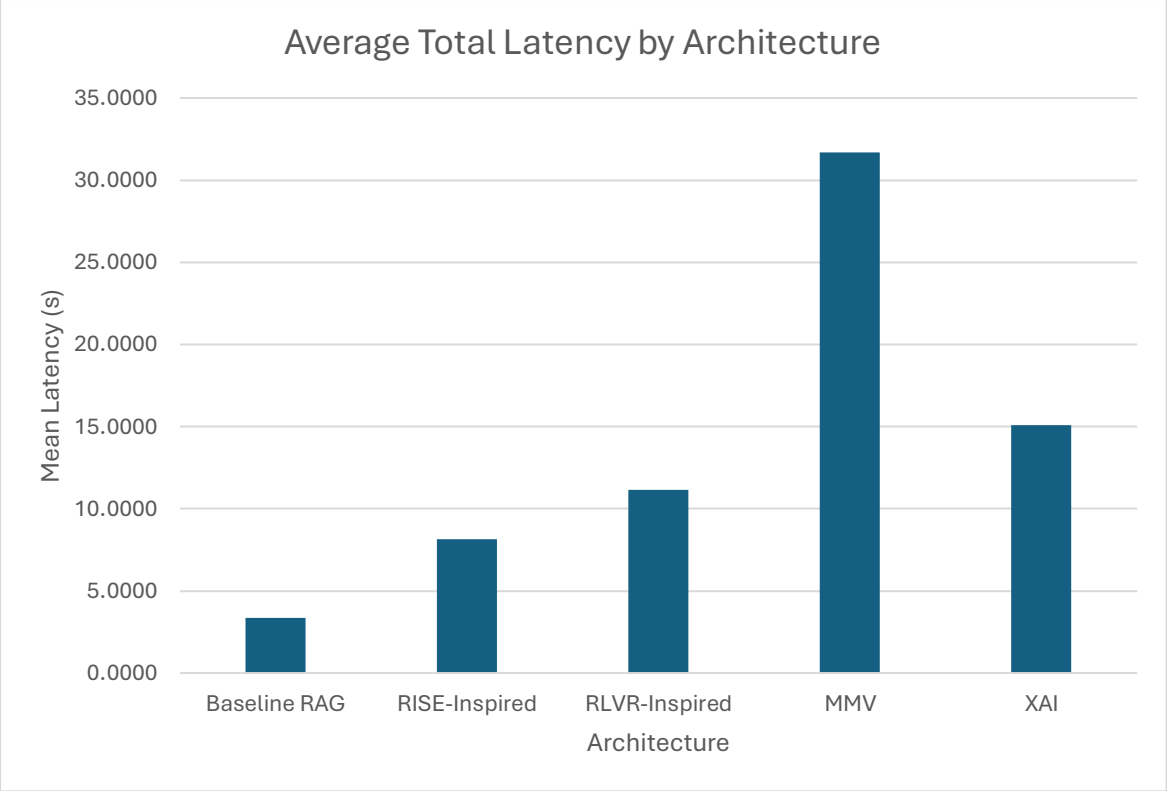


Figure 5.2: Graph of the Average total latency by architectures

The Baseline RAG architecture achieved the lowest average latency of 3.36 seconds per question, with a median latency of 2.74 seconds and a standard deviation of 2.08 seconds. The system required 110 LLM calls and 110 embedding calls across the benchmark, corresponding to one call of each type per question.

Among the enhanced architectures, the RISE-inspired approach achieved the lowest mean latency of 8.15 seconds per question, approximately 2.43 times that of the baseline. It recorded a median latency of 7.10 seconds and a standard deviation of 4.41 seconds, with 240 LLM calls and 115 embedding calls. The RLVR-inspired architecture recorded a higher mean latency of 11.14 seconds, approximately 3.31 times the baseline, requiring 280 LLM calls and 110 embedding calls.

The MMV architecture exhibited the greatest computational overhead, with a mean latency of 31.68 seconds per question, approximately 9.43 times that of the baseline. It also recorded the

highest median latency of 25.77 seconds and the largest standard deviation of 27.94 seconds. The architecture required 440 LLM calls across the benchmark, reflecting the additional processing associated with its multi-model verification mechanism.

The XAI architecture recorded a mean latency of 15.07 seconds per question, approximately 4.48 times that of the baseline. It required 220 LLM calls and 110 embedding calls. Although XAI required fewer LLM calls than RISE and RLVR, its higher mean latency demonstrates that the number of model calls does not directly correspond to total processing time.

Across all architectures, the reported initial generation latency remained relatively consistent, ranging from 2.90 to 3.36 seconds per question. In contrast, mean total latency varied substantially, reflecting differences in the additional processing associated with each architecture. The computational results therefore provide an important basis for examining the trade-offs between question-answering performance and processing requirements, which are discussed further in Chapter 6.

Table 5-5: RISE-inspired specific metric results

Metric	Result
Refinement Trigger Rate	2.73%
Final Answer Modification Rate	1.82%
Refinement Acceptance Rate	50%

5.5 Architecture Specific Results

5.5.1 RISE inspired Refinement Behaviour

The RISE-inspired architecture was further evaluated to examine the frequency and outcomes of its iterative refinement mechanism. Unlike the Baseline RAG system, the RISE-inspired implementation evaluates each initially generated response and initiates refinement only when the initial evaluation score falls below the maximum score of 3.

Across the 110 benchmark questions, 107 initial responses received the maximum evaluation score and therefore bypassed the refinement stage. The remaining three responses triggered refinement, corresponding to a refinement trigger rate of 2.73%. Of the three questions requiring refinement, two produced revised responses that received higher evaluation scores and were accepted by the system. Both accepted revisions occurred during the first refinement attempt, increasing their evaluation scores from 0 to 3 and from 2 to 3, respectively. The remaining question underwent two refinement attempts, neither of which produced an improvement in evaluation score. Therefore, the original response was then retained.

In total, four refinement attempts were performed, of which two were accepted and two were rejected. This corresponds to an attempt-level acceptance rate of 50.00% and a question-level refinement success rate of 66.67%, as determined by the automated evaluation framework. No reductions in final evaluation score were recorded, consistent with the architecture's requirement

that a revision must achieve a higher score before replacing the original response. The initial responses achieved an accuracy of 97.27%, with 107 of the 110 questions evaluated as correct. Following refinement, this increased to 99.09%, corresponding to 109 correct responses and an improvement of 1.82 percentage points within the RISE-inspired experimental run. The refinement stage introduced an average overhead of 20.68 seconds for the three questions requiring intervention. However, because the majority of responses bypassed refinement, the average refinement overhead across the complete benchmark was 0.564 seconds per question.

These results indicate that the conditional refinement mechanism was selectively activated and that the evaluation framework accepted revisions for two of the three initially lower-scoring responses. However, the accepted revisions should be interpreted as improvements according to the automated evaluation framework rather than independently established factual corrections. The reliability of these evaluation decisions and their implications for the refinement mechanism are considered further in Chapter 6.

5.5.2 RLVR Inspired Verification Behaviour

The RLVR-inspired architecture was evaluated to examine the effectiveness of its claim-level verification and targeted correction mechanism. Unlike the RISE-inspired approach, which uses the benchmark evaluation score to determine whether refinement is required, the RLVR-inspired architecture identifies potentially unsupported or incorrect claims through verification against the retrieved governance documents.

Across the 110 benchmark questions, the architecture achieved an initial accuracy of 96.36%, corresponding to 106 correct responses. Following verification and correction, the final accuracy increased to 97.27%, with 107 correct responses. This represents an improvement of 0.91 percentage points within the RLVR-inspired experimental run.

Table 5-6: RLVR-inspired specific metric results

Metric	Result
Total benchmark questions	110
Initially correct responses	106
Initially incorrect responses	4
Questions Triggering Correction	28
Answers changed	17
Incorrect-to-correct transitions	1
Correct-to-incorrect transitions	0
Final incorrect responses	3
Accuracy improvement	0.91%

The verification mechanism triggered correction for 28 questions, representing 25.45% of the benchmark. These triggers resulted in 28 additional answer generations, of which 17 produced changes to the final responses. The remaining 11 questions retained their original responses. Overall, 15.45% of the benchmark responses were modified following the verification and correction process.

Despite the relatively high number of correction attempts, only one response transitioned from incorrect to correct according to the benchmark evaluator. No correct-to-incorrect transitions were reported. The correction mechanism produced a net improvement of one correctly answered question, while the remaining response modifications did not affect the overall binary accuracy. The difference between the number of triggered corrections and the number of initially incorrect responses is also noteworthy. Although only four initial responses were classified as incorrect, the claim-level verifier triggered correction for 28 questions. This indicates that the verification

mechanism identified potential issues in responses that were otherwise classified as correct by the benchmark evaluator.

Overall, the RLVR-inspired architecture demonstrated a modest improvement in evaluated QA accuracy while maintaining the correctness of initially correct responses. However, the relatively small number of incorrect-to-correct transitions compared with the number of correction attempts highlights the distinction between identifying potential claim-level issues and achieving measurable improvements in final answer correctness. The implications of this behaviour are examined further in Chapter 6.

5.5.3 Multi-Model Verification Results

The Multi-Model Verification (MMV) architecture was evaluated to examine the effectiveness of combining independent verification decisions from three language models: Llama3, Qwen3:8B, and Mistral:7B. Unlike the RISE-inspired and RLVR-inspired architectures, MMV does not modify generated responses. Instead, each verifier assesses the generated answer against the retrieved governance evidence, with the final verification decision determined through majority voting. Table X presents the verification classification performance of the individual models and the combined ensemble. Mistral achieved the highest individual verification accuracy of 72.73%, correctly classifying 80 of the 110 responses. Qwen3 achieved an accuracy of 72.22% across 108 evaluated responses, while Llama3 recorded the lowest individual accuracy of 70.00% across 110 responses.

The MMV ensemble achieved a verification classification accuracy of 71.30%, correctly classifying 77 of the 108 responses included in the ensemble evaluation. This result was lower than the individual accuracies achieved by Mistral and Qwen3, indicating that majority voting did not produce an improvement in overall verification classification accuracy in this experiment.

Table 5-7: MMV specific results

Verifier	Accuracy	TP	TN	FP	FN
Llama3	70% (77/110)	1	76	30	3
Qwen3:8B	72.22 (78/108)	2	76	28	2
Mistral:7B	72.73% (80/110)	1	79	27	3
MMV Ensemble	71.30% (77/108)	1	76	28	3

The differing evaluation denominators should be considered when interpreting these comparisons. The agreement analysis indicated that all three models reached the same binary verification decision in 72.22% of cases, while a two-to-one disagreement occurred in the remaining 27.78%. Llama3 disagreed with both independent verifiers in 1.85% of cases. These results demonstrate that the verifiers frequently produced consistent decisions, although agreement alone does not establish whether those decisions were correct.

Verification performance also varied substantially across question categories. The ensemble achieved 100% accuracy in the Semantic Retrieval, Ambiguous, and Procedural categories, while performance was considerably lower for Hallucination Trap and Tricky questions, at 0% and 20%, respectively. In the Hallucination Trap category, Qwen3 achieved 75% verification accuracy, whereas Llama3 and Mistral each achieved 0%. The ensemble also recorded 0% accuracy in this category, illustrating a case in which majority voting favoured the decisions of the two lower-performing verifiers.

Table 5-8: Model results against specific question categories

Category	Ensemble	Llama3	Qwen3	Mistral
Semantic Retrieval	100%	90%	100%	100%
Hallucination Trap	0%	0%	75%	0%
Tricky	20%	20%	30%	20%
Multi-Hop Reasoning	90%	90%	60%	100%
Temporal	80%	80%	40%	80%
Procedural	100%	100%	100%	100%

Of the 110 benchmark questions processed by the Multi-Model Verification architecture, 108 produced valid ensemble verification outcomes. Two questions, Q066 (Multi-Hop Reasoning) and Q106 (Summary), resulted in unresolved ensemble classifications due to invalidly formatted responses from the Qwen3 verifier. In both cases, the remaining two verifiers returned a SUPPORTED verdict. However, the implemented ensemble procedure required valid outputs from all three models before determining a majority classification. Hence, ensemble verification performance was evaluated over 108 valid classifications, while the two unresolved cases were reported separately. These parsing failures did not affect the number of questions processed or the independent evaluation of generated answer correctness.

Overall, the results demonstrate that combining multiple verification models did not necessarily improve verification accuracy. Although the ensemble provided a mechanism for aggregating independent assessments, its effectiveness remained dependent on the accuracy and agreement

patterns of the individual models. The implications of these findings, particularly regarding correlated verification errors and the computational cost of multi-model verification, are examined further in Chapter 6.

5.5.4 XAI Results

The XAI architecture generated post-hoc explanations for all 110 benchmark questions, providing claim-level assessments, supporting evidence, and identification of potential errors or missing information. Three representative examples were selected to illustrate the explanations produced by the architecture.

5.5.4.1 Q013 – Correct Explanation with Supporting Evidence

Q013 examined the relationship between the Operational Manual and the Constitution and Bylaws when conflicting provisions exist. The generated response correctly identified that the Operational Manual is subordinate to the Constitution and Bylaws.

The XAI mechanism classified the response as correct and identified supporting evidence from the Operational Manual, including the following verified quotation:

"This operations manual is subordinate to the Constitution and Bylaws."

The explanation identified the relevant document hierarchy and associated the generated response with a verified source passage.

5.5.4.2 Q056 – Identification of an Incorrect Response

Q056 concerned the Treasurer's authority to approve Society expenditures. The XAI mechanism identified a discrepancy between the generated response and the reference answer regarding the allocation of financial approval authority.

The explanation classified the response as incorrect and highlighted the conflicting information, demonstrating how the mechanism reports factual discrepancies within generated answers.

5.5.4.3 Q014 – Disagreement with the Original Evaluation

Q014 asked which document explains how Technical Committees operate after their establishment. The generated response identified the SMCS Technical Manual, consistent with the reference answer.

Although the original benchmark evaluator assigned the response a correctness score of 3, the XAI mechanism classified it as incorrect. Its explanation questioned the supporting evidence despite identifying verified passages from the Technical Manual.

This example illustrates an instance in which the XAI explanation produced a different correctness judgment from the original evaluation.

5.5.4.4 Summary of Explanation Results

Across the 110 questions, 51 explanations completed without citation warnings, while 59 contained at least one warning. The mechanism produced 78 verified evidence quotations across 41 questions, while 204 proposed quotations were rejected by the implemented verification procedure.

Table 5.9 summarises the representative examples.

Table 5-9: XAI representative example results

Question	Original Evaluation	XAI Verdict	Explanation Outcome
Q013	Correct	Correct	Correct document hierarchy identified with verified supporting evidence
Q056	Incorrect	Incorrect	Discrepancy in financial approval authority identified
Q014	Correct	Incorrect	Incorrect XAI judgment despite a correct generated response

These examples illustrate the range of explanation outputs produced by the XAI architecture, including evidence-supported explanations, identification of response errors, and disagreement with the original evaluation. The reliability and implications of these findings are discussed in Chapter 6.

6 Discussion

6.1 Interpretation of Overall Performance and Question Categories

The experimental results demonstrate that the implemented RAG architectures achieved generally high levels of QA accuracy across the governance-document benchmark, although the extent of improvement varied between configurations. The Baseline RAG architecture achieved an accuracy of 94.55%, correctly answering 104 of the 110 benchmark questions. This indicates that the baseline retrieval and generation pipeline was already capable of answering most questions using the selected governance documents. Consequently, the enhancement techniques operated within a relatively narrow margin for improvement, with only six baseline questions classified as incorrect.

Among the evaluated configurations, the RISE-inspired architecture achieved the highest overall accuracy at 99.09%, followed by RLVR-inspired verification at 97.27%, Multi-Model Verification (MMV) at 96.36%, and Explainable AI (XAI) at 95.45%. Relative to the baseline, these results corresponded to improvements of 4.55, 2.73, 1.82, and 0.91 percentage points, respectively. However, these differences should not be interpreted as equivalent evidence of improvement caused by each enhancement technique, as the architectures differ in how they interact with generated responses.

In particular, the RISE-inspired and RLVR-inspired architectures incorporate mechanisms capable of revising generated answers, whereas MMV and XAI operate as post-generation verification and explanation stages. In the implemented MMV and XAI configurations, the original answers remain unchanged regardless of the subsequent verification or explanation outcome. Therefore, their differences in overall accuracy relative to Baseline RAG reflect the results of independent experimental runs rather than direct improvements introduced by these mechanisms. This distinction is important when interpreting the effectiveness of each

architecture, as answer correction, answer verification, and explanation generation represent different aspects of system reliability.

The comparatively strong baseline performance also raises questions regarding the difficulty and discriminative capability of the benchmark. While the question set included direct retrieval, cross-document reasoning, ambiguity, negation, and other challenging categories, the baseline achieved perfect accuracy in 11 of the 16 categories. This suggests that many benchmark questions were adequately addressed by the existing retrieval and generation pipeline. However, it may also indicate that the selected questions did not consistently expose the limitations of the baseline architecture. These interpretations are not mutually exclusive, and the high baseline accuracy alone is insufficient to determine whether the benchmark was appropriately challenging.

The category-level results provide further insight into where differences between the architectures were concentrated. Direct Retrieval questions were answered with relatively high accuracy across all configurations, with Baseline RAG achieving 91.67% and the enhanced architectures achieving 100%. Similarly, Semantic Retrieval, Hallucination Trap, Contradicting, Temporal, Procedural, Summary, and Differentiation questions achieved perfect accuracy across all five configurations. These results indicate that the implemented architectures performed consistently on several categories, although they provide limited evidence for distinguishing the effectiveness of individual enhancement techniques.

Greater variation was observed in the categories involving more complex interpretation or response requirements. Baseline RAG achieved 80% accuracy on Tricky, Negation, and Clarification Required questions, while the RISE-inspired architecture achieved 100% in each of these categories. RLVR also achieved 100% on Negation and Clarification Required questions, although its accuracy on Tricky questions was 90%. These differences suggest that the

configurations incorporating answer refinement or correction may be beneficial for certain questions where the initial response is incomplete, misleading, or insufficiently attentive to the wording of the question. Nevertheless, the category-level results alone do not establish which specific errors were corrected, as such conclusions require examination of the individual question outcomes.

The RISE-inspired architecture achieved perfect accuracy in 15 of the 16 categories, with Confirmatory questions representing its only category below 100%, at 80%. This demonstrates that its strong overall performance was not uniform across every question type. Similarly, RLVR achieved 90% accuracy in Cross-Document, Tricky, and Multi-Hop Reasoning questions, while MMV recorded 80% accuracy in Tricky and Conditional questions. XAI achieved 80% accuracy in Ambiguous, Tricky, and Negation questions. These variations indicate that no single configuration demonstrated uniformly superior performance across all categories, despite the differences in overall accuracy.

An important consideration is the relatively small number of questions within several benchmark categories. While Direct Retrieval contained 12 questions and categories such as Tricky and Multi-Hop Reasoning contained 10, others contained only five, three, or two questions. For a category containing five questions, a single incorrect response reduces accuracy by 20 percentage points. Hence, any apparent substantial differences in category accuracy may reflect only one or two individual responses. The category-level results should therefore be interpreted as descriptive observations within the selected benchmark rather than definitive evidence that a particular architecture is more effective for an entire class of questions.

Overall, the results suggest that the baseline RAG architecture provided a strong foundation for governance-document question answering, while the greatest measured accuracy was obtained through the RISE-inspired refinement configuration. The findings also highlight the importance

of distinguishing architectures that actively modify answers from those that provide verification or explanation without changing the original response. Although the category-level results identify areas where performance differed, the relatively small benchmark and high baseline accuracy limit the strength of conclusions that can be drawn from these differences alone. A more detailed examination of the errors addressed and those remaining unresolved is presented in Section 6.2.

6.2 Error Analysis and Effectiveness of Enhancement Techniques

Although the enhanced architectures generally achieved higher overall question-answering accuracy than Baseline RAG, their effectiveness varied according to the mechanisms employed and the types of errors they were designed to address. The RISE-inspired and RLVR-inspired architectures incorporated corrective processes capable of modifying generated responses, whereas Multi-Model Verification (MMV) and Explainable AI (XAI) focused on assessing or explaining responses without changing their content. Therefore, the effectiveness of these approaches must be considered in relation to their respective objectives rather than solely through overall accuracy.

6.2.1 RISE-Inspired Refinement

The RISE-inspired architecture demonstrated the strongest measured performance, achieving a final accuracy of 99.09%. Its selective refinement mechanism was designed to revise responses that failed the initial benchmark evaluation while retaining responses already classified as correct. This approach reduced the number of unnecessary revisions and limited the possibility of degrading initially correct answers.

Of the 110 benchmark questions, three of the responses triggered the refinement process, with two revisions accepted and one question remaining unresolved. The architecture improved its

initial accuracy from 97.27% to 99.09%, corresponding to two additional responses classified as correct. No degradation was recorded under the implemented evaluation framework.

However, examination of individual cases revealed potential limitations in the reliability of these improvements. For Q042 and Q097, revised responses received higher evaluation scores despite discrepancies between their content and the corresponding reference answers. This suggests that an improvement in the evaluator-assigned score does not necessarily establish that a revised response is more factually accurate or complete.

A further consideration is the architecture's dependence on reference-based evaluation to determine when refinement should occur and whether a revision should be accepted. Although the refinement stage itself operated using the question, generated answer, and retrieved evidence, the decision to initiate or accept refinement relied on benchmark evaluation results. As reference answers would not ordinarily be available for previously unseen questions, the observed effectiveness of this implementation cannot be assumed to transfer directly to a practical deployment environment.

6.2.2 RLVR-Inspired Verification and Correction

The RLVR-inspired architecture employed an evidence-based verification process to identify potentially unsupported claims and generate targeted corrections. Unlike RISE, its correction mechanism did not require the benchmark reference answer to determine whether a response should be revised.

Across the benchmark, 28 of the 110 questions triggered additional verification-guided generation, resulting in 17 changes to the final responses. However, only one additional response was classified as correct following correction, increasing accuracy from 96.36% to 97.27%. None of the initially correct responses were recorded as becoming incorrect.

The difference between the number of modified responses and the number of improvements in binary accuracy suggests that verification-guided correction frequently altered responses without changing their final correctness classification. Some modifications may have improved wording, completeness, or evidence alignment without affecting the evaluation outcome. Alternatively, the verification mechanism may have identified issues that were not sufficiently significant to change the benchmark score. The available aggregate results do not establish which explanation applies to each modification.

Furthermore, the verification process was triggered for 28 questions despite only four initial responses being classified as incorrect by the benchmark evaluator. This difference indicates that the document-based verifier and reference-based evaluator did not always identify the same responses as requiring correction. Such disagreement may arise from differences in their assessment criteria, although the individual verification records would require further examination to determine the precise causes.

These findings suggest that evidence-based verification can support selective answer correction without requiring reference answers during inference. However, the relatively small improvement in measured accuracy also indicates that triggering additional correction does not guarantee a meaningful improvement in the final response.

6.2.3 Multi-Model Verification

The MMV architecture introduced three independent language-model verifiers to assess whether generated answers were supported by retrieved governance evidence. Unlike RISE and RLVR, this implementation did not use verification outcomes to modify the generated responses. Its effectiveness must therefore be considered primarily in terms of verification reliability rather than answer correction.

The individual verifiers achieved classification accuracies of 70.00% for Llama3, 72.22% for Qwen3, and 72.73% for Mistral. The majority-voting ensemble achieved 71.30% across 108 valid verification outcomes, demonstrating that combining the three models did not produce a higher classification accuracy than the best-performing individual verifier.

The ensemble also exhibited substantial variation across question categories. For example, verification accuracy reached 100% for Semantic Retrieval and Procedural questions but was only 20% for Tricky questions and 0% for Hallucination Trap questions. These results indicate that the verification models were not equally effective at distinguishing supported and unsupported responses across different question types.

Additionally, two questions produced unresolved ensemble outcomes because Qwen3 failed to return a verdict in the required format. Although the other two verifiers produced valid judgments, the implemented ensemble procedure required valid outputs from all three models before calculating a majority decision. This demonstrates that verification reliability was influenced not only by the models' assessments but also by their compliance with the required output structure.

The findings suggest that independent verification does not necessarily eliminate the limitations of individual language-model judgments. Majority voting may provide a mechanism for combining assessments, but its effectiveness depends on the accuracy, consistency, and independence of the participating verifiers.

6.2.4 Explainable AI

The XAI architecture focused on producing post-hoc explanations rather than correcting generated responses. Its outputs included claim-level assessments, identification of missing information, retrieval analysis, and supporting quotations from governance documents.

The representative examples presented in Section 5.5.4 demonstrated both the capabilities and limitations of this approach. In Q013, the mechanism correctly explained the hierarchy between the Operational Manual and the Constitution and Bylaws, supported by a verified quotation. In contrast, Q014 demonstrated an inconsistency in which XAI classified a response as incorrect despite its identification of the appropriate Technical Manual and a correct benchmark evaluation.

These examples highlight the distinction between producing an explanation and ensuring that the explanation is reliable. Although the architecture incorporated exact-text verification for proposed evidence quotations, the presence of a verified quotation did not guarantee that the associated interpretation was correct. Similarly, citation verification failures did not necessarily establish that the generated explanation was factually incorrect.

Hence, the XAI mechanism provided additional visibility into generated responses and their supporting evidence, but its explanations remained dependent on language-model judgments that could themselves contain errors or inconsistencies.

6.2.5 Overall Effectiveness of the Enhancement Techniques

Taken together, the results indicate that the implemented techniques addressed different aspects of reliability. RISE achieved the largest measured improvement through selective refinement, although its dependence on benchmark evaluation limits its direct applicability to unseen questions. RLVR provided a more inference-oriented correction mechanism based on retrieved evidence, but its additional generations produced relatively few improvements in binary correctness. MMV offered independent assessments of answer support without modifying responses, while XAI provided structured explanations that could assist in examining generated answers.

A common limitation across these approaches was their dependence on the reliability of language-model judgments. In RISE, evaluation errors could influence which revisions were accepted. In RLVR, verifier decisions determined whether correction was necessary. In MMV, inconsistent judgments and invalid outputs affected ensemble reliability. In XAI, explanation errors could produce misleading assessments even when supporting quotations were available.

These findings suggest that incorporating additional verification, refinement, or explanation stages does not automatically ensure more reliable question answering. Their effectiveness depends on whether the additional stages can accurately identify errors and, where applicable, correct them without introducing new inaccuracies. The computational implications of these mechanisms are examined in Section 6.3.

6.3 Reliability and Computational Trade-offs

The introduction of additional refinement, verification, and explanation mechanisms increased the computational requirements of all enhanced RAG architectures. Although these techniques were intended to improve different aspects of reliability, their additional processing stages resulted in higher response latency and greater numbers of language-model calls. Consequently, their effectiveness must be considered not only in terms of QA accuracy but also in relation to the computational resources required to achieve their respective objectives. As presented in Section 5.4, Baseline RAG achieved the lowest average response latency at 3.36 seconds per question, requiring 110 LLM calls across the benchmark. This establishes a relatively efficient reference point against which the enhanced architectures can be compared.

6.3.1 RISE-Inspired Refinement

The RISE-inspired architecture achieved the highest measured question-answering accuracy of 99.09%, with an average latency of 8.15 seconds per question. This represented a 4.55

percentage-point improvement in accuracy relative to Baseline RAG, at approximately 2.43 times the baseline latency.

Among the evaluated configurations, RISE demonstrated the most favourable combination of measured accuracy and average latency. Its selective refinement mechanism avoided revising responses that had already achieved the maximum evaluation score, allowing additional processing to be directed towards responses identified as requiring improvement.

However, this apparent computational advantage must be considered alongside the architecture's reliance on reference-based evaluation. The benchmark evaluator was used to determine whether refinement should occur and whether a revised response should be accepted. In a practical deployment where reference answers are unavailable, an alternative verification mechanism would be required, potentially introducing additional computational costs and affecting the reliability of refinement decisions.

6.3.2 RLVR-Inspired Verification and Correction

The RLVR-inspired architecture achieved an accuracy of 97.27%, with an average latency of 11.14 seconds per question, approximately 3.31 times that of Baseline RAG. The architecture required 280 LLM calls across the benchmark, compared with 240 for RISE and 110 for Baseline RAG.

Although RLVR provided a mechanism for detecting unsupported claims and performing targeted corrections without relying on reference answers during inference, the additional verification and generation stages produced a relatively small improvement in measured accuracy. Only one additional response was classified as correct following correction, despite 28 questions triggering the correction process.

These findings suggest that verification-guided correction introduces a trade-off between additional evidence checking and computational efficiency. While the verification process may

identify issues not captured by the benchmark's binary correctness classification, the observed accuracy improvement was limited relative to the additional processing required.

Nevertheless, RLVR offers a potential practical advantage over the implemented RISE approach because its correction decisions are based on retrieved documentary evidence rather than benchmark reference answers.

6.3.3 Multi-Model Verification

MMV incurred the highest computational cost, recording an average latency of 31.68 seconds per question, approximately 9.43 times the baseline. The architecture required 440 LLM calls across the benchmark, reflecting the initial answer generation followed by independent verification using three language models.

Despite this additional computational expenditure, the ensemble achieved a verification classification accuracy of 71.30% across valid outcomes. Furthermore, majority voting did not outperform the best-performing individual verifier, Mistral, which achieved 72.73%.

These findings indicate that the additional computational cost of multi-model verification did not consistently translate into more accurate verification judgments within the implemented configuration. The use of multiple models also introduced further complexity, including disagreement between verifiers and unresolved outcomes caused by invalid model responses.

However, the computational overhead of MMV should not be evaluated solely against question-answering accuracy, as the architecture was designed to assess answer support rather than modify generated responses. Its practical value would depend on whether the additional verification information could support downstream decisions, such as flagging uncertain answers for review or preventing unsupported responses from being presented without qualification. Such applications were not directly evaluated in this study.

6.3.4 Explainable AI

The XAI architecture recorded an average latency of 15.07 seconds per question, approximately 4.48 times that of Baseline RAG, and required 220 LLM calls across the benchmark.

Unlike RISE and RLVR, the additional computational cost of XAI was associated with generating structured explanations rather than correcting responses. Its value cannot be adequately assessed through improvements in question-answering accuracy alone.

The explanations provided additional information concerning claim alignment, potential errors, missing information, and supporting documentary evidence. However, the occurrence of citation verification warnings and inconsistencies between explanation judgments and benchmark evaluations indicates that the additional computational expenditure did not guarantee reliable explanations.

The results therefore highlight a trade-off between interpretability and response efficiency. In applications where users require supporting evidence or explanations of governance-related answers, the additional latency may be acceptable. Conversely, for applications prioritising rapid response generation, the explanation stage may represent a substantial computational overhead.

6.3.5 Overall Trade-offs

The results demonstrate that increasing computational complexity does not necessarily produce proportional improvements in reliability. RISE achieved the highest measured accuracy with a lower average latency than the other enhanced architectures, although its reliance on reference-based evaluation limits direct deployment applicability. RLVR offered evidence-based correction without requiring reference answers but achieved a smaller accuracy improvement at a greater computational cost. MMV and XAI introduced verification and explanation capabilities, respectively, rather than directly improving generated responses.

These findings suggest that the selection of an appropriate architecture depends on the reliability requirements of the intended application. Where answer correctness is the primary objective, selective correction mechanisms may offer advantages over approaches that only assess or explain existing responses. However, where independent verification or evidence traceability is required, additional computational overhead may be justified despite the absence of direct improvements in answer accuracy.

Ultimately, the results indicate that reliable governance-document question answering should not be evaluated through accuracy or computational efficiency in isolation. Instead, the suitability of each architecture depends on the balance between answer correctness, verification reliability, interpretability, and the computational resources available for deployment.

6.4 Limitations and Threats to Validity

Although the experimental results provide insight into the effectiveness of different RAG enhancement techniques, several limitations must be considered when interpreting the findings. These limitations concern the evaluation framework, benchmark construction, experimental design, and generalisability of the results. Addressing these factors is important because improvements in measured question-answering accuracy do not necessarily correspond to equivalent improvements in factual reliability under practical deployment conditions.

6.4.1 Evaluation Reliability and Reference Answer Quality

A significant limitation of this study is its reliance on an LLM-based evaluator to assess the correctness of generated responses. The evaluation framework compared generated answers against predefined reference answers using a scoring rubric, with responses receiving the maximum score classified as correct. Although a consistent evaluation approach supported comparisons between configurations, the evaluator remained susceptible to incorrect judgments, inconsistent interpretations, and sensitivity to response wording.

This limitation was particularly relevant to the RISE-inspired architecture, where benchmark evaluation outcomes influenced both the selection of responses for refinement and the acceptance of revised answers. Examination of individual cases revealed that some revisions received improved evaluation scores despite apparent discrepancies with their reference answers. Hence, the reported accuracy improvements should be interpreted as improvements under the implemented evaluation framework rather than definitive evidence of increased factual correctness.

The XAI results further demonstrated the potential for disagreement between independent assessments of the same response. In certain cases, the explanation mechanism identified errors in responses that the benchmark evaluator had classified as correct. Such disagreements do not necessarily establish which assessment was accurate, but they highlight the difficulty of relying on automated judgments without additional validation.

Reference answer quality also represents a potential source of evaluation error. Governance questions may require nuanced interpretations involving multiple provisions, exceptions, or document relationships. If a reference answer is incomplete, ambiguous, or insufficiently specific, a factually reasonable generated response may be incorrectly penalised. Conversely, a response that resembles the reference answer may receive a favourable evaluation despite omitting important qualifications. Although the use of a consistent evaluation rubric supports methodological consistency, it cannot eliminate errors originating from the reference answers or evaluator itself.

6.4.2 Benchmark Size and Question Distribution

The benchmark consisted of 110 questions distributed across 16 categories intended to represent different retrieval, reasoning, and interpretation requirements. While this provided a structured

basis for comparing the architectures, the relatively small number of questions limits the statistical strength of the findings.

Several categories contained only five questions, while Summary and Differentiation contained three and two questions, respectively. Therefore, any one single incorrect response could produce a substantial change in category-level accuracy. This makes it difficult to determine whether observed differences represent consistent architectural strengths or weaknesses rather than the outcomes of a small number of individual questions.

Furthermore, Baseline RAG achieved an overall accuracy of 94.55%, leaving relatively few incorrect responses available for the enhancement techniques to address. Although this may indicate that the baseline architecture was effective for the selected governance documents, it may also suggest that the benchmark did not consistently challenge the system's capabilities. A larger benchmark containing more difficult and diverse questions would provide a stronger basis for evaluating the robustness of the observed improvements.

The benchmark was also developed around a specific collection of IEEE governance documents. As a result, the questions and reference answers may reflect the terminology, structure, and governance procedures of that collection. This limits the extent to which the findings can be generalised to other organisational or regulatory document sets.

6.4.3 Experimental Design and Comparability

The study employed a consistent document corpus, benchmark, and evaluation framework to support comparison between architectures. However, the configurations differed in their internal processing stages and the information available to their enhancement mechanisms.

In particular, RISE relied on reference-based evaluation to determine whether refinement was necessary and whether a revision should be accepted. In contrast, RLVR employed document-based verification during inference, while MMV and XAI performed post-generation assessment

without modifying the original answer. These differences mean that the architectures were not evaluated under identical inference-time information conditions.

Additionally, the architectures were executed independently rather than applying every enhancement mechanism to an identical set of initial generated responses. Differences in final accuracy, particularly for MMV and XAI, cannot be attributed exclusively to their post-generation mechanisms. Independent runs may produce different answers even when the same underlying generation model and configuration are used.

The use of deterministic generation settings where implemented may reduce output variability, but it does not necessarily guarantee identical results across separate executions. Variations in retrieval, model execution, or system conditions may still influence generated responses and computational measurements. Repeated experimental runs would therefore be beneficial for assessing the stability of the reported results.

6.4.4 Verification and Explanation Limitations

The enhancement techniques also introduced limitations associated with their own assessment mechanisms. The RLVR-inspired architecture depended on a language-model verifier to determine whether claims were supported by retrieved evidence. Although this removed the need for reference answers during correction, the verifier could still incorrectly identify valid claims as unsupported or fail to recognise factual errors. Similarly, MMV demonstrated that combining multiple verification models did not necessarily improve classification accuracy. The ensemble was affected by disagreements between models and by invalid outputs that prevented a valid majority decision. These findings suggest that using multiple language models does not guarantee independent or complementary judgments, particularly when the models are presented with the same retrieved evidence.

The XAI architecture also relied on language-model-generated explanations. Although proposed supporting quotations were checked against retrieved text, successful quotation verification established only that the quoted text appeared in the retrieved material. It did not independently establish that the quotation supported the associated claim or that the explanation's interpretation was correct. Consequently, explanations containing verified quotations could still be misleading or incomplete. These limitations are particularly important in governance-document question answering, where users may rely on supporting evidence and explanations when interpreting organisational rules and procedures.

6.4.5 Generalisability and Computational Constraints

The experimental corpus consisted of four IEEE governance documents, limiting the diversity of document structures, terminology, and organisational contexts examined. The findings may not directly transfer to larger collections, frequently updated policies, legal documents, or governance frameworks containing more extensive contradictions and exceptions.

The experiments were also conducted using locally deployed language models through Ollama and a specific hardware environment. Hence, the measured latency and computational requirements reflect the selected models, implementation, and available computing resources. Different hardware configurations, model sizes, or deployment environments could produce substantially different performance characteristics.

Furthermore, the study primarily evaluated answer correctness, textual and semantic similarity, and computational performance. Although these metrics provide useful evidence of system behaviour, they do not fully capture other practical dimensions of reliability, such as user trust, the consequences of incorrect governance advice, or the effectiveness of human review.

6.4.6 Summary of Limitations

Overall, the findings should be interpreted within the scope of the selected document corpus, benchmark, evaluation framework, and experimental environment. The most significant limitation is the dependence on automated judgments of correctness and evidence support, particularly where these judgments influence subsequent answer refinement.

Despite these constraints, the controlled comparison provides useful evidence of how different enhancement mechanisms behave when applied to governance-document question answering. The identified limitations also highlight areas requiring further investigation, including stronger evaluation validation, larger and more diverse benchmarks, repeated experiments, and assessment of the architectures under practical deployment conditions.

6.5 Implications for Governance-Document QA and Research Questions

The findings of this study provide insight into the effectiveness of different approaches for improving the reliability of retrieval-augmented generation when answering questions over governance documents. Although all evaluated architectures achieved relatively high question-answering accuracy, their results demonstrated that reliability involves more than producing an answer that satisfies an automated correctness evaluation. The ability to identify unsupported claims, correct inaccurate responses, and provide meaningful supporting evidence represents additional considerations when designing systems intended for governance-related applications.

6.5.1 Implications for Reliable Governance-Document Question Answering

The strong performance of Baseline RAG suggests that retrieval-augmented generation provides a useful foundation for answering questions over structured governance documents. By incorporating retrieved documentary evidence into the generation process, the system was able to answer most benchmark questions without requiring additional enhancement mechanisms.

However, the remaining incorrect responses demonstrate that access to relevant documents does not guarantee accurate interpretation or complete answers.

This distinction is particularly important for governance documents, where the interpretation of a provision may depend on its relationship with other rules, exceptions, or organisational procedures. An answer that is generally consistent with the retrieved evidence may still be misleading if it omits an important condition or fails to distinguish between different circumstances. Therefore, reliable governance-document question answering requires attention to both the factual content of generated responses and the completeness of their interpretation.

The results also suggest that selectively applying additional processing may be more practical than subjecting every response to increasingly complex verification procedures. The RISE-inspired architecture achieved the highest measured accuracy through targeted refinement, while the RLVR-inspired architecture demonstrated that document-based verification could identify responses requiring further examination. However, the limitations observed in both approaches indicate that the effectiveness of selective correction depends on the reliability of the mechanism used to identify and assess errors.

6.5.2 Implications of Verification and Explainability

The MMV and XAI results highlight the distinction between improving answer correctness and improving the ability to assess generated responses. Although these architectures did not directly correct answers, they introduced mechanisms for evaluating documentary support and providing explanations of generated content.

The MMV results demonstrated that multiple independent language-model assessments do not necessarily produce more reliable verification outcomes. Disagreement between verifiers, classification errors, and invalid responses affected the consistency of the ensemble. This suggests that the number of verification models alone should not be treated as an indicator of

reliability. Instead, the quality of individual verification judgments and the method used to combine them are important considerations.

Similarly, the XAI architecture demonstrated the potential value of structured explanations and evidence quotations while revealing limitations in their reliability. Providing an explanation can make a generated answer easier to examine, but it does not independently establish that the answer or explanation is correct. In governance-related applications, where users may interpret citations as evidence of authority, this distinction is particularly important.

These findings indicate that verification and explainability are potentially valuable components of reliable governance-document question answering, but they should not be regarded as substitutes for accurate answer generation or independent validation.

6.5.3 Implications for the Research Objectives

The primary objective of this study was to evaluate methods for improving the reliability of question answering over governance documents through a controlled comparison of RAG architectures and enhancement techniques.

The results demonstrate that the evaluated techniques differed substantially in their effectiveness and operational characteristics. RISE achieved the highest measured accuracy, indicating that selective refinement guided by evaluation feedback can improve benchmark performance. However, its dependence on reference-based evaluation limits the extent to which these results can be generalised to practical question-answering scenarios.

RLVR demonstrated a different approach by using retrieved evidence to verify and correct responses without requiring reference answers during inference. Although its improvement in measured accuracy was smaller, the implementation provides evidence that document-based verification can support targeted correction. The relatively large number of triggered corrections

compared with the number of improved correctness classifications also highlights the need to distinguish between identifying potential issues and successfully resolving them.

MMV and XAI addressed complementary aspects of reliability through independent verification and explanation generation. Their results demonstrated that these mechanisms can provide additional information about generated answers, while also introducing new opportunities for inconsistent or incorrect judgments. Taken together, the findings indicate that no single enhancement technique was uniformly superior across all dimensions of reliability. The most appropriate approach depends on whether the system prioritises answer correction, evidence verification, explanation, or computational efficiency.

6.5.4 Practical Considerations and Future Development

For practical governance-document QA systems, the findings suggest that a combination of reliable retrieval, selective verification, and carefully controlled correction may offer a promising direction for further development. Rather than applying computationally expensive processing to every response, a system could prioritise additional examination for answers exhibiting uncertainty, insufficient evidence, or potential inconsistencies. However, the results also demonstrate that such an approach requires dependable mechanisms for identifying these conditions. In particular, verification processes should be assessed independently of the answer-generation process to reduce the risk of accepting incorrect responses or unnecessarily modifying correct ones.

Future implementations would also benefit from stronger evidence validation, clearer handling of unresolved or contradictory information, and mechanisms for communicating uncertainty when the available governance documents do not support a definitive answer. These considerations are especially relevant where incorrect interpretations could influence organisational decisions or procedural compliance.

6.5.5 Overall Implications

Overall, this study demonstrates that enhancing RAG with refinement, verification, and explanation mechanisms can influence different aspects of question-answering reliability, but additional processing does not automatically produce more accurate or trustworthy responses.

The strongest measured QA performance was achieved through selective refinement, while document-based verification provided an alternative correction mechanism that did not depend on reference answers during inference. Multi-model verification and explainability offered additional assessment capabilities, although their reliability remained dependent on the quality of the underlying language-model judgments.

The central implication is that reliable governance-document QA requires not only effective retrieval and generation but also dependable methods for determining when an answer is incorrect, whether a correction is justified, and how supporting evidence should be interpreted. These findings provide a basis for the conclusions and future research directions presented in the following chapter.

7 Sustainability and Ethical Considerations

7.1 Sustainability Considerations

The sustainability implications of this research primarily relate to the computational resources and the energy consumption associated with operating LLMs. Although Retrieval-Augmented Generation (RAG) enables existing language models to access external information without needing any additional model training, the generation, verification, and refinement processes still require computational resources. This study evaluated five RAG architectures with varying computational requirements. The enhanced architectures introduced additional language-model calls and increased response latency in relation to the baseline. In particular, Multi-Model Verification (MMV) required three additional verification models for each generated response, resulting in a substantially greater computational overhead as compared to the baseline RAG architecture. These findings highlight the importance of considering computational efficiency when developing a reliability-enhanced AI system, as increasing the processing requirements may also contribute to a greater total energy consumption.

From a sustainability perspective, the use of a locally deployed, pre-trained language model through Ollama avoids the need to train new models from scratch. Furthermore, the selective refinement mechanisms investigated in this research demonstrate that there is a potential to limit any unnecessary processing by directing additional computational effort towards responses that are identified as requiring improvement. However, computational efficiency must be balanced against the reliability requirements of the intended application. While additional verification may increase resource consumption, incorrect governance information may also lead to inefficient decision-making or unnecessary administrative processes. Therefore, future systems should aim to minimise computational overhead while maintaining an appropriate level of answer reliability.

7.2 Ethical Considerations

The primary ethical considerations of this research concern the accuracy, transparency, and responsible use of AI-generated information. Governance documents establish organisational rules, responsibilities, and procedures, meaning that any incorrect or misleading responses could result in detrimental misunderstandings or inappropriate decisions. Although RAG incorporates retrieved documentary evidence to support generated answers, it does not guarantee factual correctness of any of its responses. The experimental findings demonstrated that automated refinement, verification, and explanation mechanisms may also produce incorrect judgments. Therefore, it is appropriate that users should not assume that an answer is reliable solely because it has undergone additional verification or includes supporting citations.

Transparency is another important consideration. The Explainable AI (XAI) architecture investigated methods for providing supporting evidence and identifying potential errors in generated responses. However, the results demonstrated that explanations could still contain inconsistencies and the presence of verified quotations did not necessarily establish that the associated interpretation was actually correct. This highlights the importance of being able to presenting supporting evidence accurately and avoiding any misleading claims of certainty. The research also raises considerations regarding accountability and human oversight. AI-generated responses should support rather than fully replace human judgment, particularly when interpreting governance provisions that may affect decisions made by the organisation. Where information is ambiguous, contradictory, or insufficient, systems should communicate these limitations rather than generate unsupported conclusions.

Finally, the study used a defined collection of governance documents and locally deployed models, allowing the experimental process to be conducted without transmitting the document corpus to an external language-model service during the inference. Nevertheless, practical deployments would require appropriate consideration of document confidentiality, access

permissions, and the handling of potentially sensitive organisational information. Overall, this research highlights the importance of developing AI systems that are both computationally responsible and ethically reliable. Improving question-answering accuracy must be accompanied by careful consideration of resource efficiency, transparency, accountability, and the potential consequences of incorrect information.

8 Conclusion and Future Work

8.1 Conclusion

This research evaluated methods for improving the reliability of Retrieval-Augmented Generation (RAG) when answering questions over governance documents. Five architectures were investigated: Baseline RAG, RISE-inspired refinement, RLVR-inspired verification and correction, Multi-Model Verification (MMV), and Explainable Artificial Intelligence (XAI). Their performance was assessed using a benchmark of 110 questions across 16 categories, considering answer correctness, textual and semantic similarity, and computational performance. The results demonstrated that Baseline RAG provided a strong foundation, achieving 94.55% accuracy. RISE-inspired refinement achieved the highest measured accuracy of 99.09%, followed by RLVR at 97.27%, MMV at 96.36%, and XAI at 95.45%. However, the findings also highlighted limitations in automated evaluation, verification consistency, and explanation reliability. In particular, improvements in evaluator-assigned scores did not always establish factual improvement, while additional verification and explanation mechanisms introduced computational overhead without necessarily improving answer correctness.

Overall, the study demonstrates that selective refinement and evidence-based verification can support more reliable governance-document question answering. However, increasing architectural complexity does not guarantee improved reliability. The findings emphasise the importance of dependable evaluation and verification mechanisms, alongside effective retrieval and generation, when developing trustworthy document-grounded question-answering systems.

8.2 Future Work

Future research could extend this study by evaluating the architectures using larger and more diverse governance-document collections and more challenging question benchmarks. Incorporating human expert assessments and independent verification methods could also improve evaluation reliability and reduce dependence on automated correctness judgments.

Further investigation could explore combining complementary enhancement techniques, such as evidence-based verification, selective refinement, and explainability, within a unified RAG architecture. Particular attention should be given to developing correction mechanisms that do not require reference answers during inference, while minimising unnecessary revisions and computational overhead. Finally, evaluating these approaches in practical governance-document question-answering environments would provide further insight into their scalability, usability, and reliability under real-world conditions.

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10 Appendix A – Repository README

The source code for all five RAG architectures, together with the supporting benchmark materials and execution instructions, is available in the accompanying GitHub repository (https://github.com/TheAsianTyrant/Thesis-Model-Files). The repository provides the implementation details required to reproduce the experimental procedures described in this thesis. Additional information regarding software dependencies, model configurations, and execution instructions is provided in the repository README, which is reproduced in Appendix A.

README

Evaluating Methods of Reliable Question Answering over Governance Documents

Research Project — Curtin University

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1. Project Overview

This repository contains the source code, experimental configurations, and supporting materials developed for the research thesis *Evaluating Methods of Reliable Question Answering over Governance Documents*.

The project investigates methods for improving the reliability of Retrieval-Augmented Generation (RAG) when answering questions using governance documents from the IEEE Systems, Man, and Cybernetics Society (SMCS).

Five architectures were implemented and evaluated:

Baseline RAG: A conventional retrieval-augmented generation pipeline using document retrieval and language-model-based answer generation.

RAG + RISE-Inspired Refinement: Introduces selective iterative refinement of responses based on evaluation feedback.

RAG + RLVR-Inspired Verification: Incorporates evidence-based claim verification and targeted correction during inference. This is an RLVR-inspired approach, not a reinforcement-learning training implementation.

RAG + Multi-Model Verification (MMV): Uses three language models to independently verify generated answers and combines their assessments through majority voting.

RAG + Explainable AI (XAI): Generates post-hoc explanations, claim-level assessments, and supporting documentary evidence without modifying the generated answers.

The architectures were evaluated using 110 benchmark questions across 16 categories, covering different retrieval, reasoning, and interpretation requirements.

2. Repository Structure

The repository is organised into the following directories:

Folder	Description
BaseRag/	Baseline RAG implementation and associated files.
RAG+RISE/	RISE-inspired refinement implementation and associated files.
RAG+RLVR/	RLVR-inspired verification implementation and associated files.
RAG+MMV/	Multi-model verification implementation and associated files.
RAG+XAI/	Explainable AI implementation and associated files.
Necessary Files/	Supporting files required to execute the architectures, including the benchmark questions and governance documents.
README.md	Project documentation and execution instructions.

Each architecture is stored in its own directory, containing the corresponding Python implementation and associated files.

The actual filenames and folder paths should match those used in the final implementation.

3. System Requirements

The project uses Python and locally deployed language models through Ollama.

Required software:

Python 3

Ollama

Python packages specified in requirements.txt

Hardware:

A computer capable of running the required language models locally.

Sufficient memory and storage for model inference, embeddings, and the document database.

The experiments were conducted using a local computing environment. Exact hardware specifications and software versions should be recorded to support reproducibility.

4. Language Models

The following Ollama models were used:

Model	Purpose
llama3	Answer generation and language-model-based processing
nomic-embed-text	Document and question embeddings
qwen3:8b	Additional MMV verifier
mistral:7b	Additional MMV verifier

Install Ollama from https://ollama.com.

Download the models using:

```
ollama pull llama3

ollama pull nomic-embed-text

ollama pull qwen3:8b

ollama pull mistral:7b
```

Ensure the Ollama service is running before executing the Python scripts.

5. Installation

Clone or download the repository and open a terminal in the project directory.

Create a Python virtual environment:

```
python -m venv .venv
```

Activate the environment on Windows:

```
.venv\Scripts\activate
```

Alternatively, on macOS or Linux:

```
source .venv/bin/activate
```

Install the required Python packages:

```
pip install -r requirements.txt
```

The implementations use libraries including ChromaDB, Ollama, pypdf, ROUGE Score, SacreBLEU, and BERTScore.

6. Governance Document Corpus

The retrieval corpus consists of four IEEE-related governance documents:

- Draft Operational Manual V5.
- IEEE Constitution and Bylaws.
- SMCS Constitution and Bylaws, August 2023.
- SMCS Technical Manual, August 2023.

Place the required PDF files in the docs/ directory using the filenames expected by the scripts.

The documents are processed and divided into overlapping text chunks before being embedded and stored in ChromaDB.

If the original documents are not included in the repository, users must obtain them from authorised sources.

7. RAG Configuration

The baseline retrieval configuration used in the experiments is summarised below.

Parameter	Value
Embedding model	nomic-embed-text
Generation model	llama3
Vector database	ChromaDB
Chunk size	1,000
Chunk overlap	200
Retrieved passages	Top 12
Evaluation questions	110
Correctness score	0–3
Correct classification	Score of 3

Additional architecture-specific settings, including refinement limits, verification procedures, and prompt configurations, are defined in the corresponding implementation scripts.

8. Running the Experiments

Each RAG architecture can be executed independently.

To run an implementation:

1. Download or clone the repository.
2. Install Python, Ollama, and the required dependencies.
3. Download the required Ollama models.
4. Open the Necessary Files/ directory.
5. Copy the required supporting files and folders into the directory of the architecture you wish to execute (e.g., BaseRag/ or RAG+RISE/).
6. Open a terminal in the selected architecture's directory.
7. Run the corresponding Python script using its actual filename.
8. For example, if the baseline implementation is named RAG.py:
9. python "RAG.py"

Important: The supporting files from Necessary Files/ must be placed inside the selected architecture directory before execution. Otherwise, the script may be unable to locate the required governance documents or benchmark questions.

Ensure that the Ollama service is running before executing the scripts.

9. Evaluation Framework

The benchmark contains 110 questions distributed across 16 categories, including direct retrieval, cross-document reasoning, semantic retrieval, hallucination traps, ambiguity, contradictions, multi-hop reasoning, temporal interpretation, negation, and conditional questions.

The architectures were assessed using:

- Question-answering performance
- Correctness accuracy.
- Evaluation rubric scores.
- Category-level accuracy.

Textual and semantic similarity

- ROUGE-1, ROUGE-2, and ROUGE-L F1.
- BLEU.
- BERTScore F1.

Computational performance

- Total response latency.
- Retrieval and generation durations.
- Language-model calls.
- Embedding calls.

Architecture-specific evaluation also examined refinement outcomes, verification judgments, and explanation behaviour.

10. Experimental Results

The final benchmark accuracies were:

Architecture	Accuracy
Baseline RAG	94.55%
RISE-Inspired	99.09%
RLVR-Inspired	97.27%
Multi-Model Verification	96.36%
Explainable AI	95.45%

These values represent correctness under the implemented benchmark evaluation framework.

The RISE and RLVR configurations can modify generated answers. In contrast, MMV and XAI perform post-generation assessment without changing the original response. Therefore, accuracy differences for MMV and XAI should not be interpreted as improvements directly caused by their verification or explanation mechanisms.

Detailed experimental outputs and summaries are provided in the results/ directory, where available.

11. Reproducibility and Limitations

The experimental findings are specific to the selected governance documents, benchmark questions, model configurations, and computing environment. Although the implementations were designed to support consistent evaluation, locally executed language models may exhibit variations across runs or hardware environments.

Furthermore, automated evaluation and verification mechanisms can produce incorrect or inconsistent judgments. Reported benchmark accuracy should therefore not be interpreted as a guarantee of factual correctness. The RISE-inspired implementation also uses reference-based benchmark evaluation to guide refinement decisions. Such reference answers would not ordinarily be available when answering new questions in a deployed system.

12. Academic Attribution

This repository contains the source code and supporting materials developed as part of the research thesis *Evaluating Methods of Reliable Question Answering over Governance Documents* at Curtin University.

The repository is provided for research transparency and reproducibility. Anyone referencing or building upon this work is expected to provide appropriate academic attribution and comply with applicable academic integrity requirements.

Suggested Citation:

Lim, L. (2026). *Evaluating Methods of Reliable Question Answering over Governance Documents* [Undergraduate research thesis]. Curtin University.

Repository: https://github.com/TheAsianTyrant/Thesis-Model-Files

13. License

No open-source license has currently been assigned to this repository. All rights are reserved unless otherwise stated.

Permission to view the repository does not imply permission to reuse, modify, or redistribute its contents beyond applicable legal rights or permissions.

The inclusion of third-party libraries and governance documents remains subject to their respective licenses and usage conditions.

11 Appendix B – Question Benchmark

---QUESTION---

ID: Q001

CATEGORY: Direct Retrieval

QUESTION: Who may become a member of the SMCS?

ANSWER: Any IEEE member of any grade, including Student Members, who has an interest in the Society's field of interest may become a member of the SMCS upon payment of the required Society membership fee.

---QUESTION---

ID: Q002

CATEGORY: Direct Retrieval

QUESTION: Who may become a Society Affiliate?

ANSWER: Individuals who are not IEEE members but are interested in the Society's field of interest and meet the criteria established in the Bylaws may become Society Affiliates upon payment of the required IEEE registration and Society affiliation fees.

---QUESTION---

ID: Q003

CATEGORY: Direct Retrieval

QUESTION: What is the objective of the SMCS?

ANSWER: The SMCS aims to advance the theory, practice, and interdisciplinary aspects of systems science and engineering, human-machine systems, and cybernetics through scientific,

literary, educational, and professional activities. It promotes technical exchange, publications, conferences, and professional development for its members.

---QUESTION---

ID: Q004

CATEGORY: Direct Retrieval

QUESTION: What constitutes the Field of Interest?

ANSWER: The Society's field of interest includes large-scale systems; optimisation, decision analysis, modelling, simulation, testing and evaluation; cybernetics; pattern recognition; adaptive and learning systems; biocybernetics; human-machine systems; and applications of systems engineering and cybernetics across biological, ecological, environmental, healthcare, socioeconomic, transportation, urban, and other national priority systems.

---QUESTION---

ID: Q005

CATEGORY: Direct Retrieval

QUESTION: Who appoints the Secretary?

ANSWER: The Secretary is appointed by the President.

---QUESTION---

ID: Q006

CATEGORY: Direct Retrieval

QUESTION: What are the Secretary's responsibilities?

ANSWER: The Secretary sends notices as directed by the Board of Governors or the Bylaws, prepares meeting agendas, records, and distributes minutes, reports Executive Committee actions, maintains Society records, and prepares reports required by the Board of Governors, the IEEE Technical Activities Board, and IEEE Bylaws.

---QUESTION---

ID: Q007

CATEGORY: Direct Retrieval

QUESTION: What are the Treasurer's responsibilities?

ANSWER: The Treasurer promotes sound financial practices and policies, assists in preparing and presenting the Society's budget, coordinates financial matters with IEEE, and processes expense reports. The Treasurer reports through the Vice President for Finance.

---QUESTION---

ID: Q008

CATEGORY: Direct Retrieval

QUESTION: How many voting members may the Board of Governors contain?

ANSWER: The Board of Governors consists of up to 27 voting members: the two most recent Past Presidents, ten Administrative Officers, and fifteen Members-at-Large.

---QUESTION---

ID: Q009

CATEGORY: Direct Retrieval

QUESTION: How long is the President's term?

ANSWER: The President serves a two-year calendar term. A President may not serve more than two consecutive years in that office.

---QUESTION---

ID: Q010

CATEGORY: Direct Retrieval

QUESTION: Who may sign authorised disbursements?

ANSWER: Authorised disbursements may be signed by the President, Treasurer, Executive Director, or other persons specifically designated by the Board of Governors.

---QUESTION---

ID: Q011

CATEGORY: Direct Retrieval

QUESTION: What is required to form a Chapter?

ANSWER: A Chapter may be formed by IEEE members in a geographic area who petition in accordance with IEEE policies. Chapters operate under the Society Constitution, Bylaws, and applicable IEEE rules. (The SMCS Constitution references IEEE Chapter formation procedures rather than defining the detailed requirements itself.)

---QUESTION---

ID: Q012

CATEGORY: Direct Retrieval

QUESTION: Under what circumstances is a BoG member removed for non-attendance?

ANSWER: A voting member of the Board of Governors who is absent from two consecutive Board meetings is removed from the Board. The same provision applies to non-voting members.

---QUESTION---

ID: Q013

CATEGORY: Cross-Document

QUESTION: How does the Operational Manual relate to the Constitution and Bylaws when a conflict exists?

ANSWER: The Operational Manual provides operational guidance and best practices for IEEE SMCS activities but is subordinate to the Constitution and Bylaws. If there is conflicting wording or ambiguity, the Constitution takes precedence, followed by the Bylaws if necessary.

---QUESTION---

ID: Q014

CATEGORY: Cross-Document

QUESTION: Which document explains how Technical Committees operate after they are established?

ANSWER: The Technical Committee Manual defines how Technical Committees operate after establishment, including the responsibilities of Area Vice Presidents, Area Committees, Area AVPs, Technical Committee Chairs, performance evaluation, reporting requirements, and

procedures for restructuring or discontinuing committees. The Constitution and Bylaws establish the governance framework but do not describe daily Technical Committee operations in the same level of detail.

---QUESTION---

ID: Q015

CATEGORY: Cross-Document

QUESTION: How do the Constitution and Technical Manual divide responsibility for creating a Technical Committee?

ANSWER: The Constitution and Bylaws define the governance structure and identify the officers and committees responsible for technical activities. The Technical Committee Manual provides the operational process for creating a Technical Committee, including proposal submission, review by the relevant Area Committee, approval by the Area Vice President, reporting to the Board of Governors, and notification to Society members.

---QUESTION---

ID: Q016

CATEGORY: Cross-Document

QUESTION: Which document specifies committee objectives, and which specifies committee procedures?

ANSWER: The Constitution and Bylaws establish the purpose, authority, and responsibilities of Society committees. The Technical Committee Manual specifies the operational procedures used to manage Technical Committees, including reporting, evaluation, restructuring, and committee administration.

---QUESTION---

ID: Q017

CATEGORY: Cross-Document

QUESTION: How are Technical Committee Chairs governed by both the Technical Manual and the Society Bylaws?

ANSWER: The Society Bylaws place Technical Committees under the governance of the appropriate Area Vice Presidents and standing committees. The Technical Committee Manual further defines the responsibilities of Technical Committee Chairs, including organising technical activities, recruiting members, submitting biannual reports, and the circumstances under which an Area Vice President may recommend replacing a Chair or restructuring or discontinuing a committee.

---QUESTION---

ID: Q018

CATEGORY: Cross-Document

QUESTION: Which document defines governance principles, and which defines operational procedures?

ANSWER: The Constitution and Bylaws define the Society's governance principles, organisational structure, officers, committees, elections, and authority. The Operational Manual

and Technical Committee Manual define the operational procedures and best practices used to implement those governance principles in day-to-day Society activities.

---QUESTION---

ID: Q019

CATEGORY: Cross-Document

QUESTION: How do the Society Constitution and IEEE Constitution differ in organisational scope?

ANSWER: The IEEE Constitution establishes the objectives, governance, and policies of the IEEE as the parent organisation. The SMCS Constitution governs the IEEE Systems, Man, and Cybernetics Society specifically, defining its field of interest, organisational structure, officers, committees, and Society-specific governance while operating within the broader IEEE framework.

---QUESTION---

ID: Q020

CATEGORY: Cross-Document

QUESTION: Which document determines precedence when governance documents disagree?

ANSWER: The Constitution has the highest precedence among the Society governance documents. The Operational Manual explicitly states that it is subordinate to the Constitution

and Bylaws, and that in cases of conflict or ambiguity the Constitution takes precedence, followed by the Bylaws where necessary.

---QUESTION---

ID: Q021

CATEGORY: Cross-Document

QUESTION: How do the Society Bylaws and Technical Manual together define reporting responsibilities?

ANSWER: The Society Bylaws assign governance responsibilities to officers and standing committees, while the Technical Committee Manual specifies the operational reporting responsibilities of Technical Committees. For example, Technical Committee Chairs must submit biannual activity reports to the appropriate Area Vice President, who in turn reports Technical Committee activities and major decisions to the Board of Governors.

---QUESTION---

ID: Q022

CATEGORY: Cross-Document

QUESTION: Which documents must be followed when establishing an SMCS committee?

ANSWER: Establishing an SMCS Technical Committee requires following the Society's governance documents and operational procedures. The Constitution and Bylaws define the governance authority responsible for technical activities, while the Technical Committee

Manual specifies the proposal, review, approval, reporting, and announcement procedures required to establish a Technical Committee. The Operational Manual also states that its guidance is subordinate to the Constitution and Bylaws.

---QUESTION---

ID: Q023

CATEGORY: Semantic Retrieval

QUESTION: What happens if the President is unable to perform their duties?

ANSWER: If the President is unable to perform the duties of the office, the President-Elect assumes the duties of the President. If both the President and President-Elect are unable to serve, the Board of Governors fills the vacancy in accordance with the Society's governing procedures.

---QUESTION---

ID: Q024

CATEGORY: Semantic Retrieval

QUESTION: How is continuity of leadership maintained?

ANSWER: Continuity of leadership is maintained through a structured succession process in which the President-Elect succeeds the President, Past Presidents remain on the Board of Governors, and newly elected officers assume office on 1 January following the Annual Meeting. This provides overlap between outgoing and incoming leadership.

---QUESTION---

ID: Q025

CATEGORY: Semantic Retrieval

QUESTION: Why must Technical Committees submit biannual reports?

ANSWER: Technical Committees submit biannual reports so the appropriate Area Vice President can monitor committee activity, evaluate performance, identify inactive committees, support planning, and report Technical Committee activities to the Board of Governors. These reports are also used when evaluating whether a committee should continue, be restructured, or be discontinued.

---QUESTION---

ID: Q026

CATEGORY: Semantic Retrieval

QUESTION: Under what conditions can a Technical Committee be discontinued?

ANSWER: A Technical Committee may be discontinued if it has been inactive for at least one year with no credible plan for revival, or if its Chair and at least two committee members jointly request its discontinuation with supporting reasons. The final decision rests with the appropriate Area Committee, after which the Area Vice President reports the decision to the Board of Governors and notifies Society members.

---QUESTION---

ID: Q027

CATEGORY: Semantic Retrieval

QUESTION: How does a proposal for a new Technical Committee become approved?

ANSWER: A proposal is submitted to the appropriate Area Vice President and must include the committee’s name, proposed Chair, members, objectives, and activity plan. The Area Committee reviews the proposal for overlap and feasibility, consulting existing Technical Committee Chairs where appropriate. If approved by the Area Vice President, the committee may begin operating, and the decision is reported to the Board of Governors and announced to Society members.

---QUESTION---

ID: Q028

CATEGORY: Semantic Retrieval

QUESTION: What responsibilities does the VP Organisation and Planning have regarding Society planning?

ANSWER: The Vice President for Organization and Planning chairs the Organization and Planning Committee and is responsible for updating the Society's Five-Year Plan, developing strategic and tactical plans, interpreting the Constitution and Bylaws when questions arise, and preparing proposed modifications to the Constitution and Bylaws for consideration by the Board of Governors.

---QUESTION---

ID: Q029

CATEGORY: Semantic Retrieval

QUESTION: What actions must occur before a Technical Committee officially begins operating?

ANSWER: Before a Technical Committee begins operating, a proposal containing the committee's name, Chair, members, objectives, and activity plan must be submitted. The Area Committee reviews the proposal, and the Area Vice President approves it. Only after this approval may the committee commence activities, after which the decision is reported to the Board of Governors and announced to Society members.

---QUESTION---

ID: Q030

CATEGORY: Semantic Retrieval

QUESTION: How are officer vacancies filled?

ANSWER: If the office of President becomes vacant, the President-Elect assumes the Presidency. Other officer vacancies are filled according to the procedures established in the Constitution and Bylaws through action of the Board of Governors. The governing documents assign the Board responsibility for filling vacancies where appropriate.

---QUESTION---

ID: Q031

CATEGORY: Semantic Retrieval

QUESTION: Why are Technical Committee evaluations conducted annually?

ANSWER: Annual evaluations measure each Technical Committee's effectiveness using criteria such as submission of biannual reports, technical activities, member acquisition, maintenance of an up-to-date webpage, and future plans. The evaluations help ensure committees remain active, contribute to Society objectives, and identify committees requiring restructuring or discontinuation.

---QUESTION---

ID: Q032

CATEGORY: Semantic Retrieval

QUESTION: What role does the Executive Committee play in Society governance?

ANSWER: The Executive Committee assists in managing the Society between meetings of the Board of Governors by conducting Society business delegated to it by the Board. Its actions are reported to the Board of Governors, ensuring governance continuity while remaining accountable to the Board.

---QUESTION---

ID: Q033

CATEGORY: Hallucination Trap

QUESTION: Who is the current IEEE President?

ANSWER: The provided governance documents do not identify the current IEEE President.

---QUESTION---

ID: Q034

CATEGORY: Hallucination Trap

QUESTION: What salary do Society officers receive?

ANSWER: The governance documents do not specify salaries, remuneration, or compensation for Society officers.

---QUESTION---

ID: Q035

CATEGORY: Hallucination Trap

QUESTION: Which programming language must committee software use?

ANSWER: The governance documents do not prescribe any programming language for committee software or technical systems.

---QUESTION---

ID: Q036

CATEGORY: Hallucination Trap

QUESTION: Which cloud platform should committees use?

ANSWER: The governance documents do not specify or recommend any cloud platform or online service for committee operations.

---QUESTION---

ID: Q037

CATEGORY: Hallucination Trap

QUESTION: What branding colours are mandatory?

ANSWER: The governance documents do not define any mandatory branding colours or visual identity requirements for the Society or its committees.

---QUESTION---

ID: Q038

CATEGORY: Hallucination Trap

QUESTION: What AI certification is required for members?

ANSWER: The governance documents do not require members to obtain any AI ethics certification or other professional certification.

---QUESTION---

ID: Q039

CATEGORY: Hallucination Trap

QUESTION: What office hours must committee chairs maintain?

ANSWER: The governance documents do not specify office hours or availability requirements for committee chairs.

---QUESTION---

ID: Q040

CATEGORY: Hallucination Trap

QUESTION: Does the Society require members to publish annually?

ANSWER: No. The governance documents do not require Society members to publish annually. While Technical Committee Chairs and Technical Committees are expected to contribute to Society activities, no annual publication requirement is imposed on Society members.

---QUESTION---

ID: Q041

CATEGORY: Ambiguous

QUESTION: Who approves changes?

ANSWER: This question cannot be answered as stated because it does not specify what type of changes are being referred to. Different changes are approved by different authorities. For example, constitutional amendments are approved through the amendment process defined in the Constitution and Bylaws, while changes to Technical Committees are approved by the appropriate Area Vice President and reported to the Board of Governors.

---QUESTION---

ID: Q042

CATEGORY: Ambiguous

QUESTION: Who manages committees?

ANSWER: This question is ambiguous because multiple committees exist within the Society. Standing committees are chaired by the appropriate Vice Presidents under the Society Bylaws, while Technical Committees are managed by the corresponding Area Vice President with support from the Area AVP and Area Committee as described in the Technical Committee Manual.

---QUESTION---

ID: Q043

CATEGORY: Ambiguous

QUESTION: What happens when someone resigns?

ANSWER: This question cannot be answered without specifying who is resigning. The governing documents contain different procedures depending on whether the person is the President, another officer, a Board of Governors member, or a Technical Committee Chair.

---QUESTION---

ID: Q044

CATEGORY: Ambiguous

QUESTION: Who oversees finances?

ANSWER: The Society's finances are overseen by multiple roles. The Vice President for Finance has responsibility for financial matters, while the Treasurer promotes sound financial practices, assists with budget preparation, and processes expense reports. Financial oversight is ultimately exercised within the Society's governance structure under the Board of Governors.

---QUESTION---

ID: Q045

CATEGORY: Ambiguous

QUESTION: What rules apply to committees?

ANSWER: This question is ambiguous because different committees are governed by different documents. Standing committees are governed by the Society Constitution and Bylaws, while Technical Committees are additionally governed by the Technical Committee Manual and applicable operational procedures.

---QUESTION---

ID: Q046

CATEGORY: Contradicting

QUESTION: Which document takes precedence over the Operational Manual?

ANSWER: The SMCS Constitution takes precedence over the Operational Manual. If there is conflicting wording or ambiguity, the Constitution has precedence. If further clarification is required, the Bylaws take precedence over the Operational Manual.

---QUESTION---

ID: Q047

CATEGORY: Contradicting

QUESTION: What should be followed if the Technical Manual conflicts with the Constitution?

ANSWER: The Constitution should be followed. The Technical Committee Manual provides operational procedures and must be applied consistently with the Society's governing documents. Where a conflict exists, the Constitution has higher authority.

---QUESTION---

ID: Q048

CATEGORY: Contradicting

QUESTION: How are conflicts between Society and IEEE governance resolved?

ANSWER: The Society Constitution states that the SMCS operates within the framework of the IEEE. However, the uploaded governance documents do not explicitly describe a formal conflict-resolution process between IEEE and Society governance documents. Therefore, no definitive precedence beyond the stated relationship can be established from the provided documents.

---QUESTION---

ID: Q049

CATEGORY: Contradicting

QUESTION: Which governance document has ultimate authority over Society procedures?

ANSWER: The SMCS Constitution has the highest authority over Society governance and procedures, with the Bylaws providing additional governance detail. Operational documents, including the Operational Manual and Technical Committee Manual, are subordinate to these governing documents.

---QUESTION---

ID: Q050

CATEGORY: Contradicting

QUESTION: Which document should be followed when committee procedures are inconsistent?

ANSWER: Committee procedures should first follow the SMCS Constitution, then the SMCS Bylaws where additional procedural guidance is provided. The Operational Manual and Technical Committee Manual should only be followed where they are consistent with these governing documents.

---QUESTION---

ID: Q051

CATEGORY: Tricky

QUESTION: Is a Vice President allowed to serve more than two terms?

ANSWER: Yes, provided the terms are not consecutive in the same Vice President position. The Bylaws limit a Vice President to serving no more than two consecutive terms in the same office, but they may serve additional non-consecutive terms or serve in a different Vice President role.

---QUESTION---

ID: Q052

CATEGORY: Tricky

QUESTION: Can a Society Affiliate vote in Board of Governors elections?

ANSWER: No. Society Affiliates may affiliate with the Society but do not possess the same governance rights as IEEE members. Board of Governors elections are conducted by eligible Society members.

---QUESTION---

ID: Q053

CATEGORY: Tricky

QUESTION: Does every Board of Governors meeting always have 27 voting members?

ANSWER: No. The Board of Governors may contain up to 27 voting members, but the actual number depends on the Society's current leadership structure (for example, whether there is a President-Elect serving during that period).

---QUESTION---

ID: Q054

CATEGORY: Tricky

QUESTION: Can a Technical Committee be created immediately after a proposal is submitted?

ANSWER: No. A proposal must first be reviewed by the appropriate Area Committee, approved by the Area Vice President, and then reported to the Board of Governors before the committee is formally established.

---QUESTION---

ID: Q055

CATEGORY: Tricky

QUESTION: Are all Technical Committee Chairs members of the Board of Governors?

ANSWER: No. Technical Committee Chairs lead their respective Technical Committees but are not automatically members of the Board of Governors.

---QUESTION---

ID: Q056

CATEGORY: Tricky

QUESTION: Is the Treasurer responsible for approving Society expenditures?

ANSWER: No. The Treasurer is responsible for promoting sound financial practices, assisting with budget preparation, coordinating financial matters, and processing expense reports. The documents do not assign sole authority for approving Society expenditures to the Treasurer.

---QUESTION---

ID: Q057

CATEGORY: Tricky

QUESTION: Can a Technical Committee continue operating without submitting activity reports?

ANSWER: No. Technical Committees are expected to submit biannual activity reports. Failure to meet reporting and activity requirements may lead to evaluation, restructuring, replacement of the Chair, or discontinuation of the committee.

---QUESTION---

ID: Q058

CATEGORY: Tricky

QUESTION: Does every officer have the same term length?

ANSWER: No. Officer term lengths vary by position. For example, the President-Elect serves a one-year term, while the President and Vice Presidents serve two-year terms.

---QUESTION---

ID: Q059

CATEGORY: Tricky

QUESTION: Does removing a Technical Committee automatically remove its Chair from Society membership?

ANSWER: No. Discontinuing or restructuring a Technical Committee affects the committee itself and its leadership role but does not affect the individual's Society membership.

---QUESTION---

ID: Q060

CATEGORY: Tricky

QUESTION: Are all governance documents equal in authority?

ANSWER: No. The governance documents have a defined hierarchy. The Constitution has the highest authority, followed by the Bylaws. The Operational Manual and Technical Committee Manual provide operational guidance and are subordinate to the governing documents.

---QUESTION---

ID: Q061

CATEGORY: Multi-Hop Reasoning

QUESTION: How does a proposal for a new Technical Committee progress from submission to official operation?

ANSWER: A proposal for a new Technical Committee is submitted to the appropriate Area Vice President and must include the proposed committee's name, Chair, members, objectives, and activity plan. The relevant Area Committee reviews the proposal for feasibility and overlap with existing committees, consulting existing Technical Committee Chairs if necessary. If approved by the Area Vice President, the committee may begin operating. The approval is then reported to the Board of Governors and announced to Society members.

---QUESTION---

ID: Q062

CATEGORY: Multi-Hop Reasoning

QUESTION: Which Society officers are involved in financial management, and what are their respective responsibilities?

ANSWER: Financial management is primarily shared between the Vice President for Finance and the Treasurer. The Vice President for Finance oversees the Society's financial affairs and chairs the Finance Committee. The Treasurer promotes sound financial practices, assists with preparing and presenting the Society budget, coordinates financial matters with IEEE, and processes expense reports. Financial oversight is exercised within the Society's governance structure under the Board of Governors.

---QUESTION---

ID: Q063

CATEGORY: Multi-Hop Reasoning

QUESTION: How does information from a Technical Committee eventually reach the Board of Governors?

ANSWER: Technical Committee Chairs submit biannual activity reports to the appropriate Area Vice President. The Area Vice President oversees Technical Committee activities and reports significant activities, decisions, and recommendations to the Board of Governors, ensuring that committee information reaches the Society's governing body.

---QUESTION---

ID: Q064

CATEGORY: Multi-Hop Reasoning

QUESTION: Which governance bodies are responsible for overseeing Technical Committees throughout their lifecycle?

ANSWER: Technical Committees are overseen by the Area Committee, the Area Vice President, and the Board of Governors. The Area Committee reviews proposals and recommends actions, the Area Vice President approves, evaluates, restructures, or recommends discontinuation of Technical Committees, and major decisions are reported to the Board of Governors as part of the Society's governance process.

---QUESTION---

ID: Q065

CATEGORY: Multi-Hop Reasoning

QUESTION: How is continuity of governance maintained when both committee leadership and Society leadership change?

ANSWER: Continuity is maintained through structured succession and governance processes. Society leadership continuity is provided by the President-Elect succeeding the President, the continued service of Past Presidents on the Board of Governors, and scheduled officer transitions. Technical Committee continuity is maintained through oversight by Area Vice Presidents and Area Committees, which may appoint new Chairs, restructure committees, or oversee transitions when leadership changes occur.

---QUESTION---

ID: Q066

CATEGORY: Multi-Hop Reasoning

QUESTION: How do the Constitution, Bylaws, and Operational Manual together establish the hierarchy of governance documents?

ANSWER: The Constitution establishes the Society's fundamental governance structure and has the highest authority. The Bylaws provide detailed governance rules that support and implement the Constitution. The Operational Manual provides operational guidance and best practices and is subordinate to both the Constitution and the Bylaws. Where conflicts or ambiguity exist, the Constitution takes precedence, followed by the Bylaws.

---QUESTION---

ID: Q067

CATEGORY: Multi-Hop Reasoning

QUESTION: How are Technical Committee evaluations used in decisions about restructuring or discontinuation?

ANSWER: Technical Committees are evaluated annually using criteria such as activity reports, technical contributions, membership growth, website maintenance, and future plans. These evaluations help determine whether a committee remains active and effective or whether it should be restructured, merged, have its Chair replaced, or be discontinued.

---QUESTION---

ID: Q068

CATEGORY: Multi-Hop Reasoning

QUESTION: How do officer appointments and Board oversight work together to manage Society operations?

ANSWER: The President appoints certain Society officers, including the Secretary and Treasurer, while the Board of Governors provides overall governance and oversight of Society

affairs. Appointed officers carry out their assigned responsibilities, report through the appropriate governance structure, and support the Board in managing the Society's operations.

---QUESTION---

ID: Q069

CATEGORY: Multi-Hop Reasoning

QUESTION: Which governance documents collectively define the responsibilities of the Vice President for Organization and Planning?

ANSWER: The SMCS Bylaws define the Vice President for Organization and Planning's responsibilities, including chairing the Organization and Planning Committee, maintaining the Society's Five-Year Plan, developing strategic and tactical plans, interpreting governance documents, and proposing amendments. These responsibilities are exercised within the governance framework established by the SMCS Constitution.

---QUESTION---

ID: Q070

CATEGORY: Multi-Hop Reasoning

QUESTION: How do the Constitution and Technical Committee Manual together ensure accountability of Technical Committees?

ANSWER: The Constitution establishes the Society's governance structure and authority for technical activities, while the Technical Committee Manual defines operational accountability through reporting requirements, annual evaluations, performance expectations, and procedures

for restructuring or discontinuing committees. Together, they ensure that Technical Committees remain active, accountable, and aligned with Society objectives.

---QUESTION---

ID: Q071

CATEGORY: Temporal

QUESTION: According to the current governance documents, how much notice must be given before a Board of Governors meeting?

ANSWER: Written notice of a Board of Governors meeting must be provided at least 20 days before the meeting.

---QUESTION---

ID: Q072

CATEGORY: Temporal

QUESTION: If a revised set of Bylaws has been approved by the Society but not yet approved by IEEE, which version applies?

ANSWER: The currently approved Bylaws remain in effect until the revised Bylaws receive the required IEEE approval. Approval by the Society alone is not sufficient for the revised Bylaws to become effective.

---QUESTION---

ID: Q073

CATEGORY: Temporal

QUESTION: When does an amendment to the Constitution take effect after adoption?

ANSWER: An amendment to the Constitution takes effect thirty days after adoption. However, if the amendment changes the status of Officers or Officers-elect, or reduces the number of Directors, each affected Officer and Director continues to serve until their existing term expires.

---QUESTION---

ID: Q074

CATEGORY: Temporal

QUESTION: At the time covered by these documents, is a President-Elect included on the Board of Governors?

ANSWER: Yes. The governing documents include the President-Elect as one of the Administrative Officers serving on the Board of Governors.

---QUESTION---

ID: Q075

CATEGORY: Temporal

QUESTION: Under the current governance rules, how many voting members may the Board of Governors have?

ANSWER: The Board of Governors may have up to 27 voting members, consisting of the two most recent Past Presidents, ten Administrative Officers, and fifteen Members-at-Large.

---QUESTION---

ID: Q076

CATEGORY: Temporal

QUESTION: Which officer assumes office immediately after being elected, and which assumes office after a transition period?

ANSWER: The governance documents state that newly elected officers assume office on 1 January following the Annual Meeting. The President-Elect serves a one-year term before succeeding to the Presidency. The documents do not identify any officer who assumes office immediately upon election.

---QUESTION---

ID: Q077

CATEGORY: Temporal

QUESTION: Which governance document should be followed before an updated version officially comes into effect?

ANSWER: The currently approved version of the applicable governance document should continue to be followed until the revised version has completed the required approval process and becomes effective.

---QUESTION---

ID: Q078

CATEGORY: Temporal

QUESTION: If the Constitution is amended, when should Operational Manual changes be applied?

ANSWER: The Operational Manual should be updated after the constitutional amendment becomes effective so that it remains consistent with the governing documents. Because the

Operational Manual is subordinate to the Constitution and Bylaws, it should reflect any approved constitutional changes.

---QUESTION---

ID: Q079

CATEGORY: Temporal

QUESTION: How would the answer to a governance question change if a newer approved Bylaw replaced the current one?

ANSWER: Once a newer Bylaw has been officially approved and becomes effective, governance questions should be answered using the revised Bylaw rather than the previous version. Until then, answers should be based on the currently effective Bylaws.

---QUESTION---

ID: Q080

CATEGORY: Temporal

QUESTION: Which governance rules apply when a document revision has been proposed but has not yet taken effect?

ANSWER: The currently effective governance documents continue to apply. Proposed revisions that have not yet completed the required approval process do not supersede the existing governance documents.

---QUESTION---

ID: Q081

CATEGORY: Negation

QUESTION: Under what circumstances does the Operational Manual not take precedence?

ANSWER: The Operational Manual does not take precedence when it conflicts with or is inconsistent with the SMCS Constitution or SMCS Bylaws. In cases of conflict or ambiguity, the Constitution takes precedence, followed by the Bylaws.

---QUESTION---

ID: Q082

CATEGORY: Negation

QUESTION: Which members are not required to be IEEE members?

ANSWER: Technical Committee members are not required to be IEEE members. Technical Committee membership is open to all scholars with an interest in the committee's technical area, regardless of IEEE membership status. However, the Society encourages Technical Committees to actively increase the proportion of SMCS members in good standing.

---QUESTION---

ID: Q083

CATEGORY: Negation

QUESTION: When is a Technical Committee not permitted to continue operating?

ANSWER: A Technical Committee is not permitted to continue operating if it is formally discontinued by the Society. This may occur if the committee has been inactive for at least one year with no credible plan for revival, or if its Chair and at least two committee members jointly request discontinuation. The decision is made by the appropriate Area Committee and reported to the Board of Governors.

---QUESTION---

ID: Q084

CATEGORY: Negation

QUESTION: Which governance documents do not define officer responsibilities?

ANSWER: The Operational Manual and the Technical Committee Manual do not define the formal responsibilities of Society officers. Officer responsibilities are defined in the SMCS Constitution and SMCS Bylaws, while the Operational Manual and Technical Committee Manual provide operational guidance and procedures.

---QUESTION---

ID: Q085

CATEGORY: Negation

QUESTION: Under what circumstances is a Board member not permitted to remain on the Board?

ANSWER: A voting member of the Board of Governors is not permitted to remain on the Board if they are absent from two consecutive Board meetings. The same attendance requirement also applies to non-voting Board members.

---QUESTION---

ID: Q086

CATEGORY: Confirmatory

QUESTION: I need a simple majority of votes to amend the Constitution, right?

ANSWER: No. Amendments to the Society Constitution require approval by at least two-thirds of the votes cast by eligible voting members, not a simple majority.

---QUESTION---

ID: Q087

CATEGORY: Confirmatory

QUESTION: A Technical Committee can begin operating as soon as the Chair submits a proposal, correct?

ANSWER: No. Submitting a proposal alone is not sufficient. The proposal must be reviewed by the appropriate Area Committee, approved by the Area Vice President, and then reported to the Board of Governors before the Technical Committee officially begins operating.

---QUESTION---

ID: Q088

CATEGORY: Confirmatory

QUESTION: The Operational Manual overrides the Constitution if it contains more recent information, doesn't it?

ANSWER: No. The Operational Manual is subordinate to the Constitution and Bylaws. If there is conflicting wording or ambiguity, the Constitution takes precedence, followed by the Bylaws, regardless of whether the Operational Manual is more recent.

---QUESTION---

ID: Q089

CATEGORY: Confirmatory

QUESTION: Society Affiliates have exactly the same membership rights as IEEE members, right?

ANSWER: No. Society Affiliates do not have the same rights as IEEE members. Affiliates may participate in the Society, but certain rights, including voting and holding elected office, are reserved for eligible IEEE members as defined in the Constitution and Bylaws.

---QUESTION---

ID: Q090

CATEGORY: Confirmatory

QUESTION: The President-Elect automatically serves a two-year term before becoming President, correct?

ANSWER: No. The President-Elect serves a one-year term before automatically succeeding to the office of President. The Presidency itself is a separate two-year term.

---QUESTION---

ID: Q091

CATEGORY: Clarification Required

QUESTION: Who approves this change?

ANSWER: This question cannot be answered without clarification. The approval authority depends on the type of change being referred to. For example, constitutional amendments, Technical Committee proposals, officer appointments, and committee restructures each have different approval processes.

---QUESTION---

ID: Q092

CATEGORY: Clarification Required

QUESTION: Which committee should I contact?

ANSWER: This question cannot be answered without clarification. The appropriate committee depends on the subject of the enquiry, such as finance, publications, conferences, governance, membership, or technical activities.

---QUESTION---

ID: Q093

CATEGORY: Clarification Required

QUESTION: Who is responsible for this decision?

ANSWER: This question cannot be answered without clarification. Different decisions are made by different governance bodies, including the Board of Governors, Executive Committee, Area Vice Presidents, Standing Committees, or Technical Committee Chairs.

---QUESTION---

ID: Q094

CATEGORY: Clarification Required

QUESTION: Which rules should I follow?

ANSWER: This question cannot be answered without clarification. The applicable rules depend on the situation. Depending on the context, the Constitution, Bylaws, Operational Manual, or Technical Committee Manual may apply.

---QUESTION---

ID: Q095

CATEGORY: Clarification Required

QUESTION: What happens if they resign?

ANSWER: This question cannot be answered without clarification. The outcome depends on who is resigning. Different procedures apply if the individual is the President, another Society officer, a Board of Governors member, a Technical Committee Chair, or another committee member.

---QUESTION---

ID: Q096

CATEGORY: Conditional

QUESTION: If the President resigns during their term, who assumes the duties of President?

ANSWER: If the President resigns or is unable to perform the duties of the office, the President-Elect assumes the duties of the President. If both the President and President-Elect are unable to serve, the Board of Governors fills the vacancy in accordance with the Society's governing procedures.

---QUESTION---

ID: Q097

CATEGORY: Conditional

QUESTION: If a Technical Committee fails to submit reports for multiple years, what actions may the Society take?

ANSWER: Failure to submit the required biannual activity reports contributes to poor annual evaluations. The Society may require corrective action, restructure the committee, replace its Chair, or discontinue the Technical Committee if it is determined to be inactive or no longer effective.

---QUESTION---

ID: Q098

CATEGORY: Conditional

QUESTION: If a proposal for a new Technical Committee overlaps with an existing committee, what happens next?

ANSWER: The Area Committee reviews the proposal to assess its scope and overlap with existing Technical Committees. Existing Technical Committee Chairs may be consulted before a recommendation is made. The Area Vice President then decides whether to approve the proposal, modify it, or reject it.

---QUESTION---

ID: Q099

CATEGORY: Conditional

QUESTION: If a governance document conflicts with the Constitution, which document should be followed?

ANSWER: The Constitution should be followed. It has the highest authority within the Society's governance hierarchy. Where conflicts or ambiguity exist, the Constitution takes precedence, followed by the Bylaws. Operational documents must remain consistent with these governing documents.

---QUESTION---

ID: Q100

CATEGORY: Conditional

QUESTION: If a Board member misses two consecutive meetings, what is the consequence?

ANSWER: A Board member who is absent from two consecutive Board of Governors meetings is removed from the Board of Governors, unless otherwise provided for under the governing procedures.

---QUESTION---

ID: Q101

CATEGORY: Procedural

QUESTION: How can the Society Constitution be amended?

ANSWER: A proposed amendment to the Constitution must first be approved by the Board of Governors. It is then submitted to the eligible voting members of the Society, where it must receive at least two-thirds of the votes cast to be adopted. The amendment subsequently becomes effective in accordance with the Society's governing procedures.

---QUESTION---

ID: Q102

CATEGORY: Procedural

QUESTION: How is a new Technical Committee established?

ANSWER: A proposal for a new Technical Committee is submitted to the appropriate Area Vice President and must include the committee's name, proposed Chair, members, objectives, and activity plan. The proposal is reviewed by the relevant Area Committee, which considers overlap with existing committees and may consult current Technical Committee Chairs. If approved by the Area Vice President, the Technical Committee may begin operating, and the approval is reported to the Board of Governors and announced to Society members.

---QUESTION---

ID: Q103

CATEGORY: Procedural

QUESTION: How is a Technical Committee discontinued?

ANSWER: A Technical Committee may be discontinued if it has been inactive for at least one year with no credible plan for revival, or if its Chair and at least two committee members jointly request discontinuation. The Area Committee reviews the recommendation, and if

approved by the appropriate Area Vice President, the discontinuation is reported to the Board of Governors and communicated to Society members.

---QUESTION---

ID: Q104

CATEGORY: Procedural

QUESTION: How are Society officers elected?

ANSWER: Society officers are elected by the eligible voting members of the Society in accordance with the procedures specified in the Constitution and Bylaws. Elections are conducted by ballot, and successful candidates assume office on 1 January following the Annual Meeting for the term specified for their office.

---QUESTION---

ID: Q105

CATEGORY: Procedural

QUESTION: How are Technical Committee Chairs evaluated?

ANSWER: Technical Committee Chairs are evaluated annually by the appropriate Area Vice President using criteria including submission of biannual activity reports, technical activities, membership growth, maintenance of an up-to-date committee website, and future planning. The evaluation results may be used to determine whether the Chair should continue in the role or whether changes to the committee's leadership or structure are required.

---QUESTION---

ID: Q106

CATEGORY: Summary

QUESTION: Summarise all responsibilities of the Treasurer.

ANSWER: The Treasurer is responsible for promoting sound financial practices throughout the Society, assisting the Vice President for Finance in preparing and presenting the annual budget, coordinating financial matters with IEEE, processing and maintaining expense reports, supporting financial planning and reporting, and carrying out other financial duties assigned by the Board of Governors or the President.

---QUESTION---

ID: Q107

CATEGORY: Summary

QUESTION: Summarise the responsibilities of a Technical Committee Chair.

ANSWER: A Technical Committee Chair provides leadership and direction for the committee by organising and conducting committee meetings, coordinating technical activities, encouraging member participation, maintaining committee membership records, submitting biannual activity reports to the appropriate Area Vice President, maintaining an up-to-date committee webpage, supporting Society conferences and publications, overseeing committee planning and performance, and ensuring the committee operates in accordance with Society policies and procedures.

---QUESTION---

ID: Q108

CATEGORY: Summary

QUESTION: Summarise the duties of the Secretary.

ANSWER: The Secretary is responsible for maintaining the Society's official records, preparing and distributing agendas and minutes for Board of Governors and Executive Committee meetings, maintaining records of governance actions, managing official correspondence and notices, certifying official Society documents when required, and performing other administrative duties assigned by the President or the Board of Governors.

---QUESTION---

ID: Q109

CATEGORY: Differentiation

QUESTION: What is the difference between the Vice President for Finance and the Treasurer?

ANSWER: The Vice President for Finance provides strategic oversight of the Society's financial affairs. This role chairs the Finance Committee, oversees financial planning and policy, and is responsible for the Society's overall financial management. The Treasurer is responsible for the day-to-day administration of the Society's finances. The Treasurer promotes sound financial practices, assists in preparing and presenting the annual budget, coordinates financial matters with IEEE, processes expense reports, and maintains financial records. In

summary, the Vice President for Finance oversees and directs the Society's financial management, while the Treasurer administers and supports the Society's financial operations.

---QUESTION---

ID: Q110

CATEGORY: Differentiation

QUESTION: What is the difference between the Operational Manual and the Technical Committee Manual?

ANSWER: The Operational Manual provides Society-wide operational guidance and best practices for areas such as governance and compliance, membership, finance, conferences, publications, and Technical Committees. It supports the implementation of the Constitution and Bylaws and is subordinate to those governing documents. The Technical Committee Manual is a specialised operational document that focuses specifically on the establishment, management, evaluation, restructuring, and discontinuation of Technical Committees, as well as the responsibilities of Technical Committee Chairs, members, and Area Vice Presidents. In summary, the Operational Manual governs the Society's general operations, whereas the Technical Committee Manual governs the operation and management of Technical Committees specifically.